Replacing Dark Matter with CPTG Geometry

Resolved Rotation Curves and Geometric–Material Mass Equivalents in 153 SPARC Galaxies

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Abstract

Observed disk-galaxy rotation curves routinely imply stronger gravitational acceleration than is accounted for by conventional estimates of the stars and gas. The usual interpretation assigns the discrepancy to an additional, unseen material halo. Curvature Polarization Transport Gravity (CPTG) investigates a different explanation: the physical material remains the source, while coupled geometric polarization and transport alter the gravitational response of spacetime without adding a halo of new matter. General Relativity (GR) is retained continuously as part of the governing equation.

This paper demonstrates how the registered, common CPTG galaxy-reconstruction procedure is applied to a large resolved sample and how its complete gravitational field can be reported as complementary material and geometric *mass equivalents*. We process 153 primary galaxies in the SPARC database, comprising 3,168 measured rotation-curve radii. For each galaxy, the observed kinematics and signed SPARC source templates enter the unchanged CPTG field solver; separately, stellar photometry and the signed gas-force template determine a modeled material inventory. The reconstructed circular field supplies a spherical gravitational mass equivalent, from which that material inventory is subtracted under an explicitly defined common convention. At the last measured radius of each galaxy, the geometric-equivalent share has a mean of 77.96%, median of 80.33%, and outer mass-equivalent-weighted share of 79.03%.

The result is a galaxy-by-galaxy quantitative realization of CPTG's geometric alternative to the inferred dark-matter halo in this rotation-curve analysis. The reported percentages are gravitational field equivalents, not fractions of physical matter. Because the registered field uses locally observed kinematics in its structural support, the exercise is a measured-field reconstruction, not a source-only prediction.

1 Introduction: why a geometric explanation matters

1.1 The galaxy rotation-curve problem

A galaxy provides two closely related kinds of information. Its starlight and gas emission reveal the distribution of its ordinary material; the orbital speeds of stars and gas reveal the total gravitational acceleration required to sustain their motion. In many disk galaxies, the acceleration inferred from measured rotation curves is larger than the acceleration attributed to the mapped stellar and gas components. The standard gravitational interpretation fills that gap with an additional dark-matter halo: a gravitating mass distribution not identified with the observed stars or gas. That inferred halo is a model-dependent gravitational component, not a direct particle census for each individual galaxy [1, 3, 4].

The question motivating this paper is therefore direct: *must the excess gravitational response be interpreted as additional material mass, or can part of the apparent gravitational mass arise from the geometry's response to the material already present?* CPTG addresses the second possibility. Its central proposition is that spacetime geometry can organize and redistribute its gravitational response through coupled curvature polarization and curvature transport. In a galaxy, the resulting circular acceleration can differ from the response of ordinary material treated with the conventional source-only reduction, without inserting a separate halo of invisible particles into the CPTG material inventory.

1.2 How CPTG explains additional gravity through geometry

Curvature Polarization Transport Gravity (CPTG) retains the Einstein curvature of General Relativity (GR) and the stress–energy of ordinary material. It adds two coupled geometric responses: *curvature polarization*, which organizes the nonlinear gravitational response to a source, and *curvature transport*, which redistributes that organization through the geometry. The completed field contains all three geometric contributions continuously. When polarization and transport become negligible, the parent equation recovers GR with the same physical material source [5].

CPTG offers a geometric explanation for the excess gravity conventionally attributed to dark-matter halos. The observed stars and gas remain the material source; the additional gravitational influence is calculated from how curvature responds and is transported. The same parent GR–polarization–transport framework is applied across gravitational scales, including galaxies and black holes, through the appropriate geometry-specific reductions [5, 7]. This shared foundation is central to CPTG's proposed alternative to adding unseen material components.

1.3 A fixed geometric method across 153 galaxies

This paper develops that alternative into a resolved numerical atlas of 153 primary SPARC galaxies and their 3,168 measured rotation-curve radii. The purpose is to calculate each galaxy's complete CPTG gravitational field, independently reconstruct its physical stellar and gas material inventory, and express the resulting gravity as complementary material and geometric *mass equivalents*. Thus, the contribution usually assigned to an additional halo is evaluated through a specified geometric accounting of the completed field.

Every galaxy is processed with *one fixed-form geometric field equation and one prescribed reconstruction procedure*: one radial functional form, one coupled polarization–transport calculation and one mass-equivalent convention. The existing registered procedure supplies three galaxy-specific reduced coefficients and a locally observed structural-support profile; the same rules are applied across all objects. This uniformity introduces no additional individually selected dark-halo density profile or halo-fitting parameter. The coefficients identify an observation-conditioned reconstructed field, rather than converting the fixed-form method into an entirely parameter-free prediction.

The atlas reports the observed rotation curves, signed physical-source force contributions, reconstructed CPTG fields and corresponding mass-equivalent shares for every galaxy. At the last measured radius, the geometric-equivalent fraction has a mean of 77.96%, median of 80.33% and mass-equivalent-weighted share of 79.03%. These are gravitational mass equivalents under the registered radial convention. SPARC's resolved rotation curves and independent photometric and gas information make the full calculation inspectable [3]. The parent-equation connection, detailed methodology and 153 individual galaxy entries that follow show how the same geometric framework is applied across diverse galaxies.

2 From the CPTG parent equation to galactic gravitational response

2.1 Physical matter remains distinct from geometric response

The parent CPTG field equation is [5]

$$G_{\mu\nu} + \Pi^{\rm pol}_{\mu\nu} + \Theta^{\rm trans}_{\mu\nu} = \frac{8\pi G}{c^4} T^{\rm material}_{\mu\nu}. \tag{1}$$

Here $G_{\mu\nu}$ is the Einstein tensor, $T^{\rm material}_{\mu\nu}$ denotes ordinary physical stress–energy, and $\Pi^{\rm pol}_{\mu\nu}$ and $\Theta^{\rm trans}_{\mu\nu}$ describe the two jointly operating geometric-response sectors. The Einstein term is never absent. Equation (1) can be rearranged exactly as

$$\underbrace{\Pi^{\rm pol}_{\mu\nu} + \Theta^{\rm trans}_{\mu\nu}}_{\text{joint geometric response}} = \frac{8\pi G}{c^4} T^{\rm material}_{\mu\nu} - G_{\mu\nu}. \tag{2}$$

In this response-isolating form, the geometric contribution appears on the left and the physical source and Einstein curvature appear together on the right. This is an algebraic rearrangement, not a modified or replacement parent law. It makes explicit why *additional gravitational response* need not be interpreted as *additional physical material*.

If one wishes to express the same dynamics in Einstein-equation bookkeeping, the combined geometry can instead be represented by an effective tensor,

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T^{\rm material}_{\mu\nu} + T^{\rm geom,eq}_{\mu\nu}\right), \qquad T^{\rm geom,eq}_{\mu\nu} = -\frac{c^4}{8\pi G}\left(\Pi^{\rm pol}_{\mu\nu} + \Theta^{\rm trans}_{\mu\nu}\right). \tag{3}$$

The superscript "eq" means *equivalent*: no second material fluid has been added to the theory. The contracted Bianchi identity constrains the combined geometrical response consistently with the ordinary material source under the parent's conservation assumptions. The polarization and transport terms need not be separately conserved material components. Where their sum becomes negligible, equation (1) recovers GR with the same source.

2.2 The measurable galaxy-limit field

The parent field equation gives the physical interpretation; the actual galaxy calculation uses its registered radial reduction [5, 6]. Write the full positive circular acceleration as $g_C(R) = V_C^2(R)/R$, and the signed SPARC unit-template input acceleration as $g_b(R)$. The reduced relation has the form

$$(1 + t_C)\, g_C = g_b + P_C, \qquad t_C = c_C \Sigma_C^2, \qquad P_C = A_C a_\star^{2/3} B_{\rm pol}\, \mathcal{S}, \tag{4}$$

where $\mathcal{S} = [1 + (g_C/a_\star)^{123/50}]^{-11/50}$ is the registered polarization suppression. Here P_C is the nonlinear polarization contribution and t_C enters the coupled transport factor; these quantities are computed together from the registered gravitational reconstruction, not treated as adjustable halo components. Equation (4) also gives the exact identity

$$g_C = g_b + \underbrace{(P_C - t_C g_C)}_{\text{net geometric field contribution}}. \tag{5}$$

Thus the completed acceleration is the registered signed source input plus the *net* geometric field response. This is the galaxy-level connection to the source–response distinction of equation (1). The net field contribution in equation (5) is *not* the same numerical quantity as the mass-equivalent remainder defined later: the former separates radial accelerations using a signed disk-force template, whereas the latter subtracts an independently reconstructed material *aperture mass* from a declared spherical gravitational field equivalent. The distinction is necessary to interpret the figures correctly.

Correspondence between the parent equation and the registered galaxy reduction. The parent equation establishes the material–geometric separation at the variational level; the procedure evaluated in this atlas is the *registered effective radial reduction*, under resolved rotation support, circular-motion interpretation and the SPARC signed source convention. The parent work derives and states the reduced carrier, polarization operator, structural factor and implicit

response in its section "Bound and Dynamical Structure," subsection "Galaxy-Scale Curvature Polarization Reduction" [5]. Their operational correspondence is:

Parent-level role	**Galaxy quantity**	**Registered radial realization**
Ordinary material source	g_b	Signed SPARC gas, unit-disk and unit-bulge force templates; equation (6)
Einstein/total gravitational field	$g_C = V_C^2/R$	Completed circular acceleration for the resolved system
Curvature polarization	$P_C = A_C a_\star^{2/3} B_{\rm pol} \mathcal{S}$	Support-derived gradient carrier and fixed nonlinear suppression
Curvature transport	$t_C = c_C \Sigma_C^2$	Coupled transport factor $(1 + t_C)g_C$ in the implicit field equation

The exact radial equation is equation (12); it is not obtained by assigning an independent spherical halo, nor does this table assert that a fully general axisymmetric component-level tensor reduction has been completed. The total field and the separately modeled material inventory enter the *mass-equivalent accounting only after* the radial solution is determined.

The compact-object relation $M_\infty = M_c + M_{\rm geom}$ arises from its own compact spacetime construction [7]. The present disk-galaxy paper makes a different, fully specified radial accounting choice. It does not transfer a compact-object enclosed-mass formula to flattened disks.

3 Methodology: Material and Geometric Mass Equivalents

The analysis consists of two complementary calculations. The first reconstructs the *complete CPTG circular gravitational field* from the registered signed source templates and locally measured structural support. The second reconstructs the *physical material inventory* from the galaxy's photometry and its signed gas-force template, independently of the completed CPTG field. The final step compares them under one declared mass-equivalent convention. The following steps are identical for all 153 galaxies.

3.1 Assemble the measured galaxy profile

The original SPARC database contains 175 galaxies [3]. Following the registered primary-catalog selection, objects with quality flag $Q = 3$ or inclination $i < 30°$ are excluded. The retained 153 original `_rotmod.dat` files contain 3,168 observed radii. There is no additional radius-level signal-to-noise cut in this atlas. The sample selection is separate from the point selection in the earlier 2016 radial-acceleration analysis [4].

At radius R_i (kpc), the eight input columns are R_i, observed circular speed $V_{{\rm obs},i}$ ($\mathrm{km\,s^{-1}}$), its stated error $\sigma_{V,i}$, signed gas velocity template $V_{{\rm gas},i}$, unit-stellar-normalization disk template $V_{{\rm disk},i}$, unit-stellar-normalization bulge template $V_{{\rm bulge},i}$, disk 3.6-μm surface brightness $I_{\rm disk}(R_i)$ and bulge surface brightness. The velocity templates encode radial gravitational *forces*, not literal independently enclosed masses. In particular, a signed gas entry is not discarded simply because a ring of gas produces a locally outward radial force. The original radius and velocity measurements are retained without altering their distance or inclination conventions.

3.2 Construct the signed source and the separately calibrated material-force display

Define the signed squared-speed operation $Q(V) = V|V|$. The *registered CPTG solver input* is the unit-template radial acceleration

$$g_b(R_i) = \frac{Q(V_{{\rm gas},i}) + Q(V_{{\rm disk},i}) + Q(V_{{\rm bulge},i})}{R_i}, \tag{6}$$

with SI units used internally. No negative baryonic field is replaced by zero. The black "Baryonic" line in the atlas is the corresponding signed speed equivalent, $\mathrm{sgn}(g_b)\sqrt{|g_b|R}$.

A different diagnostic, displayed as the blue "Material" force curve, applies the registered stellar normalizations to the input templates:

$$V^{\rm signed}_{\rm mat,force}(R) = \mathrm{sgn}(X_{\rm mat})\sqrt{|X_{\rm mat}|}, \qquad X_{\rm mat} = Q(V_{\rm gas}) + 0.5V^2_{\rm disk} + 0.7V^2_{\rm bulge}. \tag{7}$$

The 0.5 disk and 0.7 bulge coefficients are adopted 3.6-μm stellar mass-to-light ratios in solar units; they are not fitted separately to force a geometric fraction. Crucially, the black unit-template field in equation (6), the blue calibrated force curve in equation (7), and the *enclosed material mass* constructed in Section 3.4 are three different outputs. The blue force curve is not converted to a disk mass by applying the spherical expression RV^2/G.

The three uses of "material" in this paper are deliberately kept separate:

Quantity	**Source and normalization**	**Use in the calculation**
g_b	Signed SPARC unit-stellar-force gas, disk and bulge templates	Source entered in the unchanged implicit CPTG equation; black force curve
$V_{\mathrm{mat,force}}$	Signed gas and adopted 0.5 disk/0.7 bulge force templates	Blue material-force diagnostic; *not* an enclosed mass
$M_{\mathrm{material}}(< R)$	Radially integrated disk photometry, registered bulge inventory, inverted gas profile	Independently modeled cylindrical-aperture-plus-bulge inventory subtracted from $M^{\mathrm{eq}}_{\mathrm{grav}}$

Keeping these quantities distinct allows each galaxy to retain its registered SPARC input and prevents a disk-force trace from being silently treated as a spherical enclosed mass.

3.3 Reconstruct the complete CPTG circular field

For each observed radius, the registered structural acceleration scale is calculated *locally* from that radius's observed motion:

$$a_\star(R_i) = \frac{V^2_{\mathrm{obs},i}}{R_i}. \tag{8}$$

The same measurements and errors determine a smoothly varying support profile $g_{\mathrm{sup}}(R)$ through the locked uncertainty-weighted spline in $\ln R$. Where required, the registered spline-coefficient constraint enforces the admissible support shape; it does not alter the local values of $a_\star$, introduce a new galaxy parameter, or replace an unfavorable result by a floor. Primes below denote derivatives of this *same* continuous spline. The structural and polarization quantities are

$$\Sigma^2_C(R) = \frac{1}{3}\left[1 - \frac{d\ln g_{\mathrm{sup}}}{d\ln R}\right] \geq 0, \tag{9}$$

$$\mathcal{C}_C(R) = g^2_{\mathrm{sup}} + \kappa^2_C\left[(g'_{\mathrm{sup}})^2 + 2(g_{\mathrm{sup}}/R)^2\right], \tag{10}$$

$$B_{\mathrm{pol}}(R) = \frac{(R^2 + 2\kappa^2_C) g_{\mathrm{sup}}\,\mathcal{C}_C^{-2/3} - \frac{d}{dR}\left[R^2\kappa^2_C g'_{\mathrm{sup}}\mathcal{C}_C^{-2/3}\right]}{R^2}. \tag{11}$$

The derivatives are evaluated analytically from the smooth support, and Σ^2_C is *not* upper-clipped to unity. The local implicit CPTG equation, with its fixed registered exponents, is

$$\left(1 + c_C\Sigma^2_C\right) g_C = g_b + A_C a_\star^{2/3} B_{\mathrm{pol}}\left[1 + \left(\frac{g_C}{a_\star}\right)^{123/50}\right]^{-11/50}. \tag{12}$$

At every measured radius the locked solver finds the positive implicit root $g_C(R_i)$, using the coupled polarization and transport response and verifying the equation residual. The calculated circular speed is then

$$V_C(R_i) = \sqrt{R_i g_C(R_i)}. \tag{13}$$

Every galaxy uses equation (12) with the same mathematical exponents, signed-source convention, support algorithm and root procedure. The three object-specific reduced parameters κ_C (structural radial scale), A_C (polarization amplitude), and c_C (transport coupling) are *inherited from the archived registered v1.13.1 fits.* No new optimization or parameter search was conducted for the mass split. Their numerical values are not universal constants, and the local measured $a_\star$ is intentionally part of the solution. For audit purposes, the reported velocity residual statistic is $\chi^2 = \sum_i[(V_C(R_i) - V_{\mathrm{obs}}(R_i))/\sigma_{V,i}]^2$. Because measured V_{obs} participates in equation (8) and in the spline support, this is a *measured-field reconstruction* score, not a blind prediction from baryonic matter alone. This matters for the meaning of the result but does not alter the reconstruction or the mass-accounting equations.

3.4 Independently reconstruct the physical material inventory

The gravitational field alone does not determine how much physical gas or stellar material is present. We therefore reconstruct the material *separately* using the registered r51 inventory algorithm; neither the observed circular speed $V_{\rm obs}$ nor the solved CPTG speed V_C is used to fit that inventory.

Stellar disk. The measured 3.6-μm disk surface brightness $I_{\rm disk}(R)$ is interpolated using its positive measurements, with the registered constrained radial extrapolation where the sampled luminosity profile does not reach an endpoint. The disk mass within cylindrical radius R is

$$M_{\rm disk}(<R) = 2\pi\Upsilon_{\star,\rm disk}\int_0^R I_{\rm disk}(r)\, r\, dr, \qquad \Upsilon_{\star,\rm disk} = 0.5. \tag{14}$$

The required conversion of surface-brightness and radius units is included in the numerical integral. This construction respects the flattened stellar disk rather than treating its force as the field of a sphere; disk surface density and disk circular force are not interchangeable with spherical enclosed mass [2].

Stellar bulge. The separately supplied bulge unit template is interpreted through the registered spherical bulge reference, giving the enclosed model

$$M_{\rm bulge}(<R) = \mathrm{cummax}_{r\le R}\left[\frac{0.7\, r V^2_{\rm bulge}(r)}{G}\right]. \tag{15}$$

The cumulative-maximum convention enforces a nondecreasing model bulge inventory in the sampled profile; it is not an additional fitted halo.

Gas disk. A signed velocity contribution is a *force measurement template*, not a direct gas surface-density profile. The algorithm reconstructs a nonnegative axisymmetric annular surface density $\Sigma_{{\rm gas},j}$ by forward-fitting the observed *signed gas-force template*:

$$\sum_j K_{ij}\Sigma_{{\rm gas},j} \simeq V_{{\rm gas},i}|V_{{\rm gas},i}|, \qquad \Sigma_{{\rm gas},j} \ge 0. \tag{16}$$

The fixed annular kernel K_{ij} computes the signed radial disk force of each gas ring and includes the prescribed regularized softening. The nonnegative, regularized least-squares inversion penalizes strong second differences in neighboring surface densities with reference regularization parameter $\lambda = 0.12$. Gas annuli extend to $2R_{\rm max}$ to retain the influence of material exterior to the last measured rotation-curve point. The registered softening length is $\min[0.55, \max(0.09, 0.04R_{\rm max})]$ kpc. No term containing $V_{\rm obs}$ or V_C enters this gas inversion. After inversion, the gas mass inside R is computed by integrating the reconstructed surface density over annulus area.

The final physical-material estimate is therefore

$$M_{\rm material}(<R) = M_{\rm disk}(<R) + M_{\rm bulge}(<R) + M_{\rm gas}(<R). \tag{17}$$

This is a specified material inventory: the disk and gas contributions are cylindrical-aperture quantities and the bulge uses its declared spherical approximation. As in the SPARC mass-model literature, the result is conditional on the adopted stellar mass-to-light ratios and profile reconstruction. Importantly, none of these terms represents an inferred dark halo.

3.5 Turn the complete field and material inventory into two mass equivalents

The registered atlas uses one transparent gravitational accounting convention for all galaxies. At every observed radius the *complete* solved CPTG acceleration is expressed as the mass that would produce the same circular acceleration in a spherical reference field:

$$M^{\rm eq}_{\rm grav}(R) = \frac{R^2 g_C(R)}{G} = \frac{R V_C^2(R)}{G}. \tag{18}$$

This is the *gravitational mass equivalent*, not a claim that the rotating disk actually has a spherical matter distribution. We then compare it to the independently reconstructed physical material inventory at the same radius:

$$M^{\rm eq}_{\rm geom}(R) = M^{\rm eq}_{\rm grav}(R) - M_{\rm material}(< R), \tag{19}$$

$$f_{\rm material}(R) = \frac{M_{\rm material}(< R)}{M^{\rm eq}_{\rm grav}(R)}, \tag{20}$$

$$f_{\rm geom}(R) = \frac{M^{\rm eq}_{\rm geom}(R)}{M^{\rm eq}_{\rm grav}(R)} = 1 - f_{\rm material}(R). \tag{21}$$

Consequently,

$$M^{\rm eq}_{\rm grav}(R) = M_{\rm material}(< R) + M^{\rm eq}_{\rm geom}(R). \tag{22}$$

This is the *declared geometric–material mass-equivalent split*. Its scientific interpretation follows from the material–geometry distinction in equation (1); its *numerical values* follow from the completed registered galaxy field and the independently modeled material profile. The closure in equation (22) is exact arithmetic accounting, not a second gravitational equation or proof of a unique positive three-dimensional geometric mass density in an axisymmetric disk.

The percentages are calculated radially, and each galaxy entry gives the values at its own last observed radius $R_{\rm max}$. Where $M^{\rm eq}_{\rm geom}(R) \geq 0$, the red atlas curve displays the nonnegative *geometric mass-equivalent speed*

$$V^{\rm eq}_{\rm geom}(R) = \sqrt{\frac{G M^{\rm eq}_{\rm geom}(R)}{R}} = V_C(R)\sqrt{\frac{M^{\rm eq}_{\rm geom}(R)}{M^{\rm eq}_{\rm grav}(R)}}. \tag{23}$$

Where the signed arithmetic remainder is negative, the red square-root curve has a gap; the negative value remains present in the underlying data and is never clipped to zero. The red curve is *not* the signed net polarization–transport *acceleration* of equation (5) and should not be added linearly to the blue source-force speed.

3.6 Repeat without changing the law or choosing favorable galaxies

The procedure is repeated for each galaxy using its measured profile and archived coefficients, with unchanged material-inventory rules. All 153 selected galaxies appear in the atlas. The radial machine-readable table contains every reconstructed velocity, material mass, geometric equivalent and fraction; the summary contains galaxy-level values, outer-radial statistics and quality checks. The six-curve panels retain measured points and errors, the two distinct source-force curves, the red geometric mass-equivalent readout and complete orange CPTG result. A fixed-a_0 MOND green curve appears solely as an illustrative reference, not as a re-fitted competitor [6].

In this way, the atlas tests a *common geometric accounting program* across galaxies of markedly different size and surface brightness. It is an explicit alternative to placing a new physical halo into the material inventory for this reconstruction, while transparently retaining object-dependent observational support, reduced coefficients and the choice of mass-equivalent convention.

4 Shared six-curve figure key

All panels show speed equivalents in $\mathrm{km\,s^{-1}}$ versus radius (kpc).

Mark	Component	Definition
• ±	Observed SPARC	Independent blue circular points with vertical error bars and *no connecting blue line*; the observed speeds also set the local structural support.
—	Baryonic (black)	Signed unit-template gas+disk+bulge gravitational-source velocity equivalent.
—	Material (blue)	Signed material-source equivalent from gas+0.5 disk+0.7 bulge; continuous blue line.
—	Geometric (red)	Geometric mass-equivalent speed from the completed CPTG field minus reconstructed enclosed material mass. Negative mass remainders are not plotted.
—	CPTG (orange)	Full fitted CPTG rotational solution, including its complete geometric field response.
—	MOND (green)	Fixed-a_0 illustrative comparison computed from the locked unit-template input.

All five calculated traces use *solid* lines where defined. The red velocity is a spherical mass-equivalent diagnostic of the solved CPTG field, *not* an additional term to add to the black, blue, or orange speed. Its gaps mark negative geometric mass remainders and are not zeros. The black and blue source-force curves are distinct from the independently reconstructed enclosed material inventory used in the red mass-equivalent calculation.

5 Reading the result: four resolved galaxies

To make the numerical accounting concrete before the full atlas, Table 1 shows four familiar examples processed by *exactly the same registered method*. The numbers are evaluated at each galaxy's own outermost observed radius, not at a shared aperture. They illustrate the real range of structural outcomes: low-material dwarfs can have large geometric-equivalent fractions while a massive disk with a substantial modeled stellar inventory can have a smaller fraction. No geometric share is prescribed in advance.

Galaxy	$R_{\max}$ (kpc)	$M^{\mathrm{eq}}_{\mathrm{grav}}$ ($10^9 M_\odot$)	M_{material} ($10^9 M_\odot$)	$M^{\mathrm{eq}}_{\mathrm{geom}}$ ($10^9 M_\odot$)	f_{geom} (%)
DDO 154	5.92	2.8590	0.2966	2.5624	89.63
UGC 00731	10.91	14.7322	2.2621	12.4701	84.64
NGC 7331	36.31	451.9758	146.6977	305.2780	67.54
NGC 2955	35.43	451.5476	207.7179	243.8297	54.00

Table 1: Four representative outer-radius reconstructions. All three displayed mass quantities are obtained with the registered disk-galaxy mass-equivalent convention. The material column is the independently modeled physical material inventory, not the blue plotted material-force speed.

End-to-end worked example: DDO 154

The final DDO 154 SPARC row, at $R = 5.92\,\mathrm{kpc}$, reports $V_{\mathrm{obs}} = 45.5 \pm 1.3\,\mathrm{km\,s^{-1}}$, $V_{\mathrm{gas}} = 15.64\,\mathrm{km\,s^{-1}}$, the unit disk template $V_{\mathrm{disk}} = 6.26\,\mathrm{km\,s^{-1}}$, and $V_{\mathrm{bulge}} = 0$. The signed source is evaluated with the *unit* templates, not the later material-inventory calibration:

$$g_b = \frac{15.64^2 + 6.26^2}{5.92}\,\frac{(\mathrm{km\,s^{-1}})^2}{\mathrm{kpc}} = 1.553588 \times 10^{-12}\,\mathrm{m\,s^{-2}}, \tag{24}$$

$$a_\star = \frac{45.5^2}{5.92}\,\frac{(\mathrm{km\,s^{-1}})^2}{\mathrm{kpc}} = 1.133315 \times 10^{-11}\,\mathrm{m\,s^{-2}}, \tag{25}$$

where each velocity is first converted from $\mathrm{km\,s^{-1}}$ to SI units. The derivative-bearing support is reconstructed from *all 12* measured DDO 154 radii, rather than inferred from the last row alone. The following values are recovered from the archived, unchanged fitted-coefficient registry and solver, evaluated at the outermost point:

Registered step	Evaluated result	Computational role
Object coefficients	$\kappa_C = 3.259059\,\mathrm{kpc}$; $A_C = 3.007322 \times 10^{-8}$; $c_C = 1.000000 \times 10^{-10}$	Inherited reduction-specific fit; not re-optimized for the mass split
Support gradient and suppression	$\Sigma_C^2 = 0.753630$; $B_{\mathrm{pol}} = 7542.482$ (registered units); $\mathcal{S} = 0.857798$	Evaluated from the registered support and positive implicit root
Coupled polarization	$P_C = 9.817015 \times 10^{-12}\,\mathrm{m\,s^{-2}}$	$A_C a_\star^{2/3} B_{\mathrm{pol}} \mathcal{S}$
Transport multiplier	$1 + c_C \Sigma_C^2 = 1.00000000007536$	Retains transport in the implicit equation
Completed field root	$g_C = 1.137060 \times 10^{-11}\,\mathrm{m\,s^{-2}}$	$(1 + c_C \Sigma_C^2) g_C = g_b + P_C$
Reconstructed circular speed	$V_C = 45.575126\,\mathrm{km\,s^{-1}}$	$V_C = \sqrt{R g_C}$; reconstructed (not held out)

The material-inventory calculation uses the *full* disk-brightness profile and the signed gas-force inversion. At 5.92 kpc it returns $M_{\mathrm{disk}} = 0.028898$, $M_{\mathrm{bulge}} = 0$ and $M_{\mathrm{gas}} = 0.267681$, each in $10^9 M_\odot$, giving $M_{\mathrm{material}} = 0.296579 \times 10^9 M_\odot$. Evaluating the *complete* circular field under the registered convention then gives

$$M_{\mathrm{grav}}^{\mathrm{eq}} = RV_C^2/G = 2.859015 \times 10^9 M_\odot, \tag{26}$$

$$M_{\mathrm{geom}}^{\mathrm{eq}} = 2.859015 - 0.296579 = 2.562436 \quad (10^9 M_\odot), \tag{27}$$

$$f_{\mathrm{geom}} = 2.562436/2.859015 = 89.6265\%. \tag{28}$$

This is one fully specified radius within a 12-radius, observation-conditioned reconstruction; no dark-halo profile and no new mass-split fit has been introduced. The supplied companion evidence contains the raw input row, complete radial profile, coefficient registry and independently checked intermediate values.

NGC 2955, processed with the same equations but with its own measured structure and material profile, yields 54.00% at its final observed radius. This variation is the measured outcome of the calculation, not a universal fraction assigned to individual galaxies.

6 Population-scale accounting

The same registered field equation and material-accounting protocol yield a diversity of geometric mass-equivalent shares across all 153 resolved structures. The population result is the measured distribution, not a uniform percentage imposed galaxy by galaxy.

At each galaxy's last measured radius, the arithmetic mean outer geometric-equivalent share is 77.96%, the median is 80.33%, and the interquartile range is 73.01–86.46%. The survey sums of these differently sized outer apertures are $28.09368 \times 10^{12} M_\odot$ for total spherical gravitational field equivalents, $5.89138 \times 10^{12} M_\odot$ for modeled material inventories, and $22.20230 \times 10^{12} M_\odot$ for their arithmetic geometric equivalents. The mass-equivalent-weighted outer share is explicitly $\sum_g M_{\mathrm{geom},g}^{\mathrm{eq}}(R_{\mathrm{max},g}) / \sum_g M_{\mathrm{total},g}^{\mathrm{eq}}(R_{\mathrm{max},g}) = 79.03\%$. *The denominator is the total field equivalent, not the material inventory.* These cross-galaxy sums use different physical apertures, not a common enclosed volume, and are not a universal density or observed physical-matter fraction.

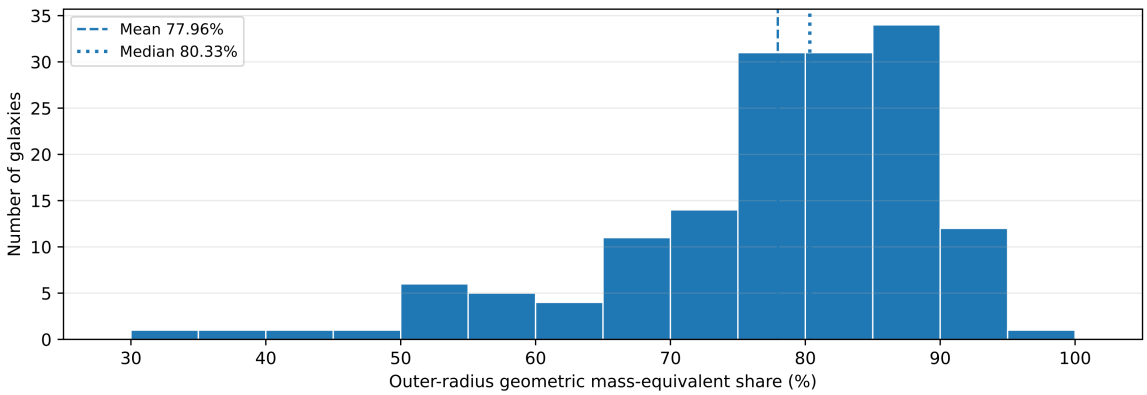


Figure 1: Distribution of outermost measured-radius geometric mass-equivalent percentages for all 153 primary SPARC galaxies. Vertical reference lines mark the 77.96% unweighted mean and 80.33% median. The separate 79.03% mass-equivalent-weighted share aggregates galaxies at different outer observed radii and should not be read as a universal cosmic matter fraction. Data: archived 153-row numerical summary; no resampling or new fitted parameters.

Numerical provenance and limitations

Among all 3,168 sampled radii, 196 in 49 galaxies have a *negative arithmetic* geometric mass-equivalent remainder. The signed values remain in the radial data and are not zero-clipped; the red mass-equivalent velocity trace is omitted at precisely those points because its real square root is undefined. This is neither evidence of negative physical material density nor a separate negative-mass component. Fourteen galaxies have reconstructed material inside their last measured radius exceeding the full-catalog $0.5L_{3.6} + 1.33M_{\rm HI}$ reference proxy by over 30%; they are flagged rather than retuned. That proxy is a diagnostic under different apertures and assumptions, not an exact mass measurement.

Galaxy-atlas diagnostic	**Result**
Primary galaxies; resolved velocity radii	153; 3,168
Outer geometric-equivalent mean; median	77.96%; 80.33%
Outer geometric-equivalent interquartile range	73.01–86.46%
Outer mass-equivalent-weighted geometric share	79.03%
Negative signed mass remainders; affected galaxies	196 radii; 49 galaxies
Material/proxy flags; replay differences $\lvert\Delta\chi^2\rvert > 0.01$	14; 6 galaxies
Full distance, inclination and material-profile systematic propagation	Not performed

The original registry contains 153 objects. On replay with the locked numerical solver, 147 have velocity χ^2 within 0.01 of the archived entry; six differ by more than 0.01. Table 2 lists the signed changes and preserves the *fresh* reconstructed values actually used in the present atlas. They are not concealed by overwriting the registry. The gas inversion has a median signed-velocity RMS of 0.120 $\mathrm{km\,s^{-1}}$ over the primary catalog. Material inventories depend on adopted stellar normalizations, assumed gas inversion geometry and regularization, photometric distances, and inclinations. The full joint effects of these systematic uncertainties have not been propagated.

Galaxy	**Archived χ^2**	**Fresh χ^2**	**Fresh minus archived**
NGC 2915	22.632154	21.841670	−0.790484
NGC 2976	26.918156	26.360171	−0.557984
NGC 3109	5.912551	5.619477	−0.293074
NGC 4051	6.703616	5.098550	−1.605066
NGC 4085	130.139599	119.406325	−10.733274
UGC 03546	14.628008	11.739516	−2.888492

Table 2: Six registered velocity-fit replay differences; $\Delta\chi^2 = \chi^2_{\rm fresh} - \chi^2_{\rm archived}$. The printed values are rounded; the machine-readable summary retains full precision.

Variation in reconstruction quality

Reconstruction accuracy is not uniform across the catalog. For example, UGC 02953 has $\chi^2 = 2331.87$ for 115 observed radii, UGC 09133 has $\chi^2 = 1946.50$ for 68, and NGC 2403 has $\chi^2 = 1811.47$ for 73. These galaxies remain in the atlas with their original measured error bars and unchanged coefficients. Overall, 33 of the 153 galaxies have $\chi^2/N_{\rm radii} > 2$, and 14 exceed 10. The ratio $\chi^2/N_{\rm radii}$ is *not* called a reduced chi-square, because no effective fitted degrees of freedom or full observational covariance matrix has been established here. Large values identify genuine error-weighted discrepancies under the present reconstruction and cannot be interpreted as satisfactory fits merely because the arithmetic mass equivalents close.

How to read the galaxy-by-galaxy atlas

Each entry pairs the original SPARC rotation measurements (blue points with error bars) with the signed source-force curves (black and blue), the completed CPTG reconstruction (orange), the geometric mass-equivalent speed (red), and a fixed-reference MOND illustration (green). Gaps in a red trace have the signed-remainder meaning defined in Section 3.5. Read the outer radius $R_{\rm max}$, observed and reconstructed speeds, and the three reported mass equivalents in $10^9 M_\odot$. Their material and geometric percentages refer to the *same outer-aperture accounting*, not fractions of newly created physical matter. Each galaxy's full radial behavior is retained in its panel and the companion 3,168-row numerical table.

7 Primary galaxy atlas: DDO through UGC

Primary SPARC galaxies 1–3 of 153 DDO 064 · DDO 154 · DDO 161

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

DDO 064 14 radii

80.48% geometric; 19.52% material at $R_{\max} = 2.98$ kpc. Outer measured speed is 46.9 ± 8.7 km s^{-1}; CPTG gives 48.8 km s^{-1} and $\chi^2 = 1.79$ for 14 points. Mass equivalents at that radius are $M_{\rm total} = 1.6485$, $M_{\rm mat} = 0.3218$, and $M_{\rm geom} = 1.3267$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

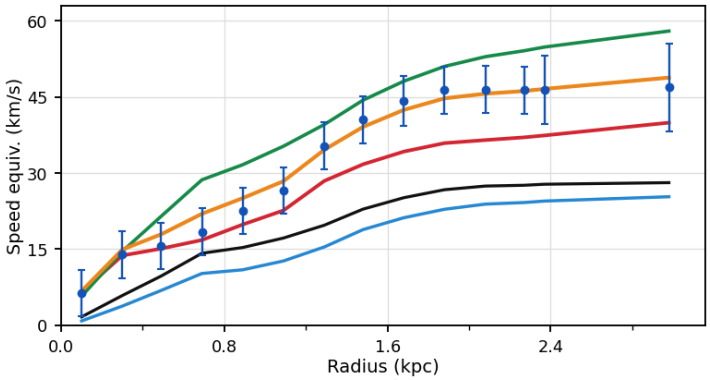


Figure 2: DDO 064.

DDO 154 12 radii

89.63% geometric; 10.37% material at $R_{\max} = 5.92$ kpc. Outer measured speed is 45.5 ± 1.3 km s^{-1}; CPTG gives 45.6 km s^{-1} and $\chi^2 = 22.29$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 2.8590$, $M_{\rm mat} = 0.2966$, and $M_{\rm geom} = 2.5624$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

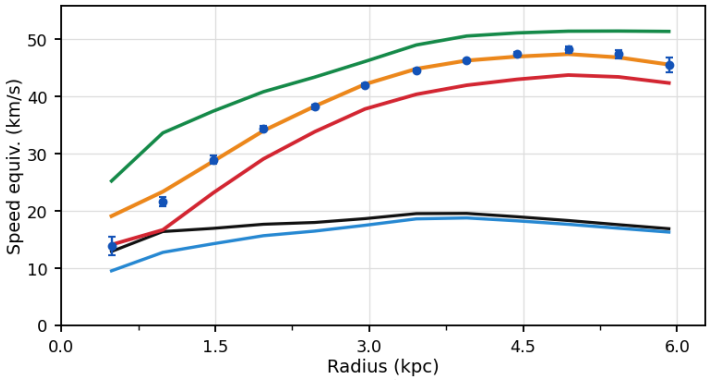


Figure 3: DDO 154.

DDO 161 31 radii

83.50% geometric; 16.50% material at $R_{\max} = 13.37$ kpc. Outer measured speed is 66.1 ± 1.3 km s^{-1}; CPTG gives 66.3 km s^{-1} and $\chi^2 = 41.87$ for 31 points. Mass equivalents at that radius are $M_{\rm total} = 13.668$, $M_{\rm mat} = 2.2551$, and $M_{\rm geom} = 11.413$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

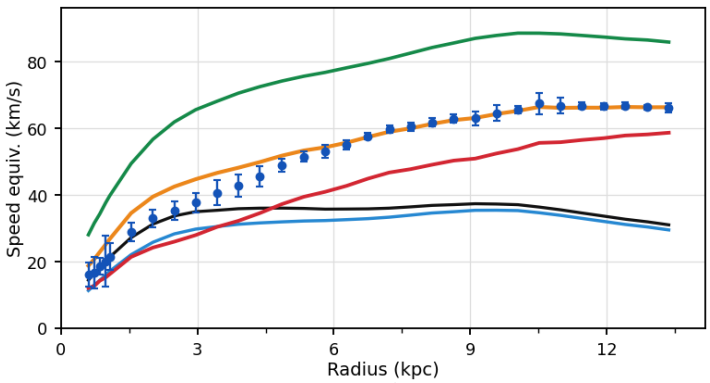


Figure 4: DDO 161.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

DDO 168 10 radii

81.73% geometric; 18.27% material at $R_{\max} = 4.12$ kpc. Outer measured speed is 52.0 ± 2.7 km s^{-1}; CPTG gives 53.2 km s^{-1} and $\chi^2 = 7.62$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 2.7146$, $M_{\rm mat} = 0.4960$, and $M_{\rm geom} = 2.2187$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

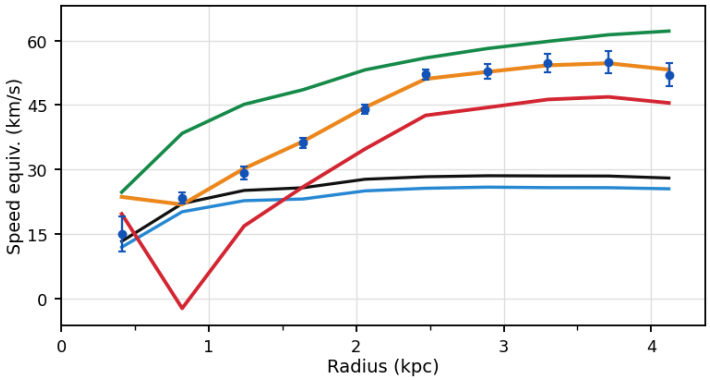


Figure 5: DDO 168.

DDO 170 8 radii

85.68% geometric; 14.32% material at $R_{\max} = 12.33$ kpc. Outer measured speed is 62.2 ± 1.0 km s^{-1}; CPTG gives 62.1 km s^{-1} and $\chi^2 = 3.29$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 11.045$, $M_{\rm mat} = 1.5815$, and $M_{\rm geom} = 9.4630$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

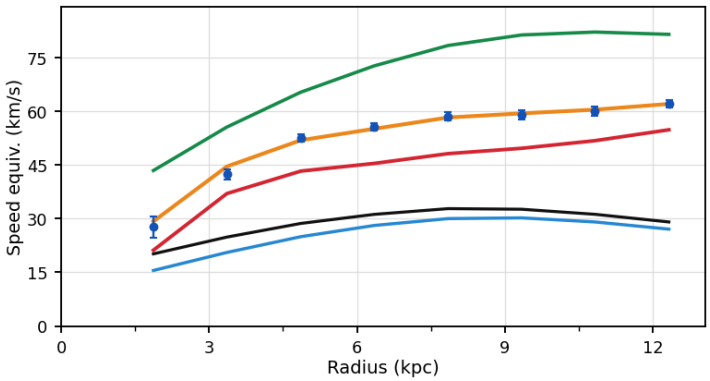


Figure 6: DDO 170.

ESO079-G014 15 radii

77.09% geometric; 22.91% material at $R_{\max} = 16.67$ kpc. Outer measured speed is 178.0 ± 2.0 km s^{-1}; CPTG gives 175.2 km s^{-1} and $\chi^2 = 31.16$ for 15 points. Mass equivalents at that radius are $M_{\rm total} = 118.99$, $M_{\rm mat} = 27.264$, and $M_{\rm geom} = 91.729$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

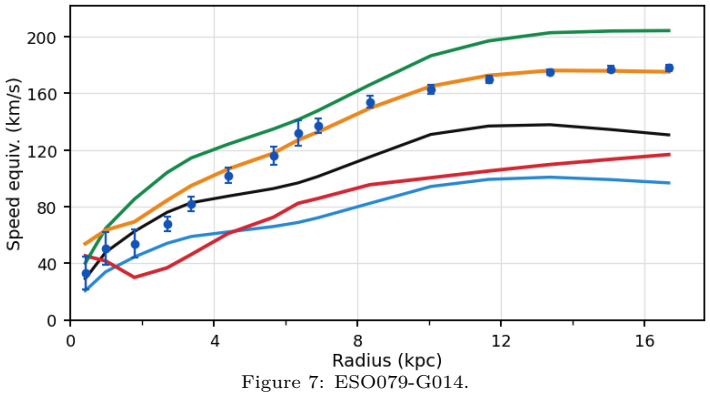


Figure 7: ESO079-G014.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

ESO116-G012 15 radii

86.99% geometric; 13.01% material at $R_{\max} = 9.86$ kpc. Outer measured speed is 112.0 ± 4.7 km s^{-1}; CPTG gives 113.0 km s^{-1} and $\chi^2 = 32.95$ for 15 points. Mass equivalents at that radius are $M_{\rm total} = 29.269$, $M_{\rm mat} = 3.8076$, and $M_{\rm geom} = 25.462$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

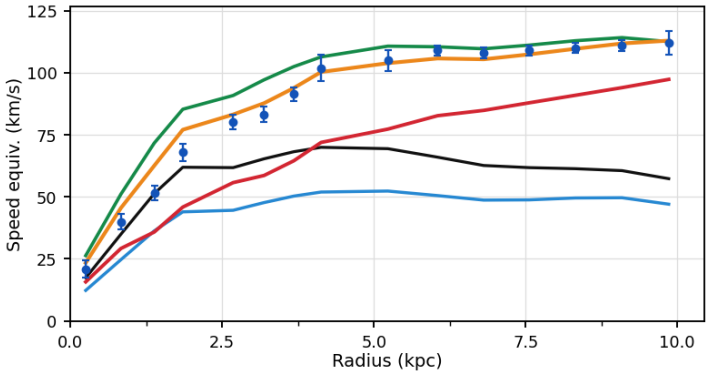


Figure 8: ESO116-G012.

ESO444-G084 7 radii

94.92% geometric; 5.08% material at $R_{\max} = 4.44$ kpc. Outer measured speed is 62.7 ± 7.4 km s^{-1}; CPTG gives 65.9 km s^{-1} and $\chi^2 = 1.19$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 4.4825$, $M_{\rm mat} = 0.2277$, and $M_{\rm geom} = 4.2548$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed.

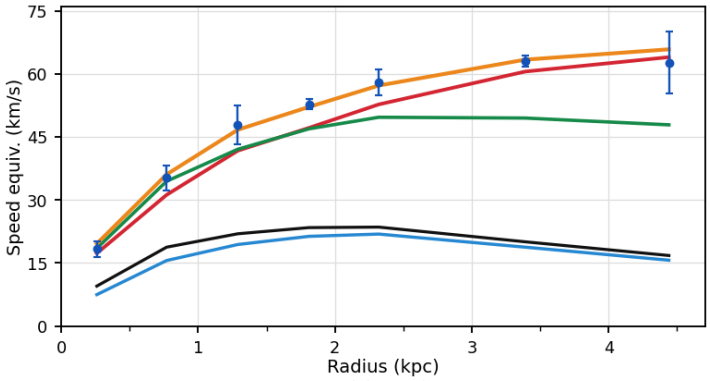


Figure 9: ESO444-G084.

ESO563-G021 30 radii

79.89% geometric; 20.11% material at $R_{\max} = 42.41$ kpc. Outer measured speed is 312.0 ± 9.1 km s^{-1}; CPTG gives 304.3 km s^{-1} and $\chi^2 = 376.19$ for 30 points. Mass equivalents at that radius are $M_{\rm total} = 913.08$, $M_{\rm mat} = 183.65$, and $M_{\rm geom} = 729.43$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

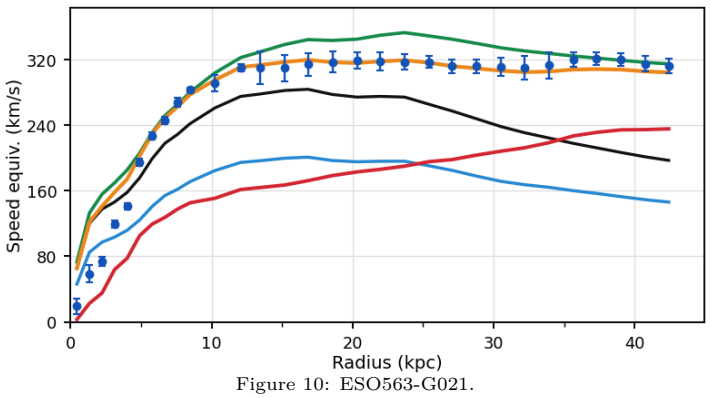


Figure 10: ESO563-G021.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

F565-V2 7 radii

89.26% geometric; 10.74% material at $R_{\max} = 8.80$ kpc. Outer measured speed is 83.1 ± 6.0 km s^{-1}; CPTG gives 82.9 km s^{-1} and $\chi^2 = 0.09$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 14.077$, $M_{\rm mat} = 1.5115$, and $M_{\rm geom} = 12.566$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

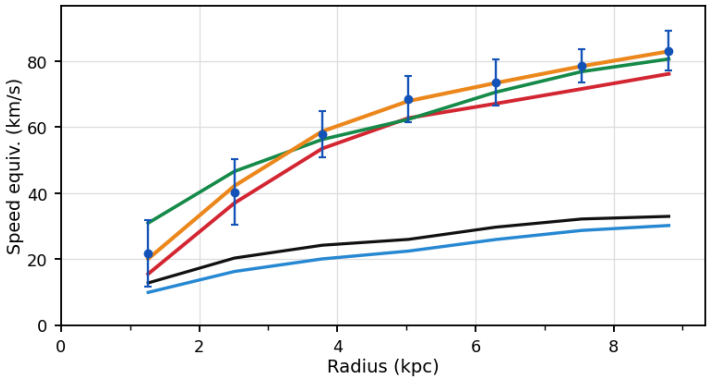


Figure 11: F565-V2.

F568-3 18 radii

84.69% geometric; 15.31% material at $R_{\max} = 17.98$ kpc. Outer measured speed is 120.0 ± 12.0 km s^{-1}; CPTG gives 114.2 km s^{-1} and $\chi^2 = 3.51$ for 18 points. Mass equivalents at that radius are $M_{\rm total} = 54.534$, $M_{\rm mat} = 8.3504$, and $M_{\rm geom} = 46.184$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

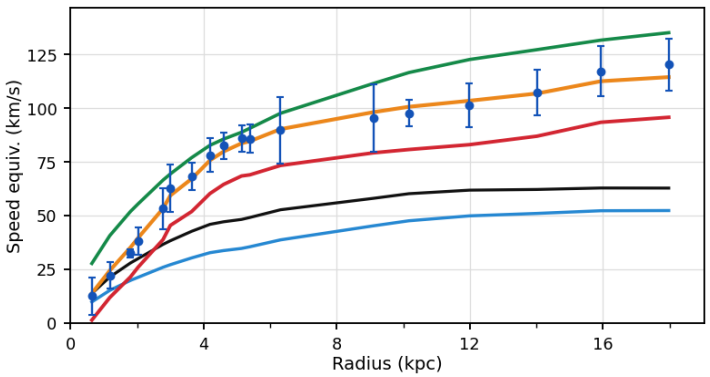


Figure 12: F568-3.

F568-V1 15 radii

90.37% geometric; 9.63% material at $R_{\max} = 17.63$ kpc. Outer measured speed is 118.0 ± 14.6 km s^{-1}; CPTG gives 116.8 km s^{-1} and $\chi^2 = 0.47$ for 15 points. Mass equivalents at that radius are $M_{\rm total} = 55.964$, $M_{\rm mat} = 5.3881$, and $M_{\rm geom} = 50.576$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

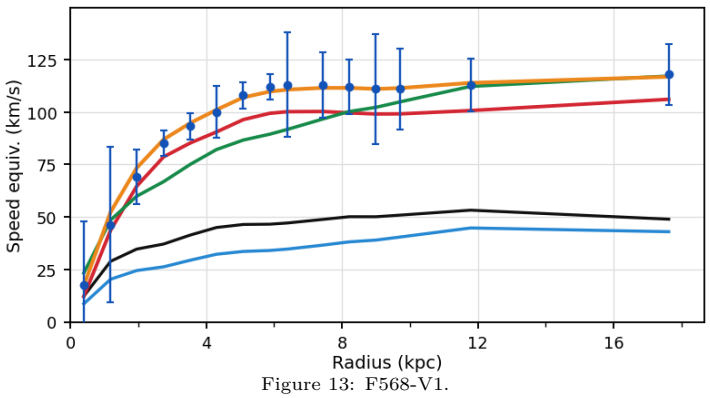


Figure 13: F568-V1.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

F571-8 13 radii

92.20% geometric; 7.80% material at $R_{\max} = 15.55$ kpc. Outer measured speed is 144.0 ± 5.0 km s^{-1}; CPTG gives 139.1 km s^{-1} and $\chi^2 = 26.33$ for 13 points. Mass equivalents at that radius are $M_{\rm total} = 69.910$, $M_{\rm mat} = 5.4510$, and $M_{\rm geom} = 64.459$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is below the last observed speed.

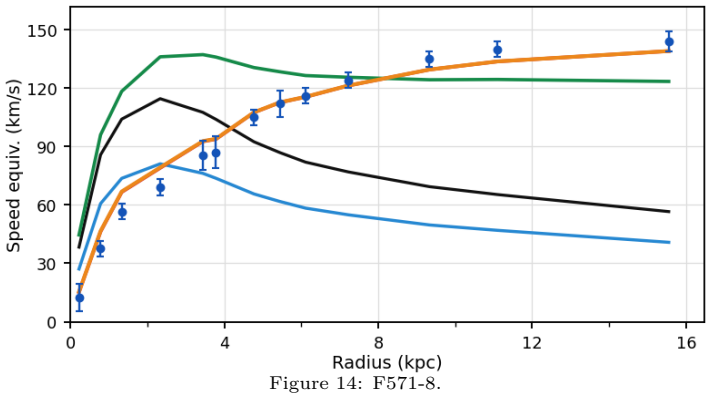


Figure 14: F571-8.

F571-V1 7 radii

87.74% geometric; 12.26% material at $R_{\max} = 13.59$ kpc. Outer measured speed is 83.9 ± 6.0 km s^{-1}; CPTG gives 84.3 km s^{-1} and $\chi^2 = 0.30$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 22.469$, $M_{\rm mat} = 2.7539$, and $M_{\rm geom} = 19.715$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

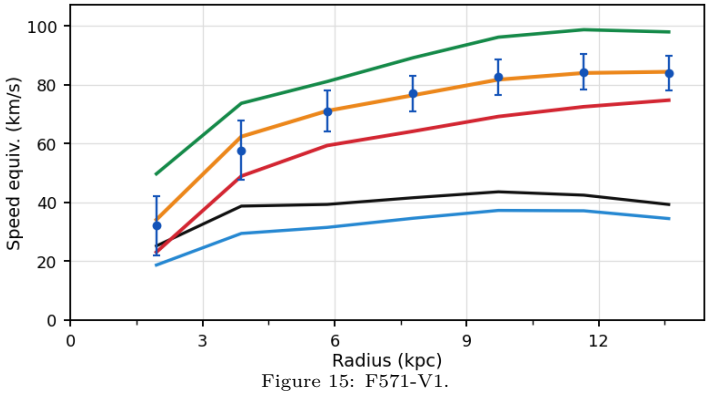


Figure 15: F571-V1.

F574-1 14 radii

81.76% geometric; 18.24% material at $R_{\max} = 12.60$ kpc. Outer measured speed is 99.7 ± 5.1 km s^{-1}; CPTG gives 99.9 km s^{-1} and $\chi^2 = 3.19$ for 14 points. Mass equivalents at that radius are $M_{\rm total} = 29.266$, $M_{\rm mat} = 5.3372$, and $M_{\rm geom} = 23.929$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

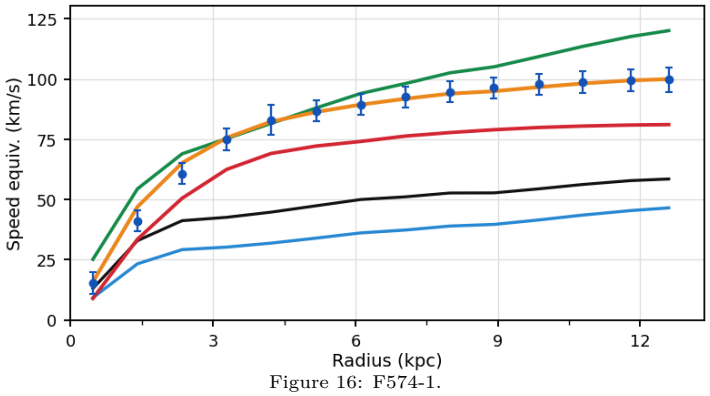


Figure 16: F574-1.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

F583-1 25 radii

82.15% geometric; 17.85% material at $R_{\max} = 16.26$ kpc. Outer measured speed is 85.8 ± 8.6 km s^{-1}; CPTG gives 86.8 km s^{-1} and $\chi^2 = 2.18$ for 25 points. Mass equivalents at that radius are $M_{\rm total} = 28.485$, $M_{\rm mat} = 5.0850$, and $M_{\rm geom} = 23.400$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

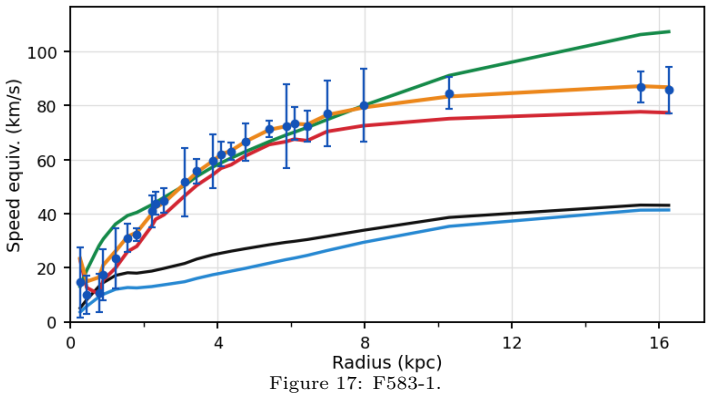


Figure 17: F583-1.

F583-4 12 radii

84.97% geometric; 15.03% material at $R_{\max} = 7.29$ kpc. Outer measured speed is 69.9 ± 4.9 km s^{-1}; CPTG gives 68.9 km s^{-1} and $\chi^2 = 0.28$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 8.0578$, $M_{\rm mat} = 1.2109$, and $M_{\rm geom} = 6.8469$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

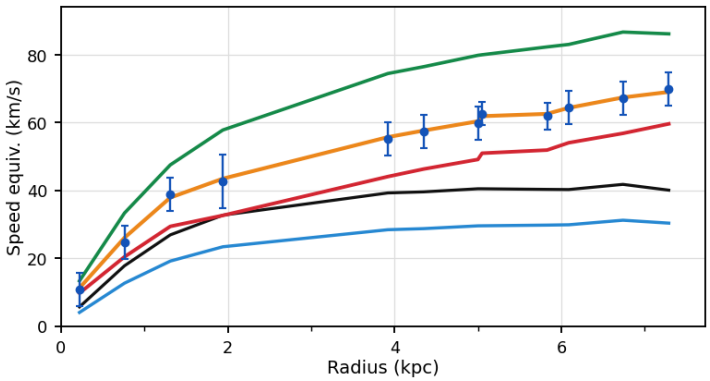


Figure 18: F583-4.

IC 2574 34 radii

53.92% geometric; 46.08% material at $R_{\max} = 10.23$ kpc. Outer measured speed is 67.5 ± 9.3 km s^{-1}; CPTG gives 59.5 km s^{-1} and $\chi^2 = 277.78$ for 34 points. Mass equivalents at that radius are $M_{\rm total} = 8.4097$, $M_{\rm mat} = 3.8756$, and $M_{\rm geom} = 4.5341$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

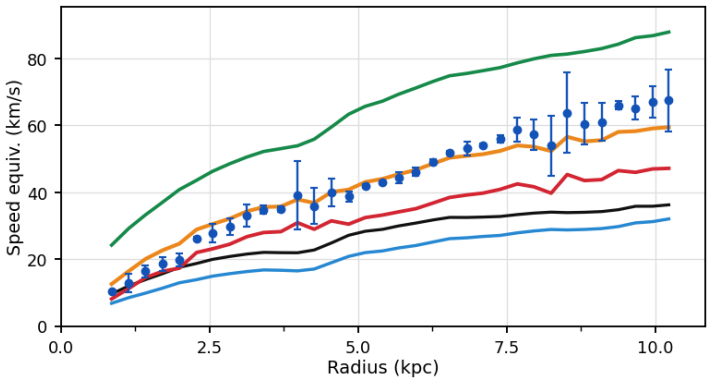


Figure 19: IC 2574.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

IC 4202 32 radii

71.36% geometric; 28.64% material at $R_{\max} = 25.90$ kpc. Outer measured speed is 247.0 ± 6.1 km s^{-1}; CPTG gives 253.1 km s^{-1} and $\chi^2 = 1262.78$ for 32 points. Mass equivalents at that radius are $M_{\rm total} = 385.66$, $M_{\rm mat} = 110.47$, and $M_{\rm geom} = 275.19$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

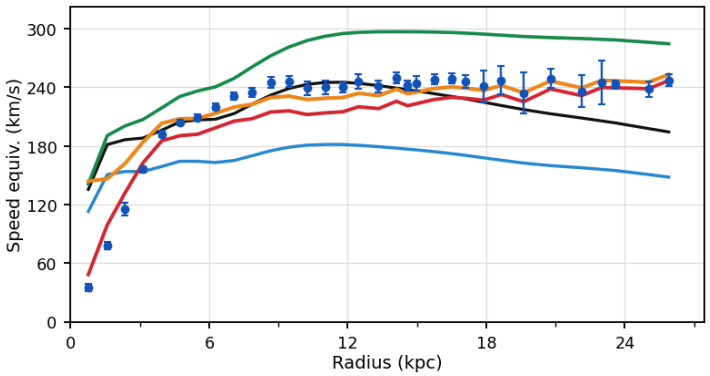


Figure 20: IC 4202.

KK98-251 15 radii

76.92% geometric; 23.08% material at $R_{\max} = 3.13$ kpc. Outer measured speed is 34.2 ± 2.0 km s^{-1}; CPTG gives 34.3 km s^{-1} and $\chi^2 = 1.19$ for 15 points. Mass equivalents at that radius are $M_{\rm total} = 0.8585$, $M_{\rm mat} = 0.1981$, and $M_{\rm geom} = 0.6604$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

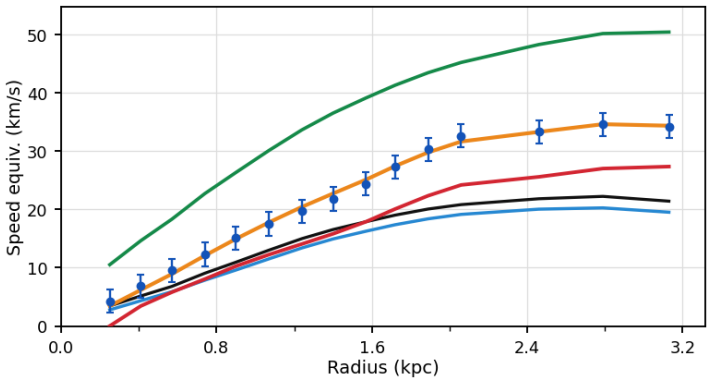


Figure 21: KK98-251.

NGC 0024 29 radii

91.02% geometric; 8.98% material at $R_{\max} = 11.27$ kpc. Outer measured speed is 110.0 ± 2.8 km s^{-1}; CPTG gives 111.2 km s^{-1} and $\chi^2 = 3.00$ for 29 points. Mass equivalents at that radius are $M_{\rm total} = 32.421$, $M_{\rm mat} = 2.9119$, and $M_{\rm geom} = 29.509$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed.

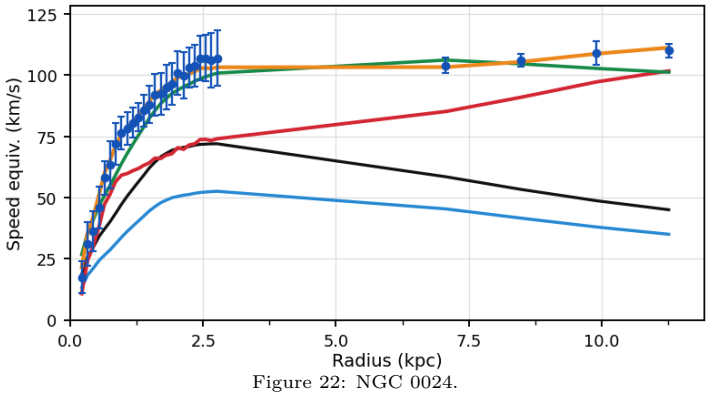


Figure 22: NGC 0024.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 0055 21 radii

77.69% geometric; 22.31% material at $R_{\max} = 13.50$ kpc. Outer measured speed is 86.5 ± 5.5 km s^{-1}; CPTG gives 84.6 km s^{-1} and $\chi^2 = 33.94$ for 21 points. Mass equivalents at that radius are $M_{\rm total} = 22.489$, $M_{\rm mat} = 5.0179$, and $M_{\rm geom} = 17.471$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

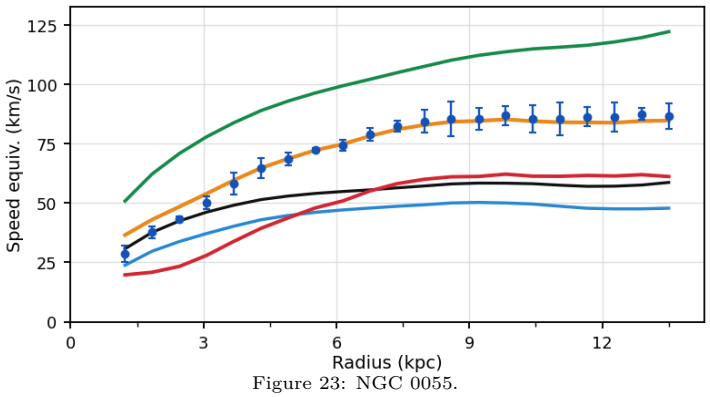


Figure 23: NGC 0055.

NGC 0100 21 radii

83.62% geometric; 16.38% material at $R_{\max} = 9.62$ kpc. Outer measured speed is 91.2 ± 13.7 km s^{-1}; CPTG gives 85.9 km s^{-1} and $\chi^2 = 26.23$ for 21 points. Mass equivalents at that radius are $M_{\rm total} = 16.518$, $M_{\rm mat} = 2.7060$, and $M_{\rm geom} = 13.812$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

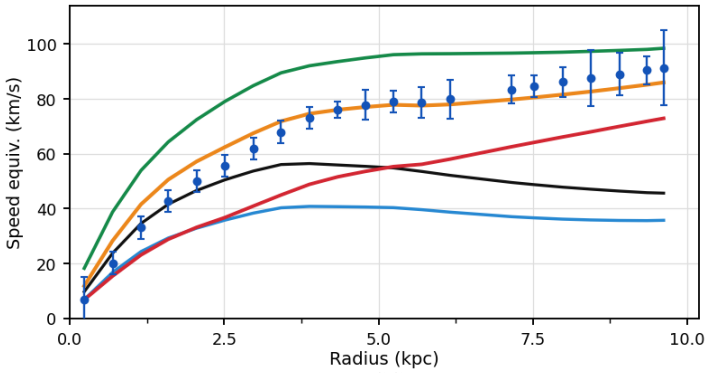


Figure 24: NGC 0100.

NGC 0247 26 radii

84.36% geometric; 15.64% material at $R_{\max} = 14.54$ kpc. Outer measured speed is 107.0 ± 9.2 km s^{-1}; CPTG gives 104.3 km s^{-1} and $\chi^2 = 12.47$ for 26 points. Mass equivalents at that radius are $M_{\rm total} = 36.759$, $M_{\rm mat} = 5.7504$, and $M_{\rm geom} = 31.009$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

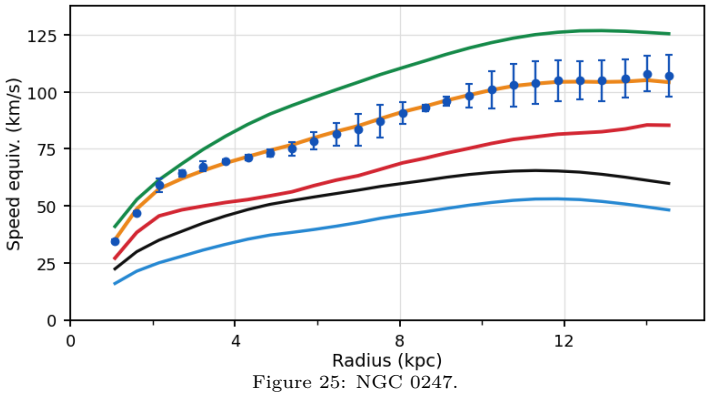


Figure 25: NGC 0247.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 0289 28 radii

83.63% geometric; 16.37% material at $R_{\max} = 71.12$ kpc. Outer measured speed is 165.0 ± 8.8 km s^{-1}; CPTG gives 161.0 km s^{-1} and $\chi^2 = 26.47$ for 28 points. Mass equivalents at that radius are $M_{\rm total} = 428.80$, $M_{\rm mat} = 70.182$, and $M_{\rm geom} = 358.62$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

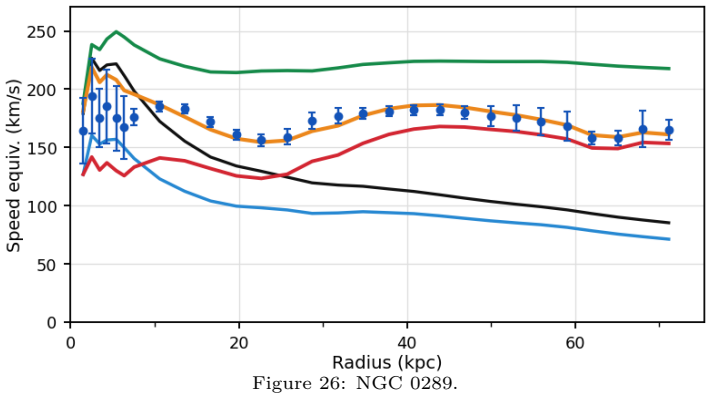


Figure 26: NGC 0289.

NGC 0300 25 radii

86.68% geometric; 13.32% material at $R_{\max} = 11.80$ kpc. Outer measured speed is 93.5 ± 12.1 km s^{-1}; CPTG gives 94.8 km s^{-1} and $\chi^2 = 9.62$ for 25 points. Mass equivalents at that radius are $M_{\rm total} = 24.638$, $M_{\rm mat} = 3.2824$, and $M_{\rm geom} = 21.355$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

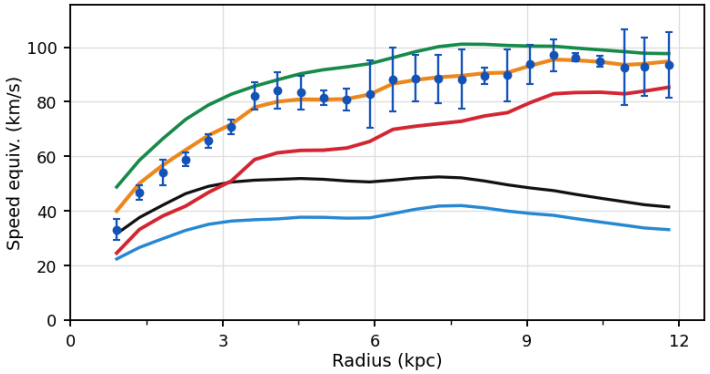


Figure 27: NGC 0300.

NGC 0801 13 radii

65.83% geometric; 34.17% material at $R_{\max} = 59.82$ kpc. Outer measured speed is 216.0 ± 8.8 km s^{-1}; CPTG gives 211.5 km s^{-1} and $\chi^2 = 259.11$ for 13 points. Mass equivalents at that radius are $M_{\rm total} = 622.29$, $M_{\rm mat} = 212.62$, and $M_{\rm geom} = 409.67$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

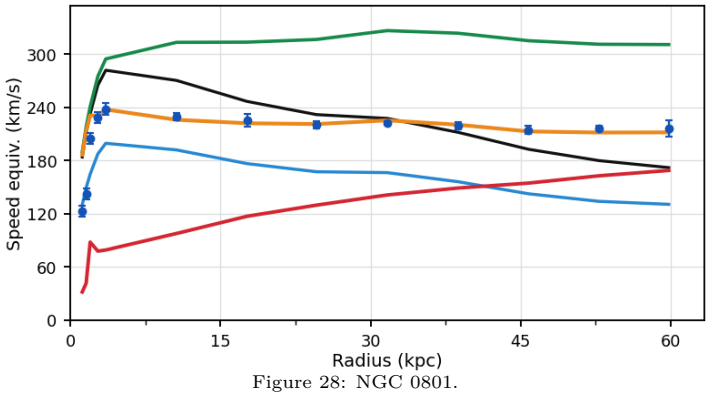


Figure 28: NGC 0801.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 0891 18 radii

55.92% geometric; 44.08% material at $R_{\max} = 17.11$ kpc. Outer measured speed is 208.0 ± 4.3 km s^{-1}; CPTG gives 212.8 km s^{-1} and $\chi^2 = 11.80$ for 18 points. Mass equivalents at that radius are $M_{\rm total} = 180.18$, $M_{\rm mat} = 79.430$, and $M_{\rm geom} = 100.75$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

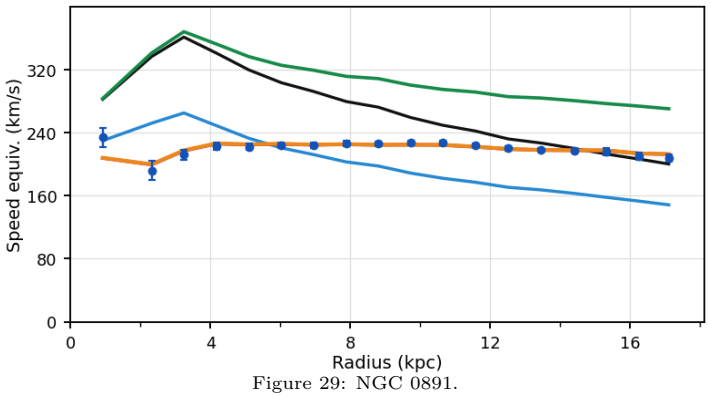


Figure 29: NGC 0891.

NGC 1003 36 radii

90.95% geometric; 9.05% material at $R_{\max} = 30.24$ kpc. Outer measured speed is 115.0 ± 1.6 km s^{-1}; CPTG gives 117.8 km s^{-1} and $\chi^2 = 75.59$ for 36 points. Mass equivalents at that radius are $M_{\rm total} = 97.504$, $M_{\rm mat} = 8.8264$, and $M_{\rm geom} = 88.677$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

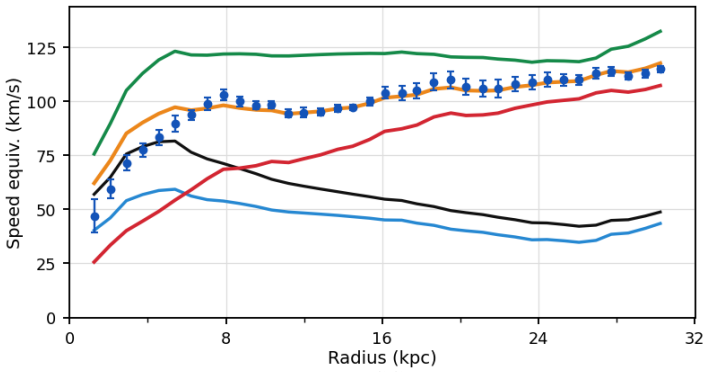


Figure 30: NGC 1003.

NGC 1090 24 radii

72.24% geometric; 27.76% material at $R_{\max} = 30.09$ kpc. Outer measured speed is 160.0 ± 2.2 km s^{-1}; CPTG gives 164.1 km s^{-1} and $\chi^2 = 88.32$ for 24 points. Mass equivalents at that radius are $M_{\rm total} = 188.38$, $M_{\rm mat} = 52.294$, and $M_{\rm geom} = 136.09$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

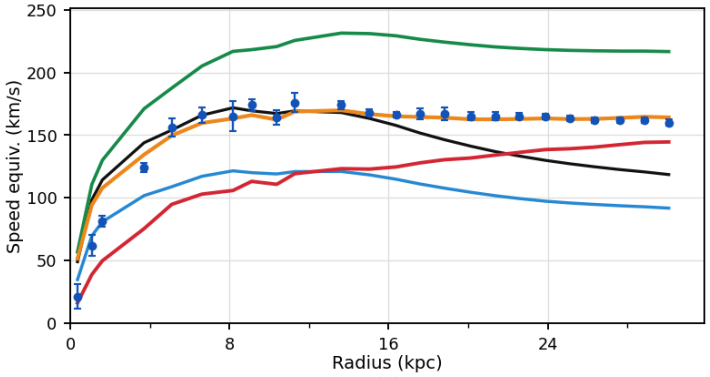


Figure 31: NGC 1090.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 2403 73 radii

87.23% geometric; 12.77% material at $R_{\max} = 20.87$ kpc. Outer measured speed is 134.0 ± 3.1 km s^{-1}; CPTG gives 141.5 km s^{-1} and $\chi^2 = 1811.47$ for 73 points. Mass equivalents at that radius are $M_{\rm total} = 97.143$, $M_{\rm mat} = 12.405$, and $M_{\rm geom} = 84.739$ in $10^9 M_\odot$. The large error-weighted residuals and material/proxy flag remain visible; neither has been corrected by rescaling the material inventory. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

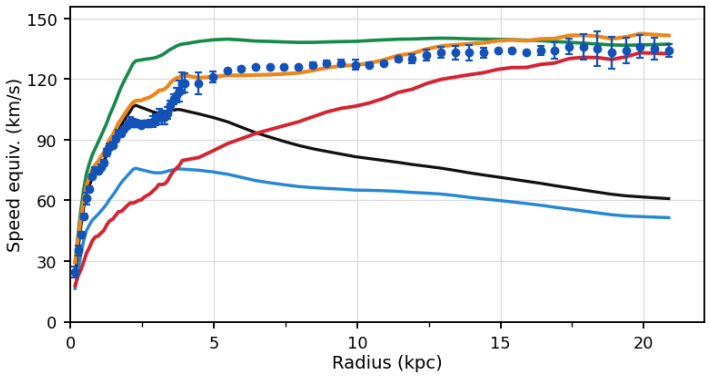


Figure 32: NGC 2403.

NGC 2683 11 radii

79.53% geometric; 20.47% material at $R_{\max} = 34.62$ kpc. Outer measured speed is 151.0 ± 12.0 km s^{-1}; CPTG gives 163.9 km s^{-1} and $\chi^2 = 17.20$ for 11 points. Mass equivalents at that radius are $M_{\rm total} = 216.36$, $M_{\rm mat} = 44.279$, and $M_{\rm geom} = 172.08$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

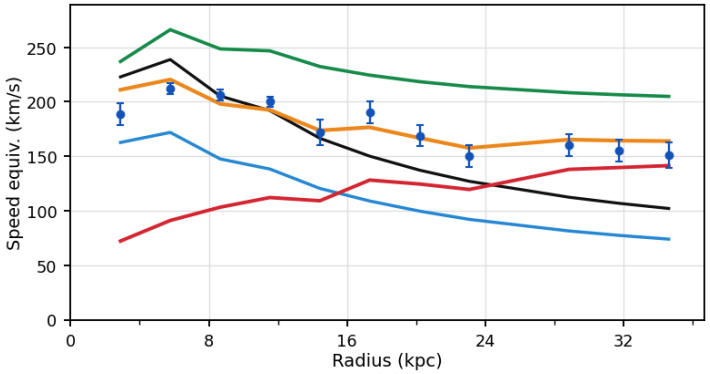


Figure 33: NGC 2683.

NGC 2841 50 radii

88.06% geometric; 11.94% material at $R_{\max} = 63.64$ kpc. Outer measured speed is 294.0 ± 7.1 km s^{-1}; CPTG gives 303.0 km s^{-1} and $\chi^2 = 72.35$ for 50 points. Mass equivalents at that radius are $M_{\rm total} = 1,358.25$, $M_{\rm mat} = 162.24$, and $M_{\rm geom} = 1,196.01$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

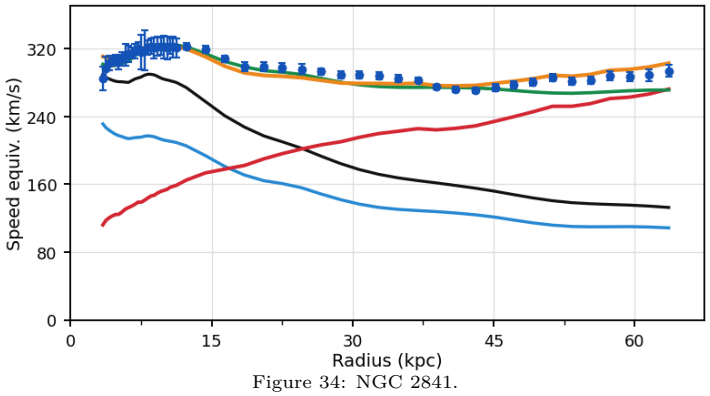


Figure 34: NGC 2841.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 2903 34 radii

77.78% geometric; 22.22% material at $R_{\rm max} = 24.96$ kpc. Outer measured speed is 180.0 ± 8.0 km s^{-1}; CPTG gives 184.0 km s^{-1} and $\chi^2 = 931.29$ for 34 points. Mass equivalents at that radius are $M_{\rm total} = 196.56$, $M_{\rm mat} = 43.676$, and $M_{\rm geom} = 152.88$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

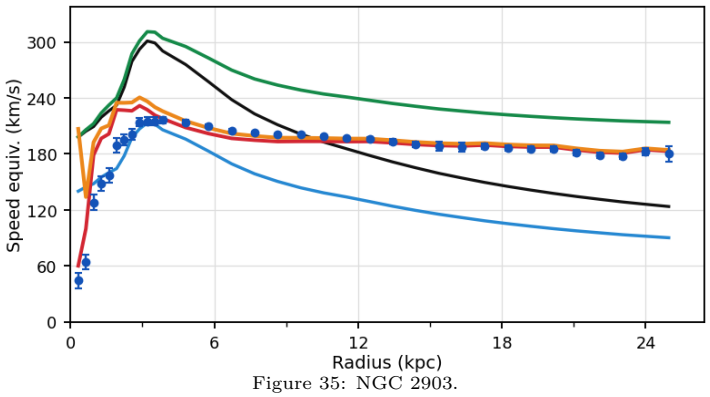


Figure 35: NGC 2903.

NGC 2915 30 radii

95.19% geometric; 4.81% material at $R_{\rm max} = 10.04$ kpc. Outer measured speed is 86.5 ± 6.5 km s^{-1}; CPTG gives 89.2 km s^{-1} and $\chi^2 = 21.84$ for 30 points. Mass equivalents at that radius are $M_{\rm total} = 18.575$, $M_{\rm mat} = 0.8927$, and $M_{\rm geom} = 17.682$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed.

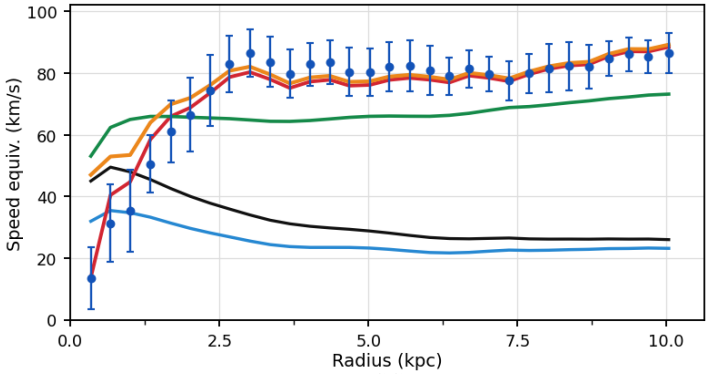


Figure 36: NGC 2915.

NGC 2955 24 radii

54.00% geometric; 46.00% material at $R_{\rm max} = 35.43$ kpc. Outer measured speed is 227.0 ± 11.2 km s^{-1}; CPTG gives 234.1 km s^{-1} and $\chi^2 = 22.77$ for 24 points. Mass equivalents at that radius are $M_{\rm total} = 451.55$, $M_{\rm mat} = 207.72$, and $M_{\rm geom} = 243.83$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

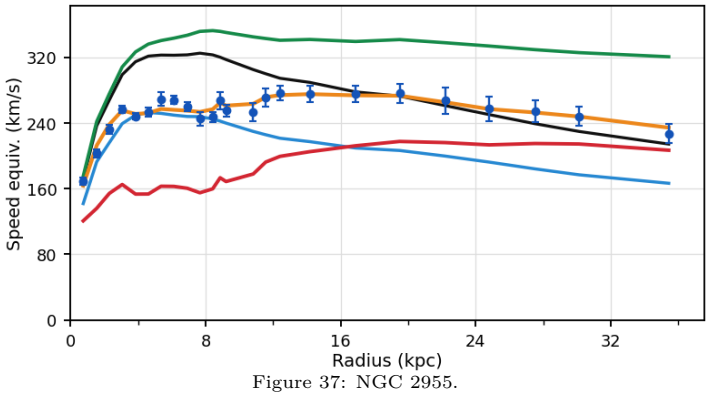


Figure 37: NGC 2955.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 2976 27 radii

52.49% geometric; 47.51% material at $R_{\rm max} = 2.27$ kpc. Outer measured speed is 85.3 ± 5.4 km s^{-1}; CPTG gives 77.1 km s^{-1} and $\chi^2 = 26.36$ for 27 points. Mass equivalents at that radius are $M_{\rm total} = 3.1374$, $M_{\rm mat} = 1.4907$, and $M_{\rm geom} = 1.6467$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

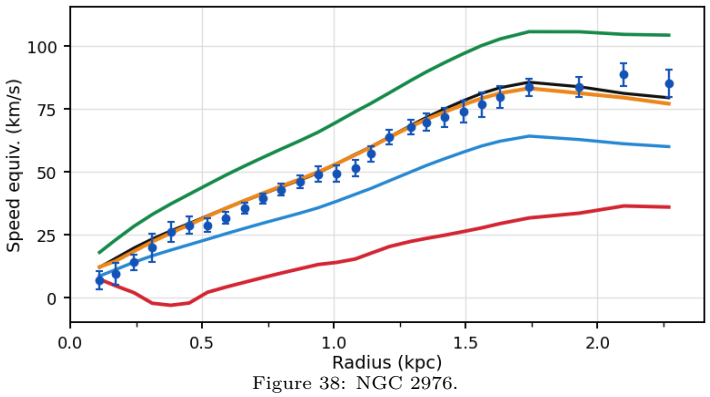


Figure 38: NGC 2976.

NGC 2998 13 radii

75.05% geometric; 24.95% material at $R_{\rm max} = 42.28$ kpc. Outer measured speed is 203.0 ± 3.0 km s^{-1}; CPTG gives 209.2 km s^{-1} and $\chi^2 = 10.36$ for 13 points. Mass equivalents at that radius are $M_{\rm total} = 430.27$, $M_{\rm mat} = 107.33$, and $M_{\rm geom} = 322.94$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

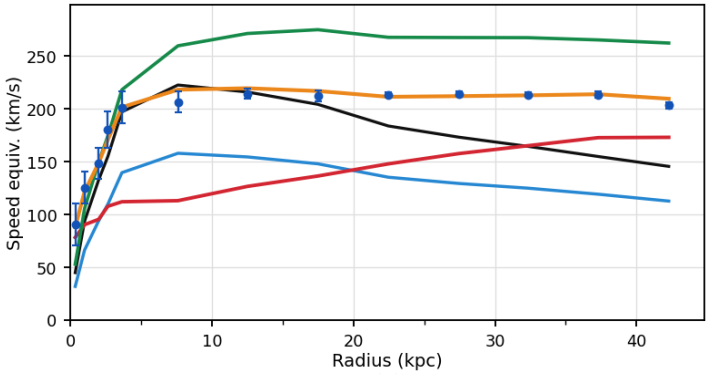


Figure 39: NGC 2998.

NGC 3109 25 radii

92.71% geometric; 7.29% material at $R_{\rm max} = 6.45$ kpc. Outer measured speed is 67.3 ± 3.0 km s^{-1}; CPTG gives 65.5 km s^{-1} and $\chi^2 = 5.62$ for 25 points. Mass equivalents at that radius are $M_{\rm total} = 6.4247$, $M_{\rm mat} = 0.4686$, and $M_{\rm geom} = 5.9560$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed.

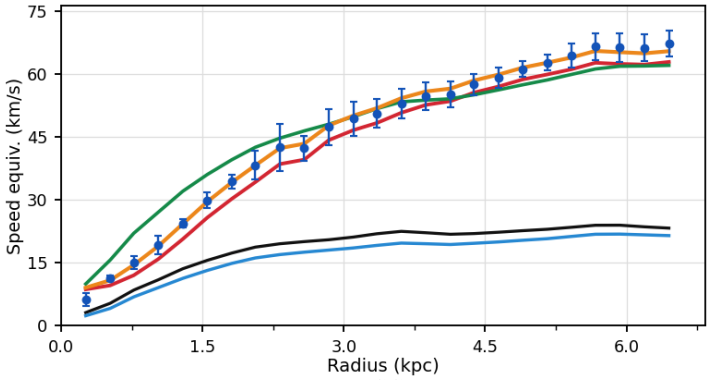


Figure 40: NGC 3109.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 3198 43 radii

87.31% geometric; 12.69% material at $R_{\max} = 44.08$ kpc. Outer measured speed is 149.0 ± 3.0 km s^{-1}; CPTG gives 160.1 km s^{-1} and $\chi^2 = 232.86$ for 43 points. Mass equivalents at that radius are $M_{\rm total} = 262.78$, $M_{\rm mat} = 33.360$, and $M_{\rm geom} = 229.42$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

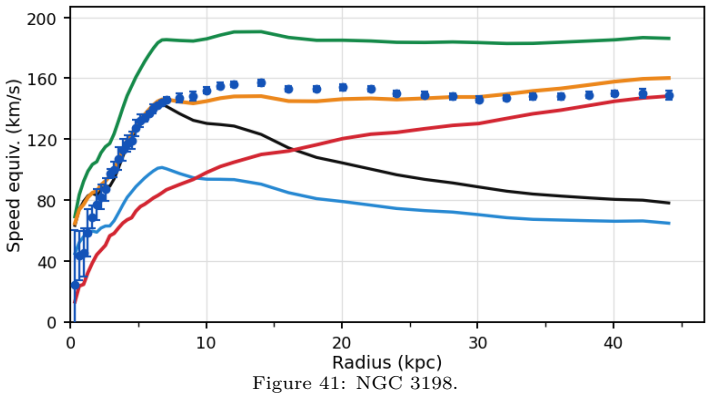


Figure 41: NGC 3198.

NGC 3521 41 radii

74.17% geometric; 25.83% material at $R_{\max} = 17.74$ kpc. Outer measured speed is 206.0 ± 10.7 km s^{-1}; CPTG gives 210.8 km s^{-1} and $\chi^2 = 4.00$ for 41 points. Mass equivalents at that radius are $M_{\rm total} = 183.30$, $M_{\rm mat} = 47.344$, and $M_{\rm geom} = 135.96$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

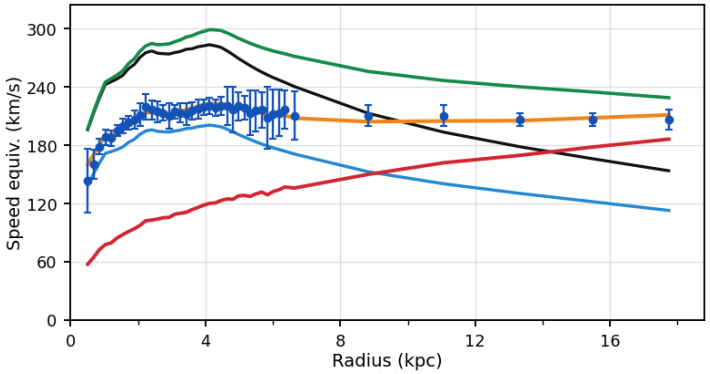


Figure 42: NGC 3521.

NGC 3726 12 radii

79.81% geometric; 20.19% material at $R_{\max} = 32.52$ kpc. Outer measured speed is 167.0 ± 15.2 km s^{-1}; CPTG gives 172.7 km s^{-1} and $\chi^2 = 85.72$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 225.55$, $M_{\rm mat} = 45.540$, and $M_{\rm geom} = 180.01$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

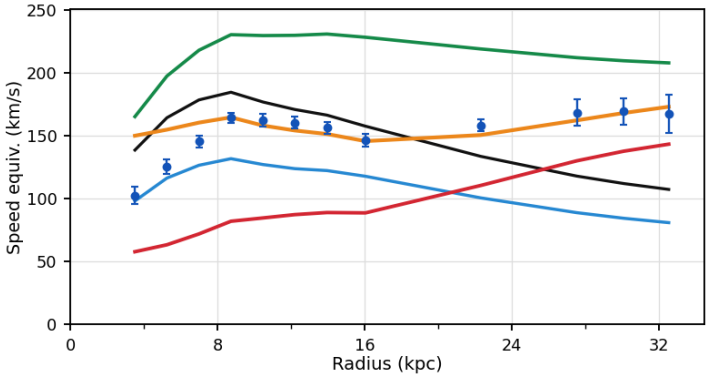


Figure 43: NGC 3726.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 3741 21 radii

94.53% geometric; 5.47% material at $R_{\max} = 7.00$ kpc. Outer measured speed is 51.6 ± 3.2 km s^{-1}; CPTG gives 53.0 km s^{-1} and $\chi^2 = 1.46$ for 21 points. Mass equivalents at that radius are $M_{\rm total} = 4.5677$, $M_{\rm mat} = 0.2501$, and $M_{\rm geom} = 4.3177$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

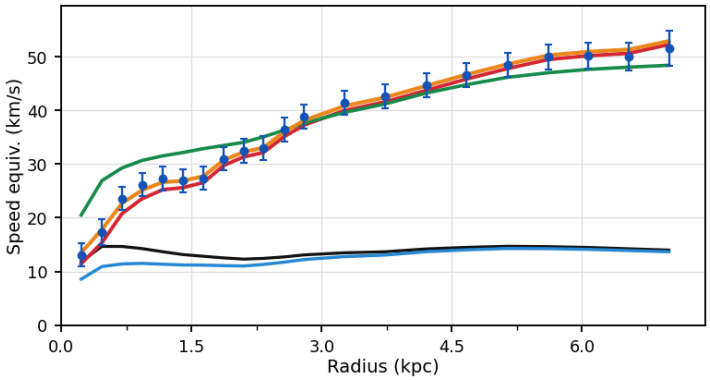


Figure 44: NGC 3741.

NGC 3769 12 radii

85.83% geometric; 14.17% material at $R_{\max} = 37.16$ kpc. Outer measured speed is 113.0 ± 10.5 km s^{-1}; CPTG gives 110.8 km s^{-1} and $\chi^2 = 23.10$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 106.00$, $M_{\rm mat} = 15.017$, and $M_{\rm geom} = 90.978$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

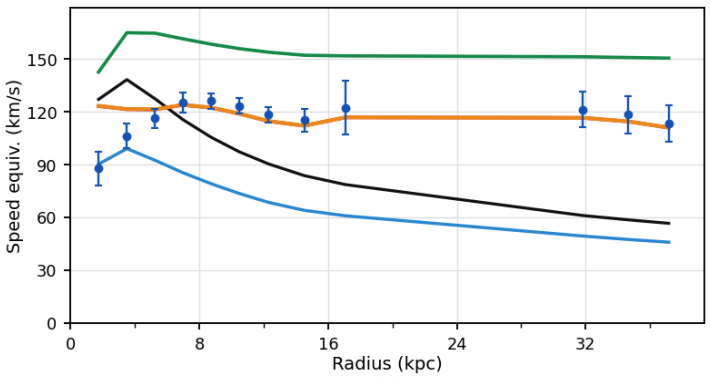


Figure 45: NGC 3769.

NGC 3877 13 radii

48.16% geometric; 51.84% material at $R_{\max} = 11.35$ kpc. Outer measured speed is 169.0 ± 7.0 km s^{-1}; CPTG gives 160.2 km s^{-1} and $\chi^2 = 71.66$ for 13 points. Mass equivalents at that radius are $M_{\rm total} = 67.718$, $M_{\rm mat} = 35.105$, and $M_{\rm geom} = 32.613$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

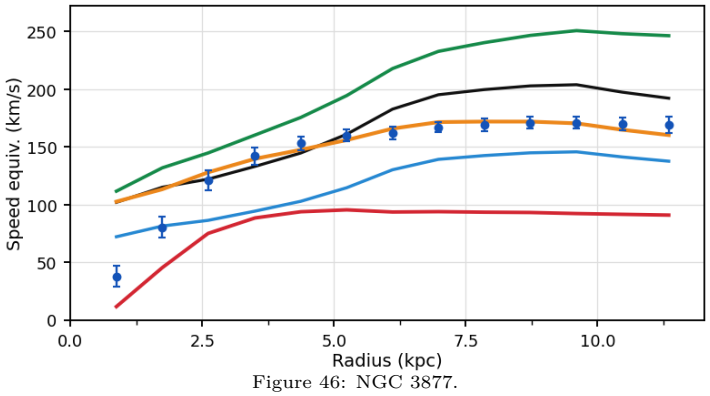


Figure 46: NGC 3877.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 3893 10 radii

77.32% geometric; 22.68% material at $R_{\max} = 19.05$ kpc. Outer measured speed is 167.0 ± 13.5 km s^{-1}; CPTG gives 187.8 km s^{-1} and $\chi^2 = 14.61$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 156.19$, $M_{\rm mat} = 35.428$, and $M_{\rm geom} = 120.76$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

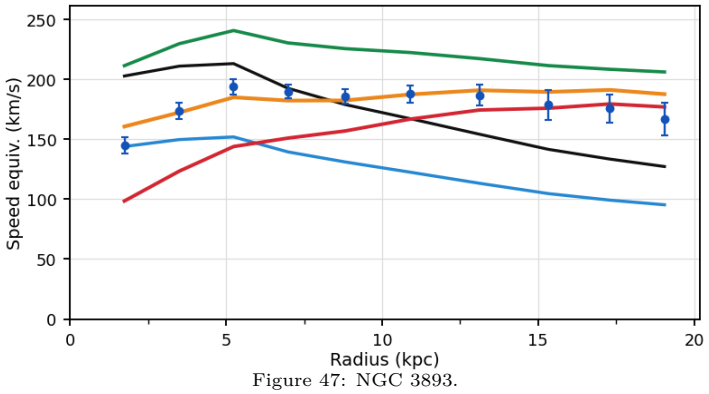


Figure 47: NGC 3893.

NGC 3917 17 radii

78.43% geometric; 21.57% material at $R_{\max} = 14.86$ kpc. Outer measured speed is 137.0 ± 7.5 km s^{-1}; CPTG gives 132.0 km s^{-1} and $\chi^2 = 16.61$ for 17 points. Mass equivalents at that radius are $M_{\rm total} = 60.214$, $M_{\rm mat} = 12.988$, and $M_{\rm geom} = 47.226$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

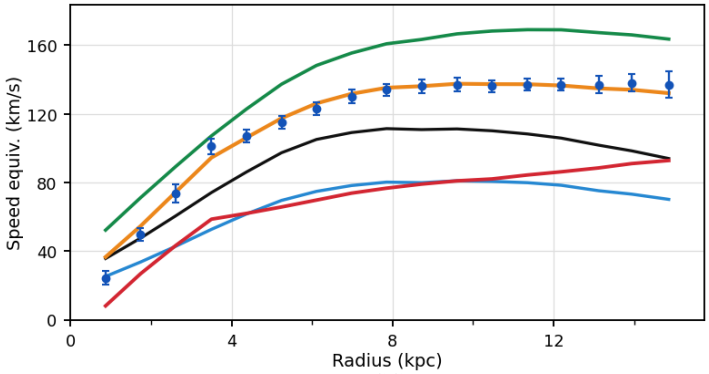


Figure 48: NGC 3917.

NGC 3949 7 radii

57.44% geometric; 42.56% material at $R_{\max} = 7.07$ kpc. Outer measured speed is 169.0 ± 25.5 km s^{-1}; CPTG gives 168.7 km s^{-1} and $\chi^2 = 12.60$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 46.789$, $M_{\rm mat} = 19.913$, and $M_{\rm geom} = 26.876$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

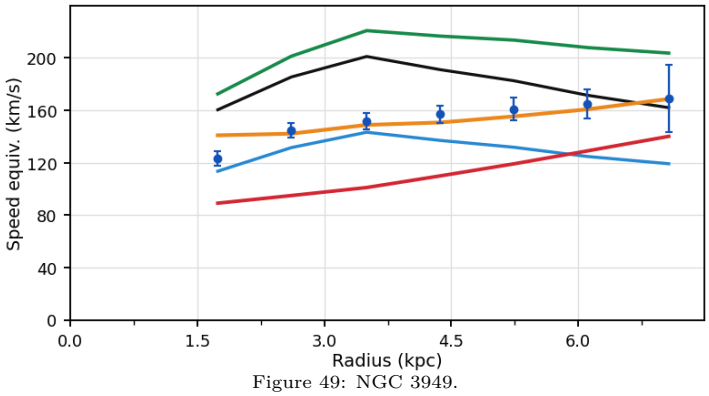


Figure 49: NGC 3949.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 3953 8 radii

63.05% geometric; 36.95% material at $R_{\max} = 15.68$ kpc. Outer measured speed is 215.0 ± 9.7 km s^{-1}; CPTG gives 217.3 km s^{-1} and $\chi^2 = 0.82$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 172.16$, $M_{\rm mat} = 63.620$, and $M_{\rm geom} = 108.54$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

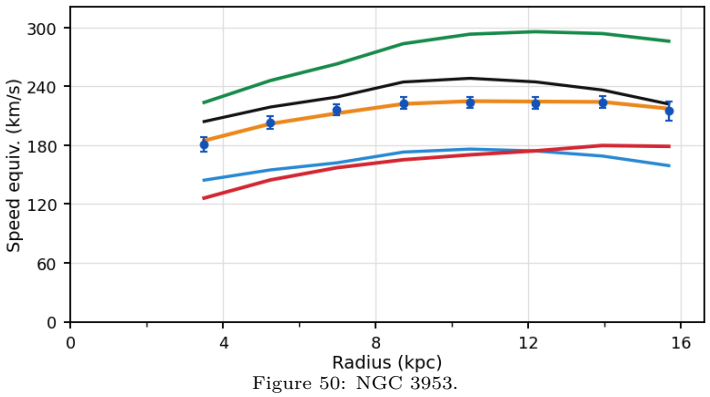


Figure 50: NGC 3953.

NGC 3972 10 radii

76.47% geometric; 23.53% material at $R_{\max} = 8.72$ kpc. Outer measured speed is 134.0 ± 4.9 km s^{-1}; CPTG gives 126.8 km s^{-1} and $\chi^2 = 12.80$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 32.621$, $M_{\rm mat} = 7.6753$, and $M_{\rm geom} = 24.945$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

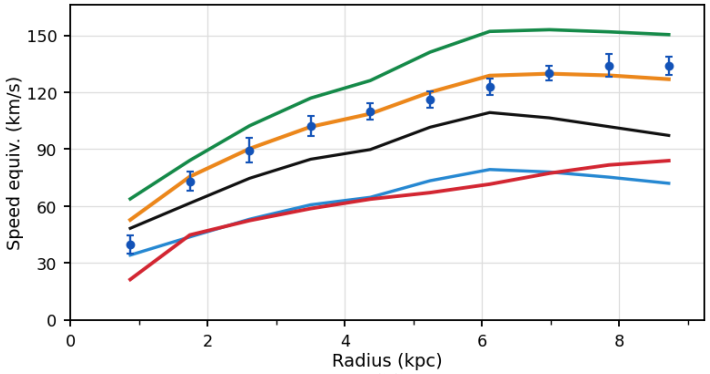


Figure 51: NGC 3972.

NGC 3992 9 radii

78.76% geometric; 21.24% material at $R_{\max} = 46.02$ kpc. Outer measured speed is 237.0 ± 8.6 km s^{-1}; CPTG gives 240.9 km s^{-1} and $\chi^2 = 1.82$ for 9 points. Mass equivalents at that radius are $M_{\rm total} = 620.72$, $M_{\rm mat} = 131.85$, and $M_{\rm geom} = 488.87$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

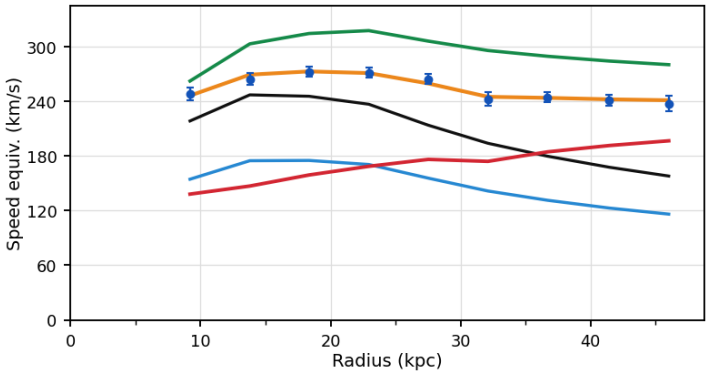


Figure 52: NGC 3992.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 4010 12 radii

69.79% geometric; 30.21% material at $R_{\max} = 10.47$ kpc. Outer measured speed is 122.0 ± 3.9 km s^{-1}; CPTG gives 116.4 km s^{-1} and $\chi^2 = 34.18$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 32.997$, $M_{\rm mat} = 9.9671$, and $M_{\rm geom} = 23.030$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

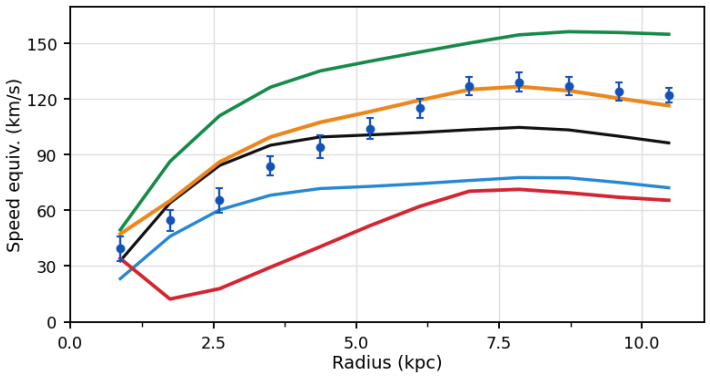


Figure 53: NGC 4010.

NGC 4013 36 radii

76.09% geometric; 23.91% material at $R_{\max} = 31.01$ kpc. Outer measured speed is 170.0 ± 10.0 km s^{-1}; CPTG gives 181.3 km s^{-1} and $\chi^2 = 31.34$ for 36 points. Mass equivalents at that radius are $M_{\rm total} = 237.09$, $M_{\rm mat} = 56.695$, and $M_{\rm geom} = 180.39$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

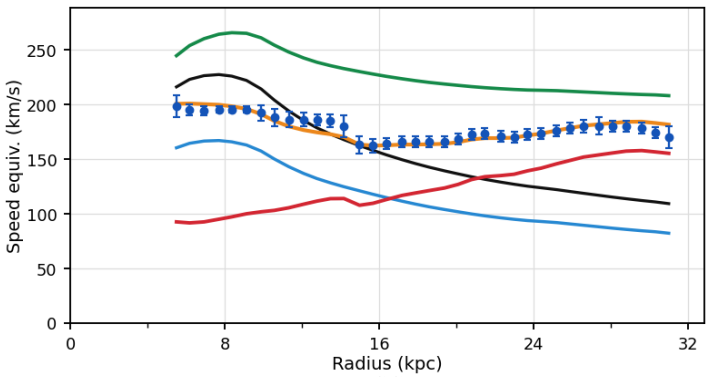


Figure 54: NGC 4013.

NGC 4051 7 radii

34.10% geometric; 65.90% material at $R_{\max} = 12.19$ kpc. Outer measured speed is 153.0 ± 9.8 km s^{-1}; CPTG gives 136.6 km s^{-1} and $\chi^2 = 5.10$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 52.864$, $M_{\rm mat} = 34.840$, and $M_{\rm geom} = 18.024$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

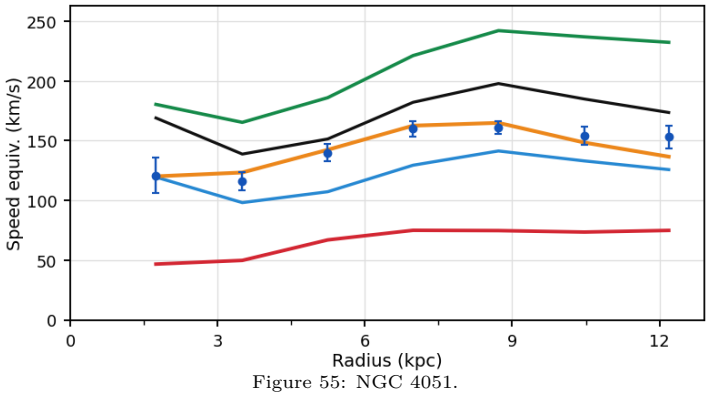


Figure 55: NGC 4051.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 4068 6 radii

64.21% geometric; 35.79% material at $R_{\max} = 2.33$ kpc. Outer measured speed is 41.9 ± 5.0 km s^{-1}; CPTG gives 37.9 km s^{-1} and $\chi^2 = 4.75$ for 6 points. Mass equivalents at that radius are $M_{\rm total} = 0.7800$, $M_{\rm mat} = 0.2791$, and $M_{\rm geom} = 0.5009$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

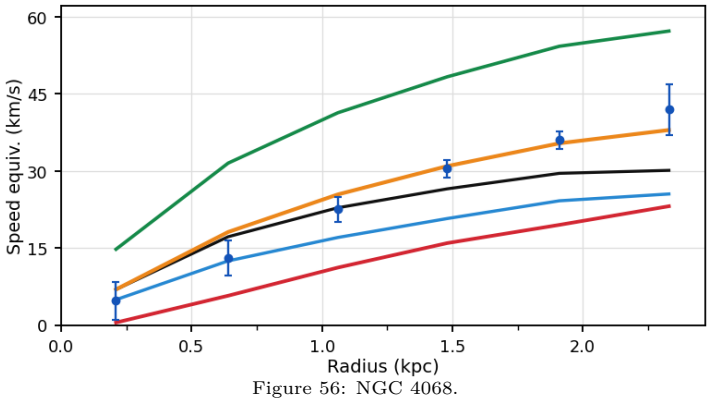


Figure 56: NGC 4068.

NGC 4085 7 radii

38.81% geometric; 61.19% material at $R_{\max} = 6.20$ kpc. Outer measured speed is 136.0 ± 7.4 km s^{-1}; CPTG gives 111.5 km s^{-1} and $\chi^2 = 119.41$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 17.913$, $M_{\rm mat} = 10.961$, and $M_{\rm geom} = 6.9521$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

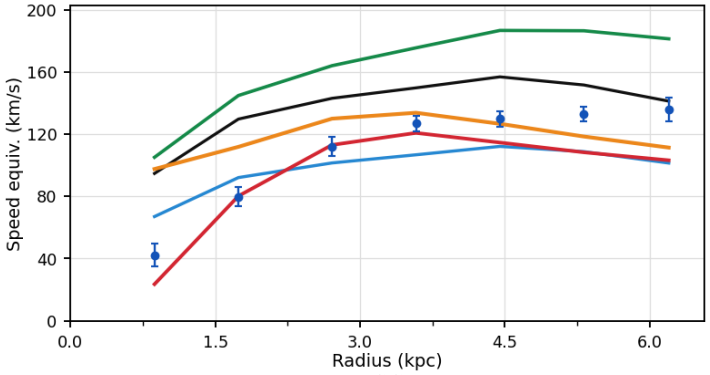


Figure 57: NGC 4085.

NGC 4088 12 radii

57.05% geometric; 42.95% material at $R_{\max} = 21.48$ kpc. Outer measured speed is 174.0 ± 7.6 km s^{-1}; CPTG gives 169.9 km s^{-1} and $\chi^2 = 46.87$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 144.23$, $M_{\rm mat} = 61.949$, and $M_{\rm geom} = 82.277$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

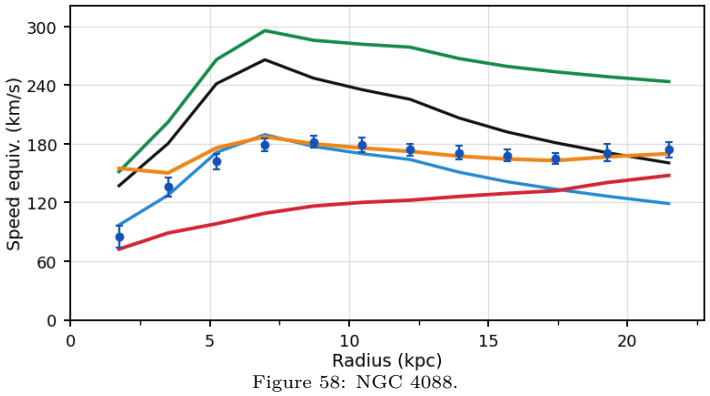


Figure 58: NGC 4088.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 4100 24 radii

79.23% geometric; 20.77% material at $R_{\max} = 22.76$ kpc. Outer measured speed is 159.0 ± 9.0 km s^{-1}; CPTG gives 168.6 km s^{-1} and $\chi^2 = 32.37$ for 24 points. Mass equivalents at that radius are $M_{\rm total} = 150.45$, $M_{\rm mat} = 31.241$, and $M_{\rm geom} = 119.21$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

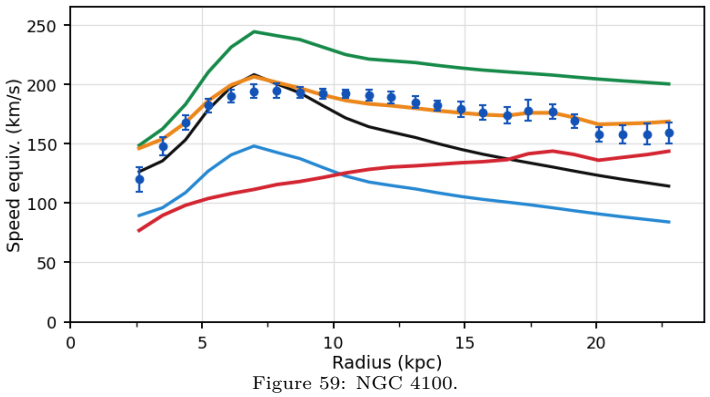


Figure 59: NGC 4100.

NGC 4138 7 radii

75.75% geometric; 24.25% material at $R_{\max} = 18.58$ kpc. Outer measured speed is 150.0 ± 21.0 km s^{-1}; CPTG gives 152.9 km s^{-1} and $\chi^2 = 12.10$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 100.98$, $M_{\rm mat} = 24.490$, and $M_{\rm geom} = 76.493$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

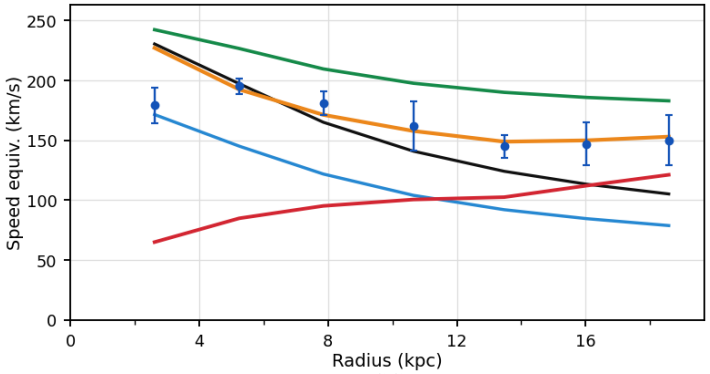


Figure 60: NGC 4138.

NGC 4157 17 radii

74.80% geometric; 25.20% material at $R_{\max} = 29.61$ kpc. Outer measured speed is 185.0 ± 13.6 km s^{-1}; CPTG gives 191.8 km s^{-1} and $\chi^2 = 38.33$ for 17 points. Mass equivalents at that radius are $M_{\rm total} = 253.36$, $M_{\rm mat} = 63.850$, and $M_{\rm geom} = 189.51$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

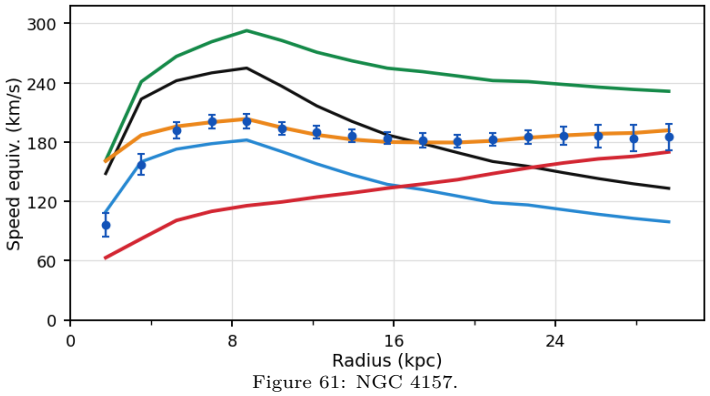


Figure 61: NGC 4157.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 4183 23 radii

82.44% geometric; 17.56% material at $R_{\max} = 21.02$ kpc. Outer measured speed is 113.0 ± 11.5 km s^{-1}; CPTG gives 112.4 km s^{-1} and $\chi^2 = 2.15$ for 23 points. Mass equivalents at that radius are $M_{\rm total} = 61.788$, $M_{\rm mat} = 10.848$, and $M_{\rm geom} = 50.939$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

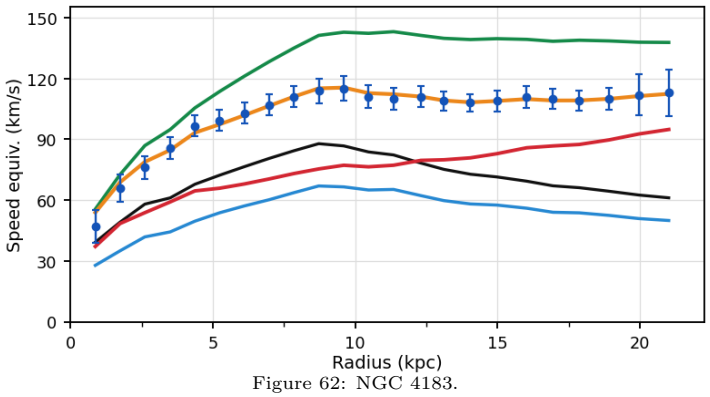


Figure 62: NGC 4183.

NGC 4217 19 radii

53.34% geometric; 46.66% material at $R_{\max} = 16.72$ kpc. Outer measured speed is 178.0 ± 12.0 km s^{-1}; CPTG gives 183.4 km s^{-1} and $\chi^2 = 316.67$ for 19 points. Mass equivalents at that radius are $M_{\rm total} = 130.79$, $M_{\rm mat} = 61.032$, and $M_{\rm geom} = 69.761$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

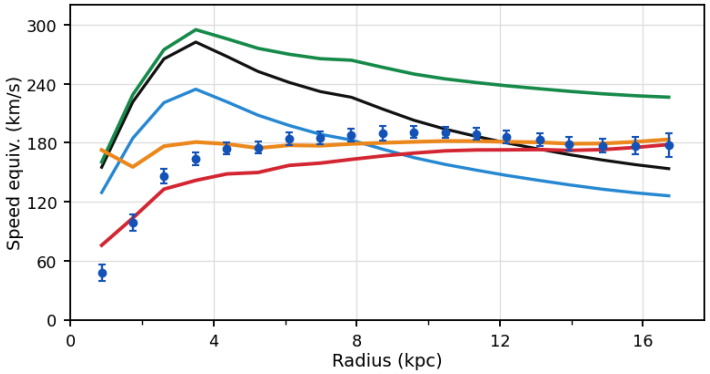


Figure 63: NGC 4217.

NGC 4559 32 radii

75.38% geometric; 24.62% material at $R_{\max} = 20.97$ kpc. Outer measured speed is 119.0 ± 6.1 km s^{-1}; CPTG gives 126.0 km s^{-1} and $\chi^2 = 8.42$ for 32 points. Mass equivalents at that radius are $M_{\rm total} = 77.397$, $M_{\rm mat} = 19.053$, and $M_{\rm geom} = 58.344$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

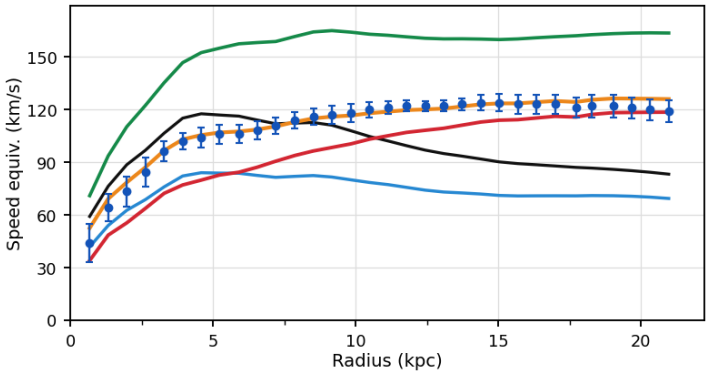


Figure 64: NGC 4559.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 5005 18 radii

50.41% geometric; 49.59% material at $R_{\max} = 11.47$ kpc. Outer measured speed is 265.0 ± 22.0 km s^{-1}; CPTG gives 264.7 km s^{-1} and $\chi^2 = 0.44$ for 18 points. Mass equivalents at that radius are $M_{\rm total} = 186.82$, $M_{\rm mat} = 92.640$, and $M_{\rm geom} = 94.185$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

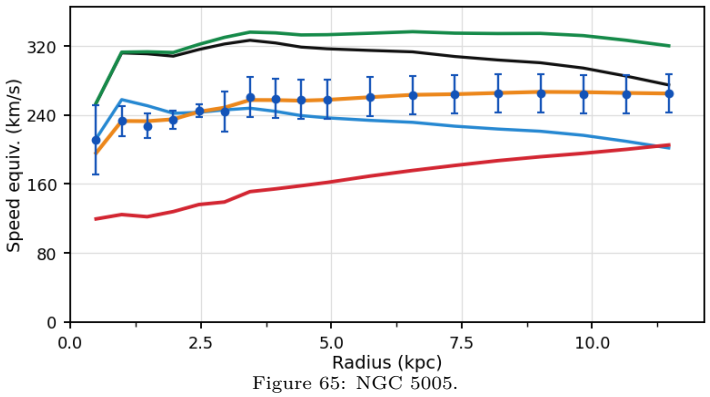


Figure 65: NGC 5005.

NGC 5033 22 radii

81.55% geometric; 18.45% material at $R_{\max} = 44.59$ kpc. Outer measured speed is 196.0 ± 5.0 km s^{-1}; CPTG gives 201.1 km s^{-1} and $\chi^2 = 34.38$ for 22 points. Mass equivalents at that radius are $M_{\rm total} = 419.42$, $M_{\rm mat} = 77.375$, and $M_{\rm geom} = 342.05$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

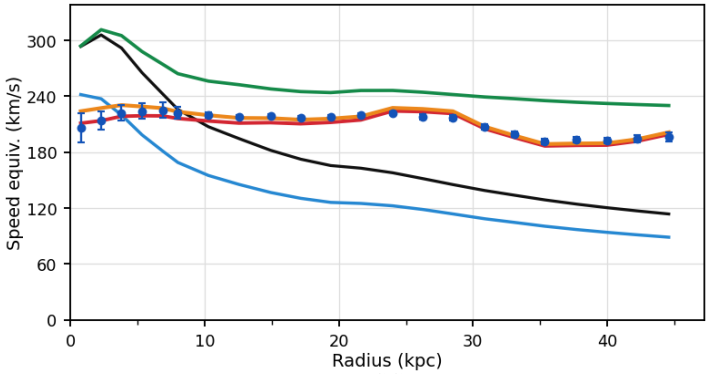


Figure 66: NGC 5033.

NGC 5055 28 radii

77.52% geometric; 22.48% material at $R_{\max} = 54.59$ kpc. Outer measured speed is 172.0 ± 4.8 km s^{-1}; CPTG gives 181.3 km s^{-1} and $\chi^2 = 1257.16$ for 28 points. Mass equivalents at that radius are $M_{\rm total} = 416.99$, $M_{\rm mat} = 93.755$, and $M_{\rm geom} = 323.23$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

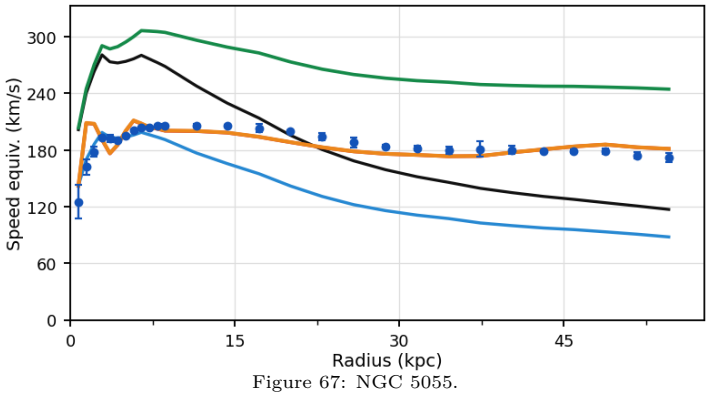


Figure 67: NGC 5055.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 5371 19 radii

65.71% geometric; 34.29% material at $R_{\max} = 46.24$ kpc. Outer measured speed is 213.0 ± 6.0 km s^{-1}; CPTG gives 220.7 km s^{-1} and $\chi^2 = 44.31$ for 19 points. Mass equivalents at that radius are $M_{\rm total} = 523.83$, $M_{\rm mat} = 179.61$, and $M_{\rm geom} = 344.22$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

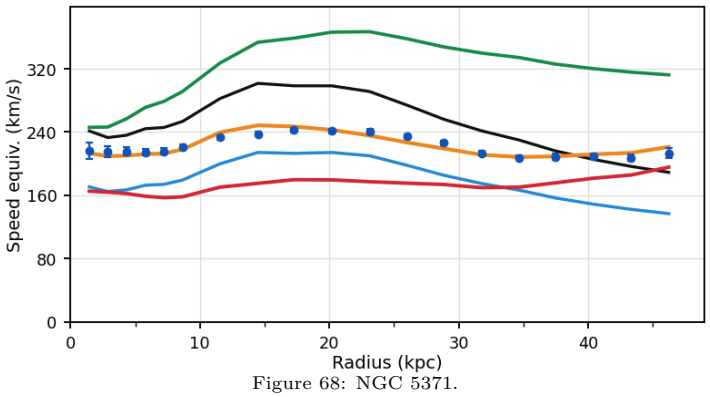


Figure 68: NGC 5371.

NGC 5585 24 radii

81.73% geometric; 18.27% material at $R_{\max} = 10.96$ kpc. Outer measured speed is 89.4 ± 2.7 km s^{-1}; CPTG gives 83.6 km s^{-1} and $\chi^2 = 296.77$ for 24 points. Mass equivalents at that radius are $M_{\rm total} = 17.810$, $M_{\rm mat} = 3.2534$, and $M_{\rm geom} = 14.557$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

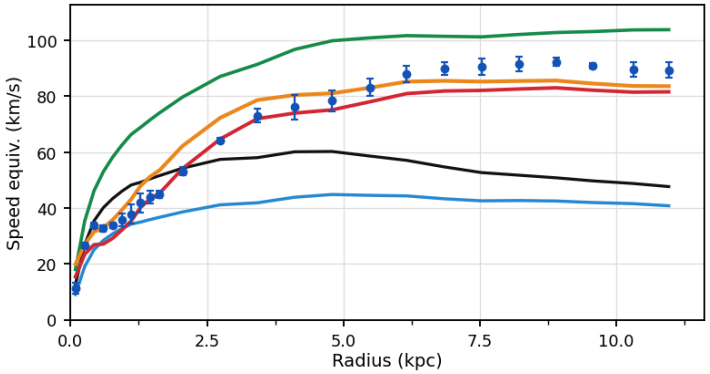


Figure 69: NGC 5585.

NGC 5907 19 radii

76.24% geometric; 23.76% material at $R_{\max} = 50.33$ kpc. Outer measured speed is 214.0 ± 5.0 km s^{-1}; CPTG gives 217.0 km s^{-1} and $\chi^2 = 37.86$ for 19 points. Mass equivalents at that radius are $M_{\rm total} = 551.08$, $M_{\rm mat} = 130.93$, and $M_{\rm geom} = 420.15$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

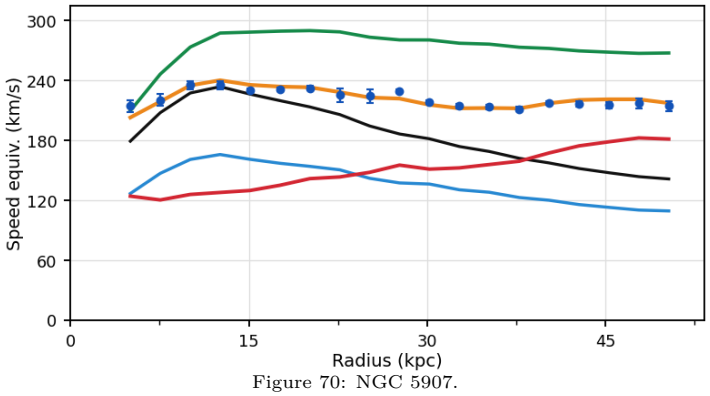


Figure 70: NGC 5907.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 5985 33 radii

82.02% geometric; 17.98% material at $R_{\max} = 34.72$ kpc. Outer measured speed is 285.0 ± 4.0 km s^{-1}; CPTG gives 287.3 km s^{-1} and $\chi^2 = 34.77$ for 33 points. Mass equivalents at that radius are $M_{\rm total} = 666.34$, $M_{\rm mat} = 119.84$, and $M_{\rm geom} = 546.50$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

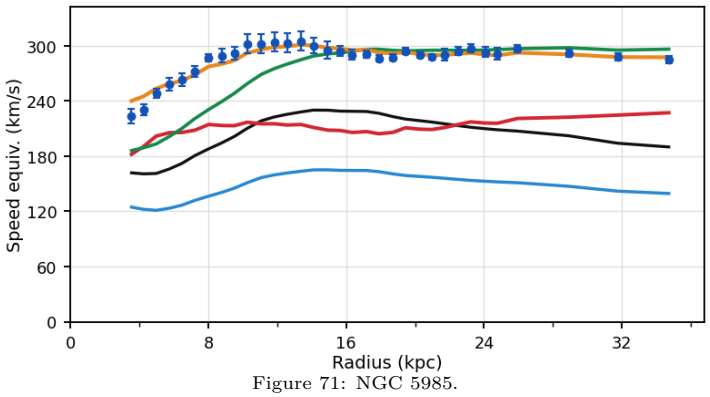


Figure 71: NGC 5985.

NGC 6015 44 radii

71.48% geometric; 28.52% material at $R_{\max} = 29.23$ kpc. Outer measured speed is 152.0 ± 3.4 km s^{-1}; CPTG gives 153.4 km s^{-1} and $\chi^2 = 65.42$ for 44 points. Mass equivalents at that radius are $M_{\rm total} = 159.85$, $M_{\rm mat} = 45.586$, and $M_{\rm geom} = 114.26$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

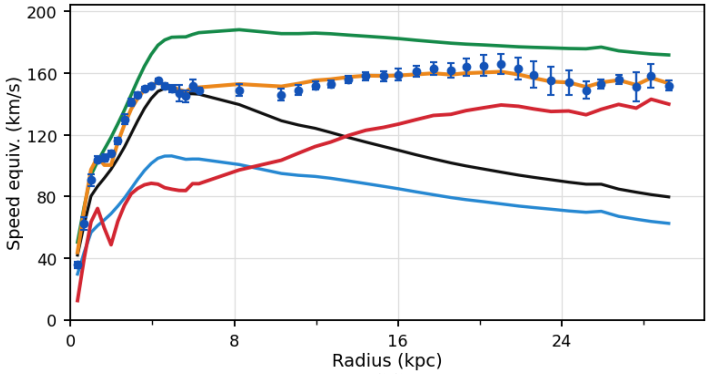


Figure 72: NGC 6015.

NGC 6195 23 radii

53.46% geometric; 46.54% material at $R_{\max} = 36.43$ kpc. Outer measured speed is 246.0 ± 4.2 km s^{-1}; CPTG gives 245.6 km s^{-1} and $\chi^2 = 14.95$ for 23 points. Mass equivalents at that radius are $M_{\rm total} = 510.95$, $M_{\rm mat} = 237.77$, and $M_{\rm geom} = 273.18$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

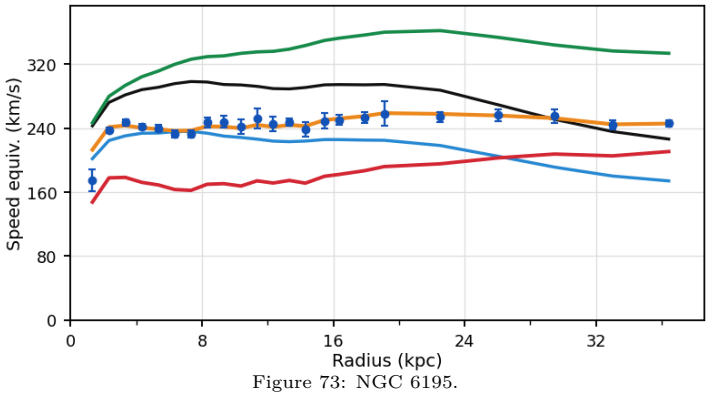


Figure 73: NGC 6195.

● ± Observed	— Baryonic	— Material	— Geometric	— CPTG	— MOND

NGC 6503 31 radii

87.24% geometric; 12.76% material at $R_{\max} = 23.50$ kpc. Outer measured speed is 115.0 ± 10.0 km s^{-1}; CPTG gives 114.8 km s^{-1} and $\chi^2 = 67.76$ for 31 points. Mass equivalents at that radius are $M_{\rm total} = 72.058$, $M_{\rm mat} = 9.1953$, and $M_{\rm geom} = 62.863$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

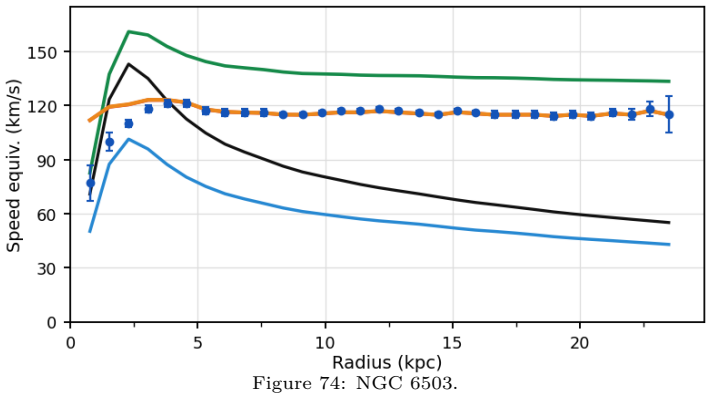


Figure 74: NGC 6503.

NGC 6674 15 radii

86.30% geometric; 13.70% material at $R_{\max} = 72.41$ kpc. Outer measured speed is 242.0 ± 5.0 km s^{-1}; CPTG gives 245.0 km s^{-1} and $\chi^2 = 3.39$ for 15 points. Mass equivalents at that radius are $M_{\rm total} = 1,010.36$, $M_{\rm mat} = 138.41$, and $M_{\rm geom} = 871.95$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

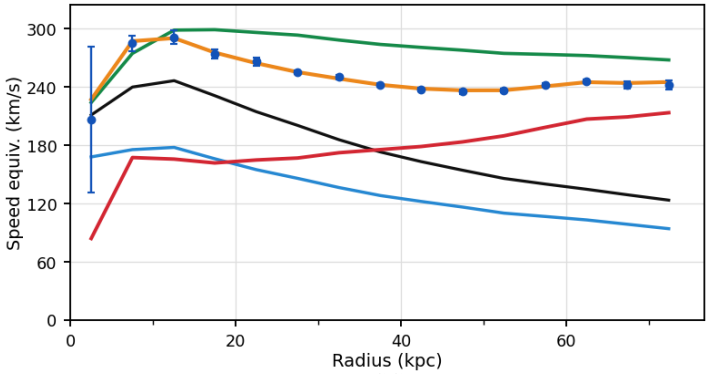


Figure 75: NGC 6674.

NGC 6789 4 radii

89.46% geometric; 10.54% material at $R_{\max} = 0.71$ kpc. Outer measured speed is 60.4 ± 9.4 km s^{-1}; CPTG gives 55.2 km s^{-1} and $\chi^2 = 0.44$ for 4 points. Mass equivalents at that radius are $M_{\rm total} = 0.5025$, $M_{\rm mat} = 0.0530$, and $M_{\rm geom} = 0.4495$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed.

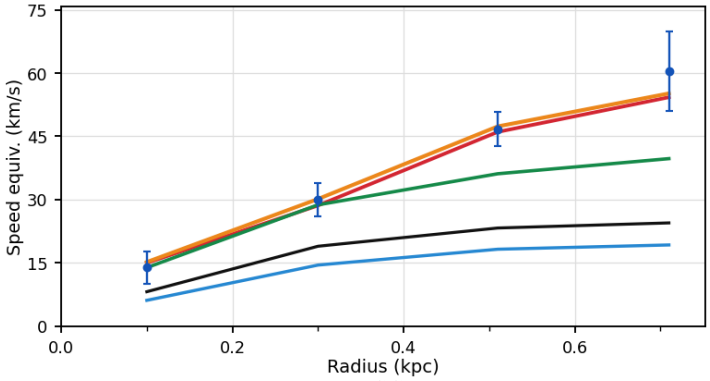


Figure 76: NGC 6789.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 6946 58 radii

69.89% geometric; 30.11% material at $R_{\rm max} = 20.40$ kpc. Outer measured speed is 154.0 ± 21.0 km s^{-1}; CPTG gives 180.8 km s^{-1} and $\chi^2 = 194.22$ for 58 points. Mass equivalents at that radius are $M_{\rm total} = 155.04$, $M_{\rm mat} = 46.690$, and $M_{\rm geom} = 108.35$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

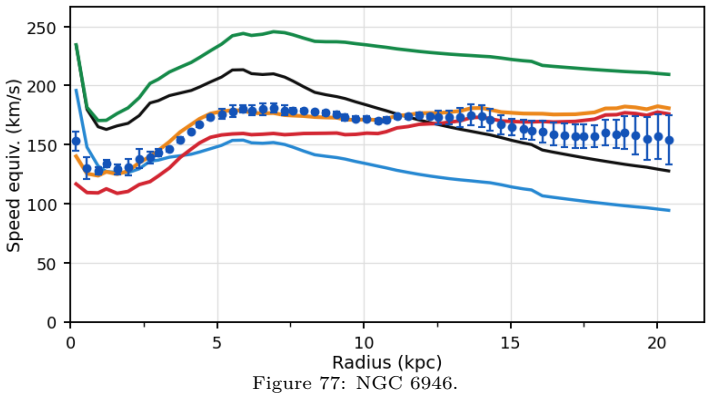


Figure 77: NGC 6946.

NGC 7331 36 radii

67.54% geometric; 32.46% material at $R_{\rm max} = 36.31$ kpc. Outer measured speed is 238.0 ± 9.0 km s^{-1}; CPTG gives 231.4 km s^{-1} and $\chi^2 = 48.90$ for 36 points. Mass equivalents at that radius are $M_{\rm total} = 451.98$, $M_{\rm mat} = 146.70$, and $M_{\rm geom} = 305.28$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

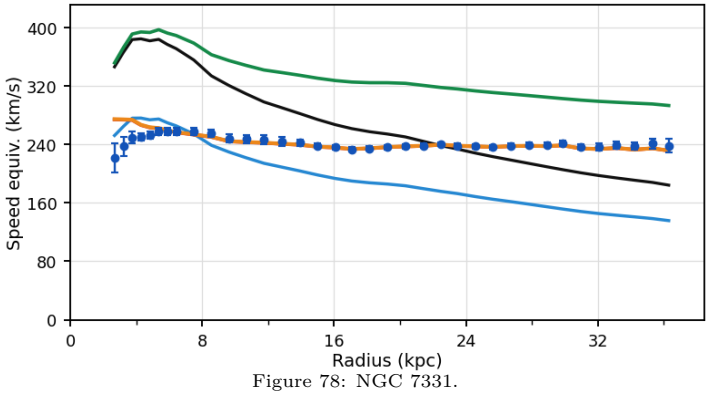


Figure 78: NGC 7331.

NGC 7793 46 radii

71.50% geometric; 28.50% material at $R_{\rm max} = 7.87$ kpc. Outer measured speed is 90.8 ± 6.1 km s^{-1}; CPTG gives 95.6 km s^{-1} and $\chi^2 = 11.70$ for 46 points. Mass equivalents at that radius are $M_{\rm total} = 16.732$, $M_{\rm mat} = 4.7693$, and $M_{\rm geom} = 11.963$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

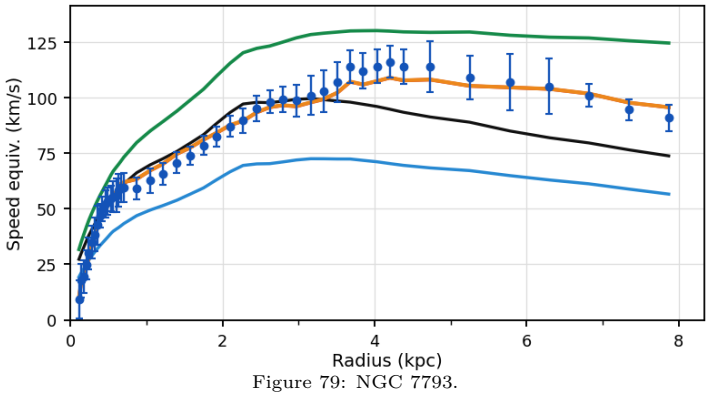


Figure 79: NGC 7793.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

NGC 7814 18 radii

76.54% geometric; 23.46% material at $R_{\max} = 19.53$ kpc. Outer measured speed is 214.0 ± 4.9 km s^{-1}; CPTG gives 214.1 km s^{-1} and $\chi^2 = 0.78$ for 18 points. Mass equivalents at that radius are $M_{\rm total} = 208.24$, $M_{\rm mat} = 48.852$, and $M_{\rm geom} = 159.39$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

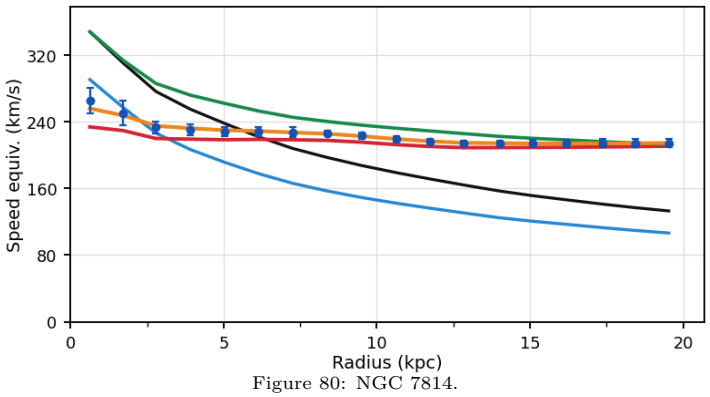


Figure 80: NGC 7814.

UGC 00128 22 radii

92.03% geometric; 7.97% material at $R_{\max} = 53.75$ kpc. Outer measured speed is 125.0 ± 6.0 km s^{-1}; CPTG gives 133.8 km s^{-1} and $\chi^2 = 626.86$ for 22 points. Mass equivalents at that radius are $M_{\rm total} = 223.76$, $M_{\rm mat} = 17.840$, and $M_{\rm geom} = 205.92$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

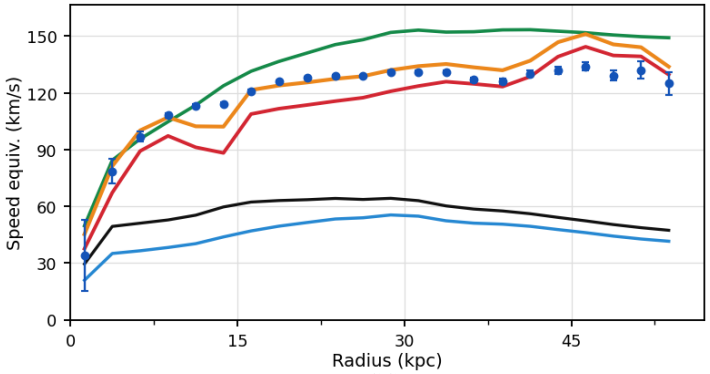


Figure 81: UGC 00128.

UGC 00191 9 radii

82.72% geometric; 17.28% material at $R_{\max} = 9.98$ kpc. Outer measured speed is 83.8 ± 1.0 km s^{-1}; CPTG gives 84.8 km s^{-1} and $\chi^2 = 12.36$ for 9 points. Mass equivalents at that radius are $M_{\rm total} = 16.695$, $M_{\rm mat} = 2.8847$, and $M_{\rm geom} = 13.810$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

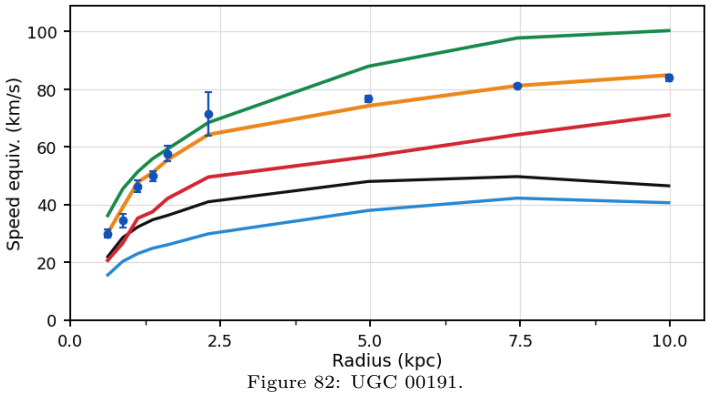


Figure 82: UGC 00191.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 00634 4 radii

88.50% geometric; 11.50% material at $R_{\max} = 18.01$ kpc. Outer measured speed is 107.5 ± 2.3 km s^{-1}; CPTG gives 105.5 km s^{-1} and $\chi^2 = 1.08$ for 4 points. Mass equivalents at that radius are $M_{\rm total} = 46.580$, $M_{\rm mat} = 5.3579$, and $M_{\rm geom} = 41.222$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

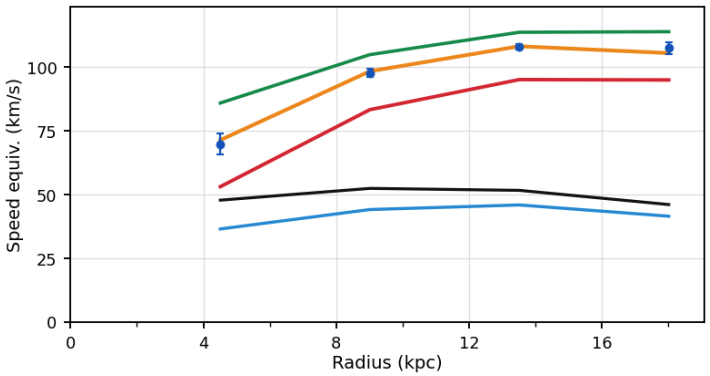


Figure 83: UGC 00634.

UGC 00731 12 radii

84.64% geometric; 15.36% material at $R_{\max} = 10.91$ kpc. Outer measured speed is 73.9 ± 3.4 km s^{-1}; CPTG gives 76.2 km s^{-1} and $\chi^2 = 2.14$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 14.732$, $M_{\rm mat} = 2.2621$, and $M_{\rm geom} = 12.470$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

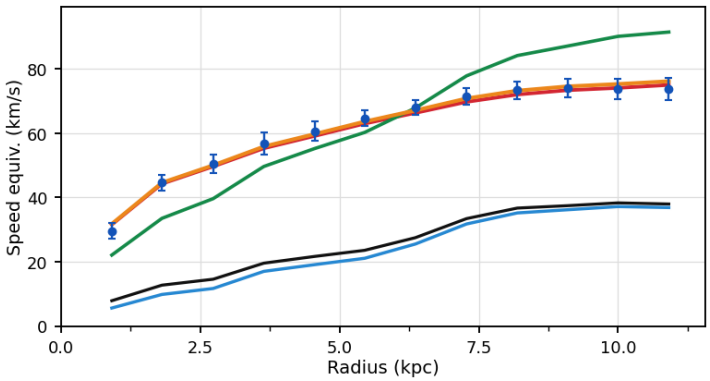


Figure 84: UGC 00731.

UGC 00891 5 radii

88.08% geometric; 11.92% material at $R_{\max} = 7.39$ kpc. Outer measured speed is 63.8 ± 0.9 km s^{-1}; CPTG gives 63.5 km s^{-1} and $\chi^2 = 8.45$ for 5 points. Mass equivalents at that radius are $M_{\rm total} = 6.9345$, $M_{\rm mat} = 0.8263$, and $M_{\rm geom} = 6.1082$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

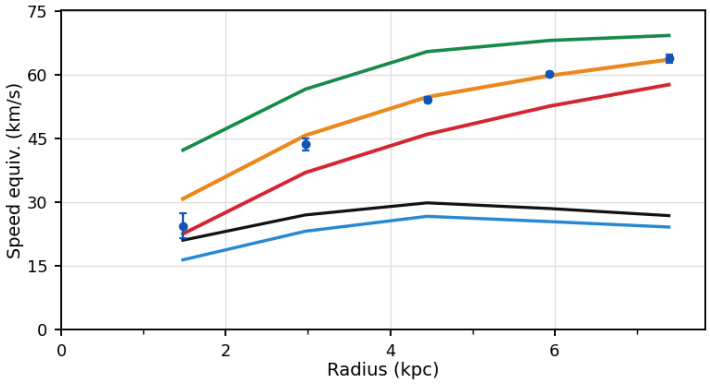


Figure 85: UGC 00891.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 01281 25 radii

61.56% geometric; 38.44% material at $R_{\max} = 4.99$ kpc. Outer measured speed is 56.9 ± 4.0 km s^{-1}; CPTG gives 54.1 km s^{-1} and $\chi^2 = 17.57$ for 25 points. Mass equivalents at that radius are $M_{\rm total} = 3.3975$, $M_{\rm mat} = 1.3058$, and $M_{\rm geom} = 2.0917$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

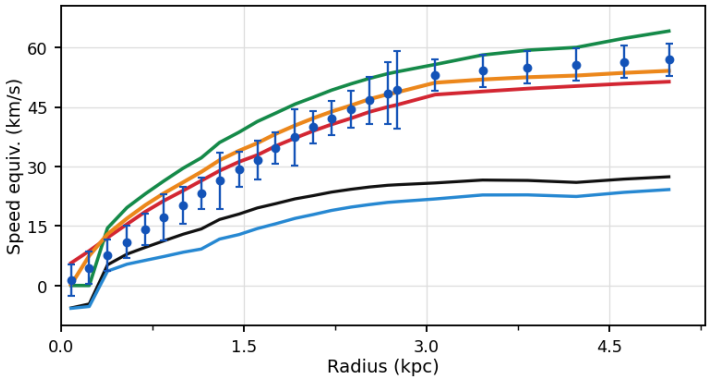


Figure 86: UGC 01281.

UGC 02259 8 radii

89.19% geometric; 10.81% material at $R_{\max} = 8.14$ kpc. Outer measured speed is 90.0 ± 3.0 km s^{-1}; CPTG gives 90.2 km s^{-1} and $\chi^2 = 1.62$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 15.403$, $M_{\rm mat} = 1.6645$, and $M_{\rm geom} = 13.739$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

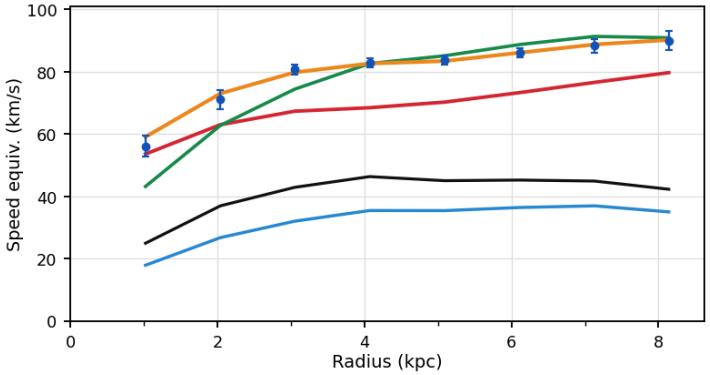


Figure 87: UGC 02259.

UGC 02487 17 radii

86.29% geometric; 13.71% material at $R_{\max} = 80.38$ kpc. Outer measured speed is 333.0 ± 15.3 km s^{-1}; CPTG gives 342.3 km s^{-1} and $\chi^2 = 21.73$ for 17 points. Mass equivalents at that radius are $M_{\rm total} = 2,190.04$, $M_{\rm mat} = 300.33$, and $M_{\rm geom} = 1,889.71$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

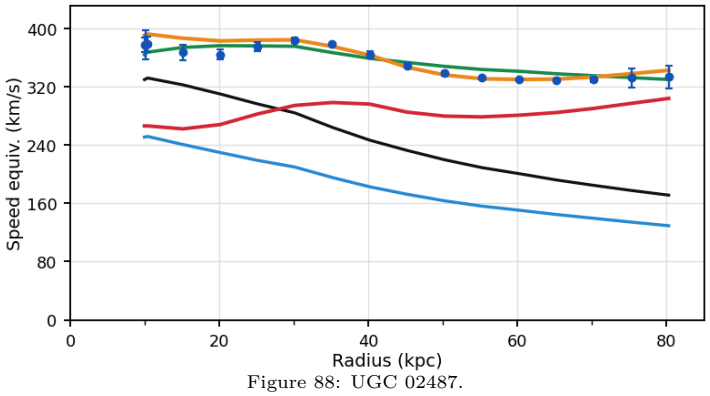


Figure 88: UGC 02487.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 02885 19 radii

80.49% geometric; 19.51% material at $R_{\max} = 74.07$ kpc. Outer measured speed is 298.0 ± 10.0 km s^{-1}; CPTG gives 302.7 km s^{-1} and $\chi^2 = 12.90$ for 19 points. Mass equivalents at that radius are $M_{\text{total}} = 1,578.19$, $M_{\text{mat}} = 307.89$, and $M_{\text{geom}} = 1,270.30$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

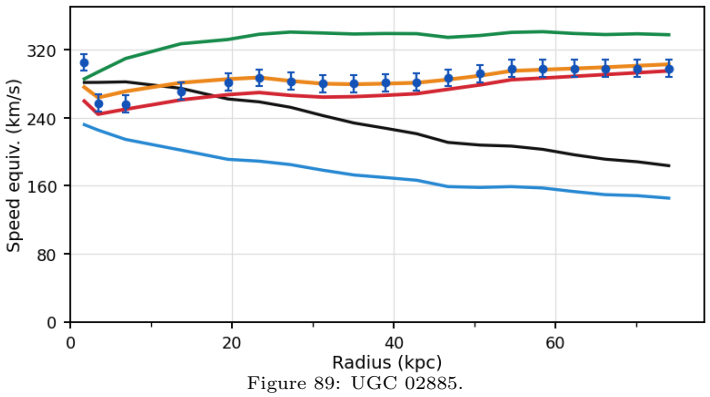


Figure 89: UGC 02885.

UGC 02916 43 radii

68.51% geometric; 31.49% material at $R_{\max} = 38.00$ kpc. Outer measured speed is 181.0 ± 10.3 km s^{-1}; CPTG gives 198.2 km s^{-1} and $\chi^2 = 39.45$ for 43 points. Mass equivalents at that radius are $M_{\text{total}} = 346.94$, $M_{\text{mat}} = 109.26$, and $M_{\text{geom}} = 237.68$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

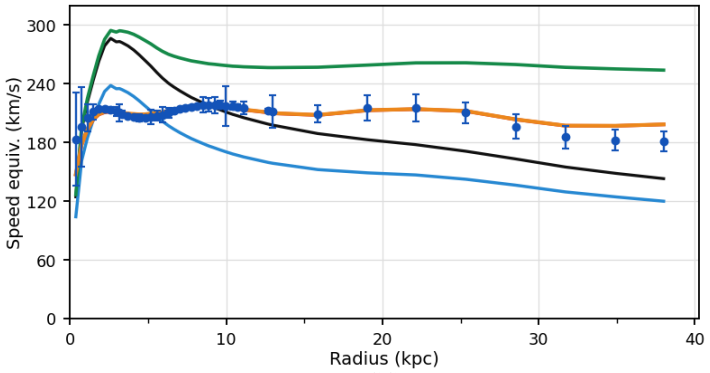


Figure 90: UGC 02916.

UGC 02953 115 radii

87.63% geometric; 12.37% material at $R_{\max} = 62.39$ kpc. Outer measured speed is 272.0 ± 5.0 km s^{-1}; CPTG gives 312.6 km s^{-1} and $\chi^2 = 2331.87$ for 115 points. Mass equivalents at that radius are $M_{\text{total}} = 1,417.96$, $M_{\text{mat}} = 175.44$, and $M_{\text{geom}} = 1,242.53$ in $10^9 M_\odot$. The catalog-wide χ^2/N_{radii} diagnostic exceeds 20; a large geometric mass equivalent does not resolve these significant velocity residuals. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

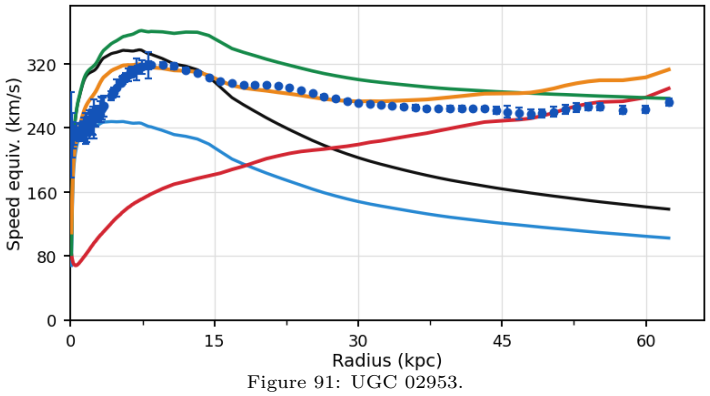


Figure 91: UGC 02953.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 03205 48 radii

84.38% geometric; 15.62% material at $R_{\max} = 40.04$ kpc. Outer measured speed is 220.0 ± 6.3 km s^{-1}; CPTG gives 224.7 km s^{-1} and $\chi^2 = 94.53$ for 48 points. Mass equivalents at that radius are $M_{\rm total} = 470.14$, $M_{\rm mat} = 73.456$, and $M_{\rm geom} = 396.68$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

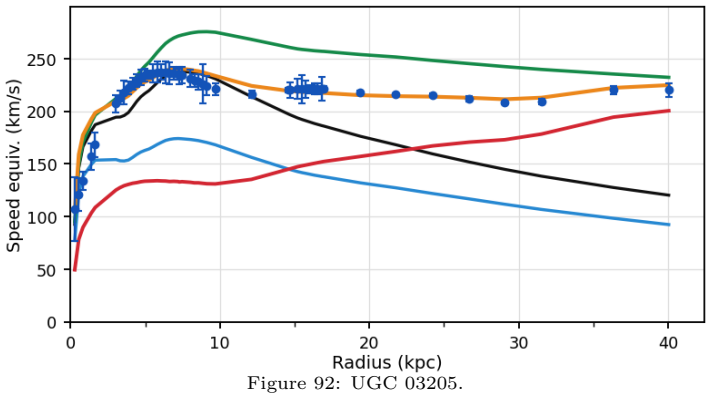


Figure 92: UGC 03205.

UGC 03546 30 radii

77.64% geometric; 22.36% material at $R_{\max} = 29.23$ kpc. Outer measured speed is 193.0 ± 18.6 km s^{-1}; CPTG gives 198.4 km s^{-1} and $\chi^2 = 11.74$ for 30 points. Mass equivalents at that radius are $M_{\rm total} = 267.64$, $M_{\rm mat} = 59.857$, and $M_{\rm geom} = 207.79$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

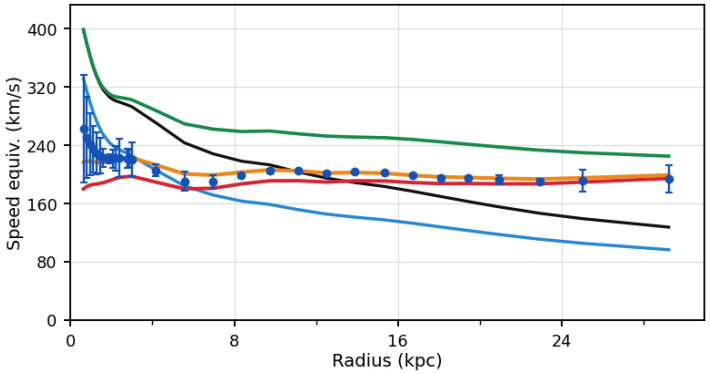


Figure 93: UGC 03546.

UGC 03580 47 radii

86.63% geometric; 13.37% material at $R_{\max} = 27.06$ kpc. Outer measured speed is 124.0 ± 1.9 km s^{-1}; CPTG gives 127.2 km s^{-1} and $\chi^2 = 248.17$ for 47 points. Mass equivalents at that radius are $M_{\rm total} = 101.83$, $M_{\rm mat} = 13.611$, and $M_{\rm geom} = 88.221$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

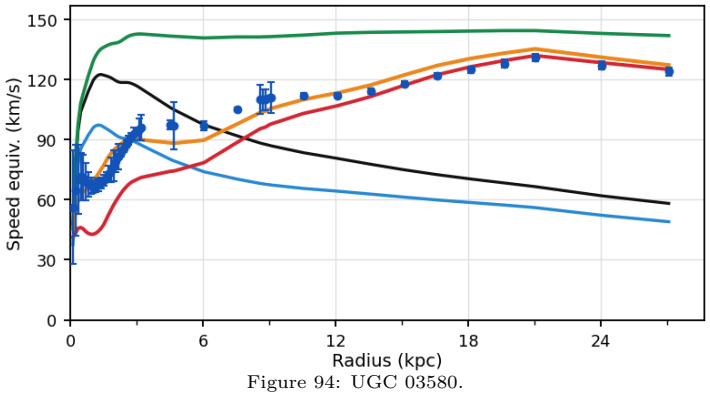


Figure 94: UGC 03580.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 04278 25 radii

84.66% geometric; 15.34% material at $R_{\max} = 6.69$ kpc. Outer measured speed is 92.8 ± 8.8 km s^{-1}; CPTG gives 96.5 km s^{-1} and $\chi^2 = 3.20$ for 25 points. Mass equivalents at that radius are $M_{\rm total} = 14.473$, $M_{\rm mat} = 2.2197$, and $M_{\rm geom} = 12.254$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

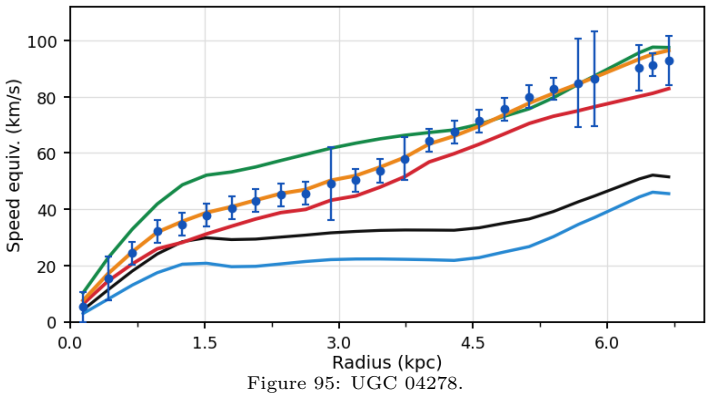


Figure 95: UGC 04278.

UGC 04325 8 radii

82.99% geometric; 17.01% material at $R_{\max} = 5.59$ kpc. Outer measured speed is 91.5 ± 3.0 km s^{-1}; CPTG gives 93.1 km s^{-1} and $\chi^2 = 2.10$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 11.269$, $M_{\rm mat} = 1.9170$, and $M_{\rm geom} = 9.3516$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

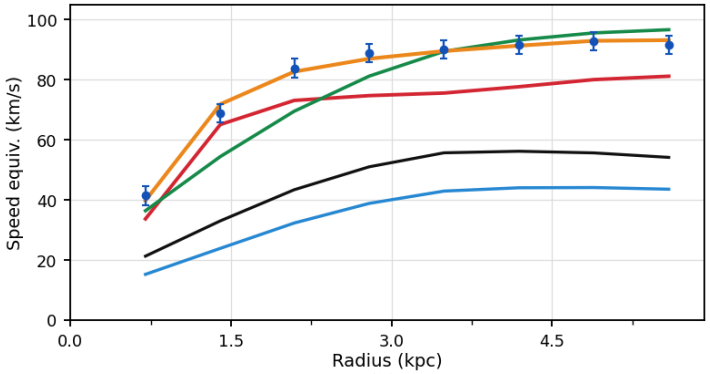


Figure 96: UGC 04325.

UGC 04483 8 radii

72.72% geometric; 27.28% material at $R_{\max} = 1.21$ kpc. Outer measured speed is 24.2 ± 1.2 km s^{-1}; CPTG gives 23.6 km s^{-1} and $\chi^2 = 1.35$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 0.1571$, $M_{\rm mat} = 0.0429$, and $M_{\rm geom} = 0.1142$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

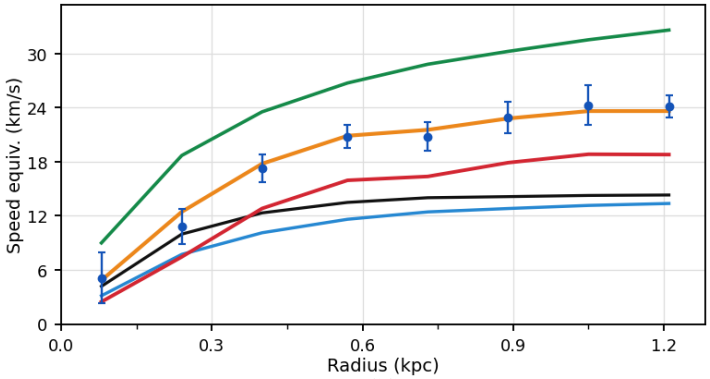


Figure 97: UGC 04483.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 04499 9 radii

79.91% geometric; 20.09% material at $R_{\max} = 8.18$ kpc. Outer measured speed is 74.3 ± 2.7 km s^{-1}; CPTG gives 75.4 km s^{-1} and $\chi^2 = 6.53$ for 9 points. Mass equivalents at that radius are $M_{\rm total} = 10.803$, $M_{\rm mat} = 2.1701$, and $M_{\rm geom} = 8.6332$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

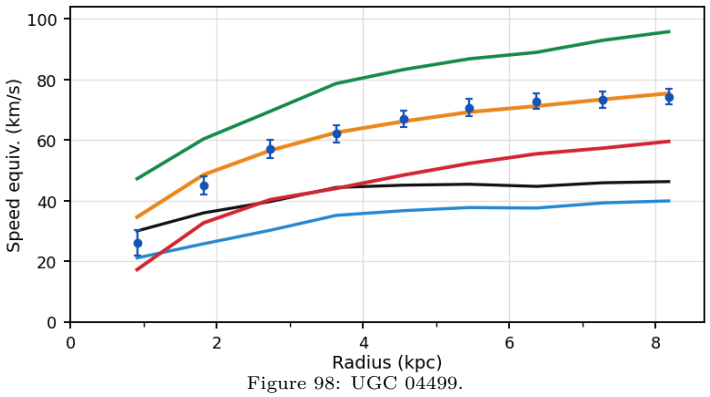


Figure 98: UGC 04499.

UGC 05005 11 radii

85.18% geometric; 14.82% material at $R_{\max} = 28.61$ kpc. Outer measured speed is 99.1 ± 10.0 km s^{-1}; CPTG gives 100.9 km s^{-1} and $\chi^2 = 2.20$ for 11 points. Mass equivalents at that radius are $M_{\rm total} = 67.748$, $M_{\rm mat} = 10.042$, and $M_{\rm geom} = 57.706$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

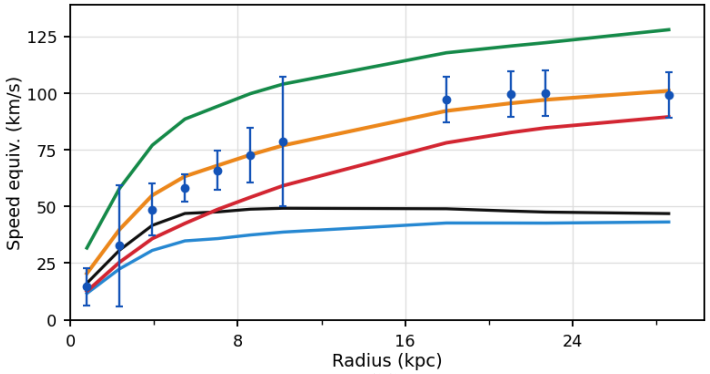


Figure 99: UGC 05005.

UGC 05253 73 radii

80.01% geometric; 19.99% material at $R_{\max} = 53.29$ kpc. Outer measured speed is 218.0 ± 6.5 km s^{-1}; CPTG gives 221.3 km s^{-1} and $\chi^2 = 30.62$ for 73 points. Mass equivalents at that radius are $M_{\rm total} = 606.96$, $M_{\rm mat} = 121.30$, and $M_{\rm geom} = 485.66$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

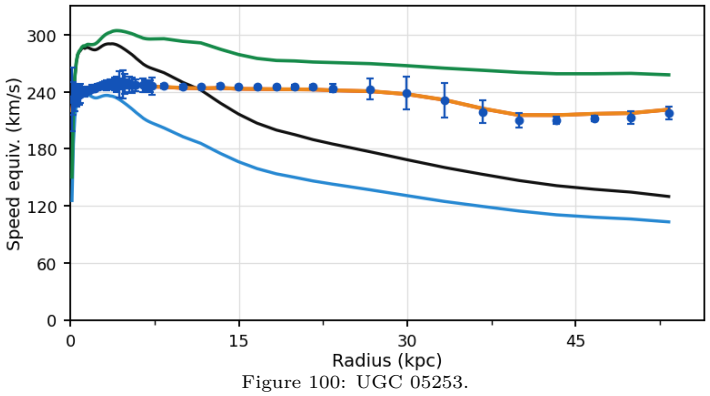


Figure 100: UGC 05253.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 05414 6 radii

72.40% geometric; 27.60% material at $R_{\max} = 4.11$ kpc. Outer measured speed is 61.4 ± 2.4 km s^{-1}; CPTG gives 58.3 km s^{-1} and $\chi^2 = 6.30$ for 6 points. Mass equivalents at that radius are $M_{\rm total} = 3.2453$, $M_{\rm mat} = 0.8958$, and $M_{\rm geom} = 2.3494$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

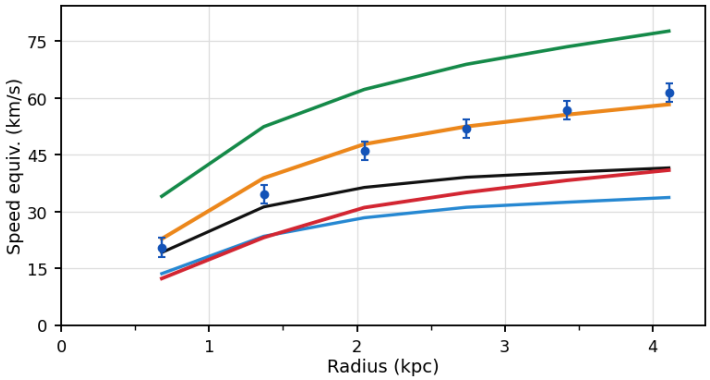


Figure 101: UGC 05414.

UGC 05716 12 radii

88.34% geometric; 11.66% material at $R_{\max} = 12.37$ kpc. Outer measured speed is 74.7 ± 2.1 km s^{-1}; CPTG gives 73.7 km s^{-1} and $\chi^2 = 1.00$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 15.621$, $M_{\rm mat} = 1.8212$, and $M_{\rm geom} = 13.800$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

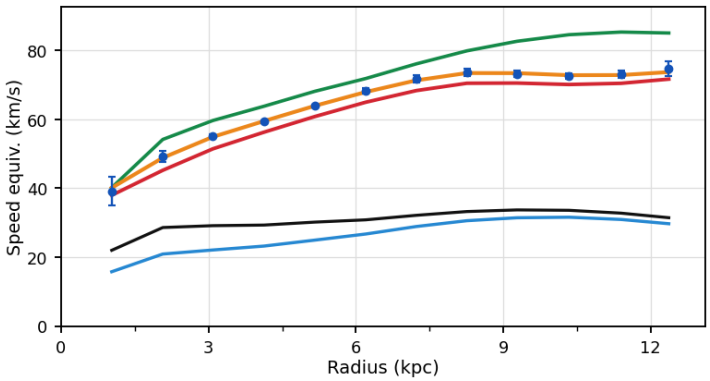


Figure 102: UGC 05716.

UGC 05721 23 radii

88.54% geometric; 11.46% material at $R_{\max} = 6.74$ kpc. Outer measured speed is 79.5 ± 4.6 km s^{-1}; CPTG gives 77.3 km s^{-1} and $\chi^2 = 28.80$ for 23 points. Mass equivalents at that radius are $M_{\rm total} = 9.3729$, $M_{\rm mat} = 1.0741$, and $M_{\rm geom} = 8.2987$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

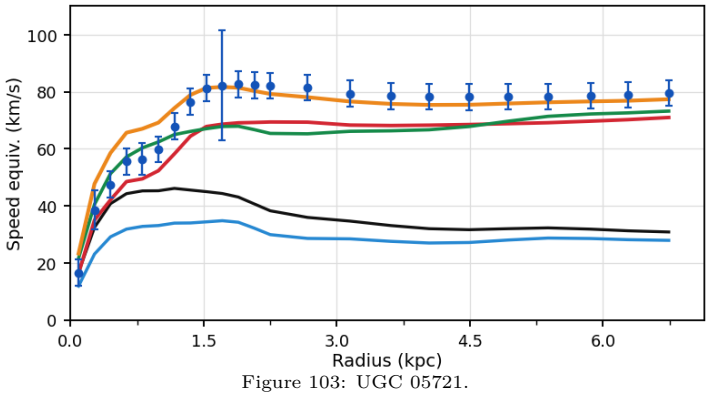


Figure 103: UGC 05721.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 05750 11 radii

76.70% geometric; 23.30% material at $R_{\max} = 22.85$ kpc. Outer measured speed is 78.9 ± 11.0 km s^{-1}; CPTG gives 77.1 km s^{-1} and $\chi^2 = 0.70$ for 11 points. Mass equivalents at that radius are $M_{\rm total} = 31.588$, $M_{\rm mat} = 7.3601$, and $M_{\rm geom} = 24.228$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

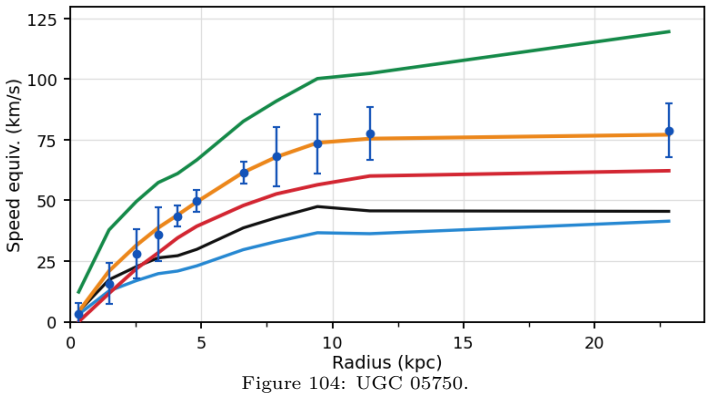


Figure 104: UGC 05750.

UGC 05764 10 radii

86.42% geometric; 13.58% material at $R_{\max} = 3.62$ kpc. Outer measured speed is 49.9 ± 2.8 km s^{-1}; CPTG gives 50.5 km s^{-1} and $\chi^2 = 6.27$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 2.1489$, $M_{\rm mat} = 0.2918$, and $M_{\rm geom} = 1.8571$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

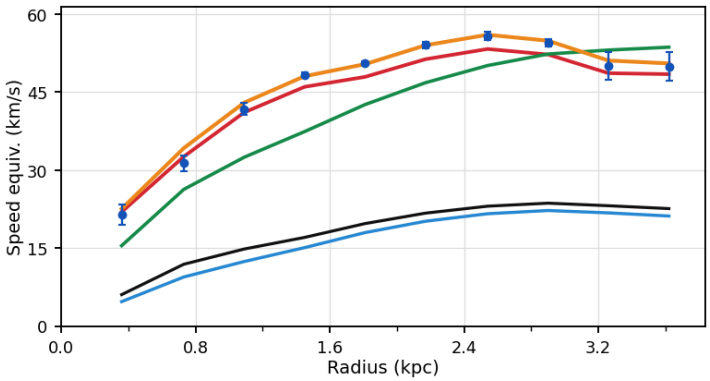


Figure 105: UGC 05764.

UGC 05829 11 radii

76.96% geometric; 23.04% material at $R_{\max} = 6.91$ kpc. Outer measured speed is 68.6 ± 5.1 km s^{-1}; CPTG gives 67.7 km s^{-1} and $\chi^2 = 0.12$ for 11 points. Mass equivalents at that radius are $M_{\rm total} = 7.3592$, $M_{\rm mat} = 1.6953$, and $M_{\rm geom} = 5.6639$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

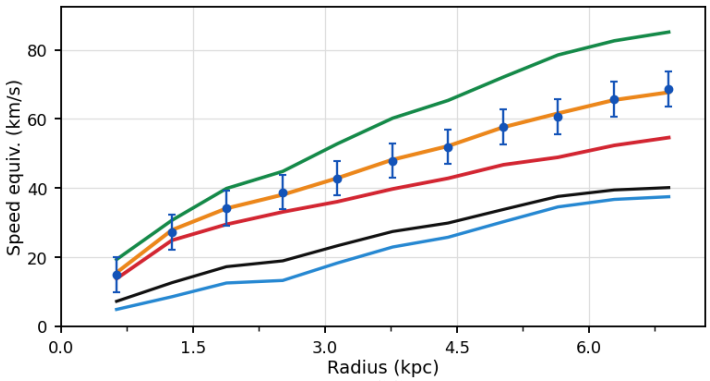


Figure 106: UGC 05829.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 05918 8 radii

82.87% geometric; 17.13% material at $R_{\max} = 4.46$ kpc. Outer measured speed is 44.5 ± 3.9 km s^{-1}; CPTG gives 44.9 km s^{-1} and $\chi^2 = 0.13$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 2.0946$, $M_{\rm mat} = 0.3588$, and $M_{\rm geom} = 1.7357$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

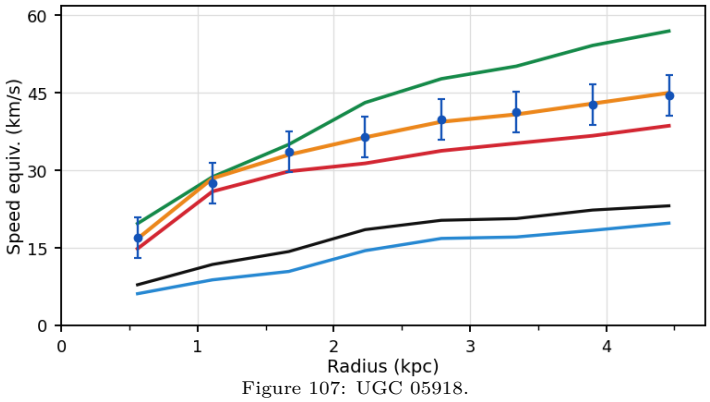


Figure 107: UGC 05918.

UGC 05986 15 radii

80.33% geometric; 19.67% material at $R_{\max} = 9.41$ kpc. Outer measured speed is 107.0 ± 6.4 km s^{-1}; CPTG gives 102.9 km s^{-1} and $\chi^2 = 25.05$ for 15 points. Mass equivalents at that radius are $M_{\rm total} = 23.179$, $M_{\rm mat} = 4.5585$, and $M_{\rm geom} = 18.620$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

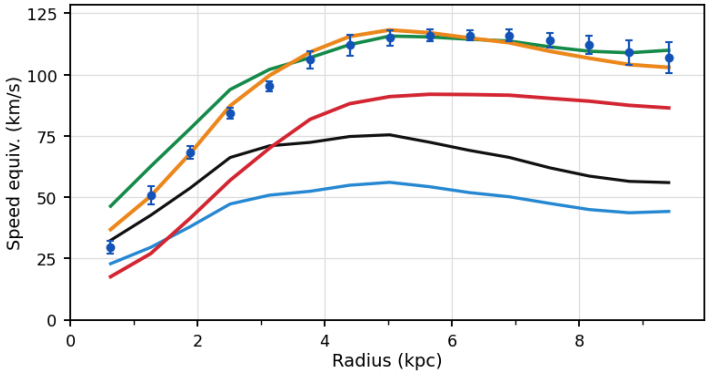


Figure 108: UGC 05986.

UGC 06399 9 radii

82.69% geometric; 17.31% material at $R_{\max} = 7.85$ kpc. Outer measured speed is 87.6 ± 4.9 km s^{-1}; CPTG gives 85.3 km s^{-1} and $\chi^2 = 0.53$ for 9 points. Mass equivalents at that radius are $M_{\rm total} = 13.277$, $M_{\rm mat} = 2.2981$, and $M_{\rm geom} = 10.979$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

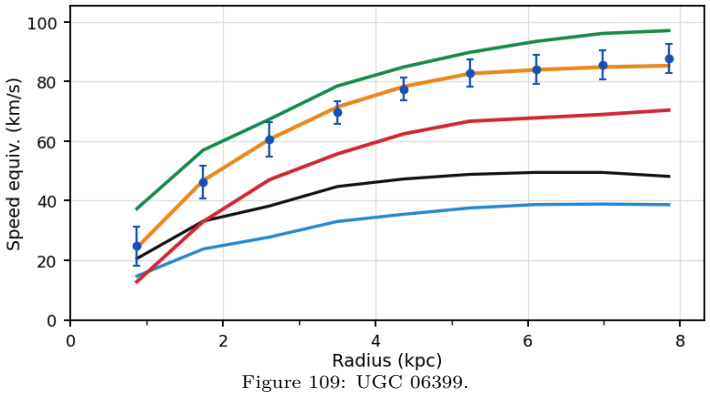


Figure 109: UGC 06399.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 06446 17 radii

87.38% geometric; 12.62% material at $R_{\max} = 10.22$ kpc. Outer measured speed is 80.1 ± 10.8 km s^{-1}; CPTG gives 85.6 km s^{-1} and $\chi^2 = 4.18$ for 17 points. Mass equivalents at that radius are $M_{\rm total} = 17.404$, $M_{\rm mat} = 2.1969$, and $M_{\rm geom} = 15.207$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

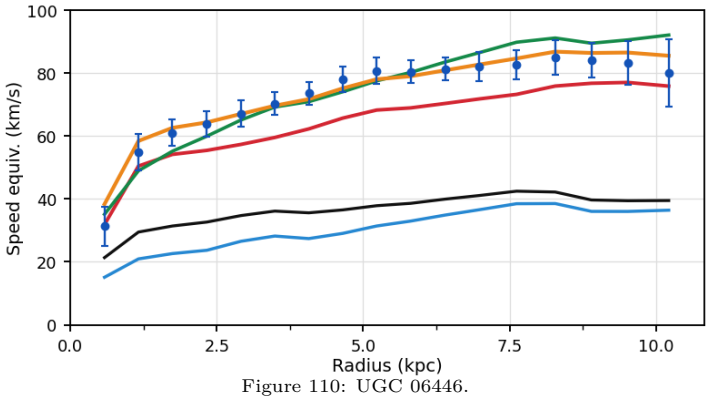


Figure 110: UGC 06446.

UGC 06614 13 radii

83.27% geometric; 16.73% material at $R_{\max} = 64.59$ kpc. Outer measured speed is 204.0 ± 16.0 km s^{-1}; CPTG gives 213.0 km s^{-1} and $\chi^2 = 1.37$ for 13 points. Mass equivalents at that radius are $M_{\rm total} = 681.38$, $M_{\rm mat} = 113.99$, and $M_{\rm geom} = 567.40$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

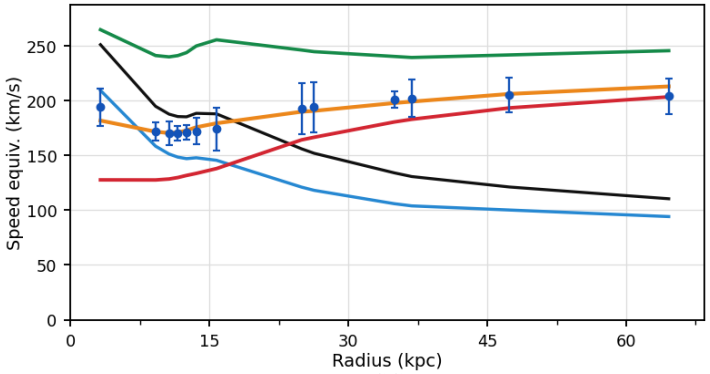


Figure 111: UGC 06614.

UGC 06667 9 radii

91.60% geometric; 8.40% material at $R_{\max} = 7.85$ kpc. Outer measured speed is 85.7 ± 3.4 km s^{-1}; CPTG gives 86.3 km s^{-1} and $\chi^2 = 0.31$ for 9 points. Mass equivalents at that radius are $M_{\rm total} = 13.602$, $M_{\rm mat} = 1.1425$, and $M_{\rm geom} = 12.460$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed.

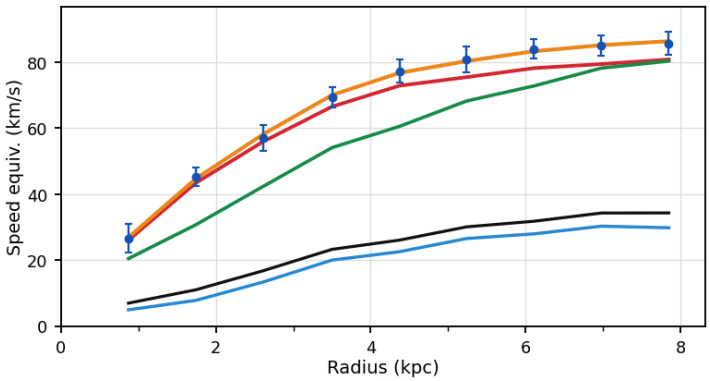


Figure 112: UGC 06667.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 06786 45 radii

89.22% geometric; 10.78% material at $R_{\max} = 34.05$ kpc. Outer measured speed is 211.0 ± 7.0 km s^{-1}; CPTG gives 232.2 km s^{-1} and $\chi^2 = 115.38$ for 45 points. Mass equivalents at that radius are $M_{\rm total} = 426.71$, $M_{\rm mat} = 46.015$, and $M_{\rm geom} = 380.69$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

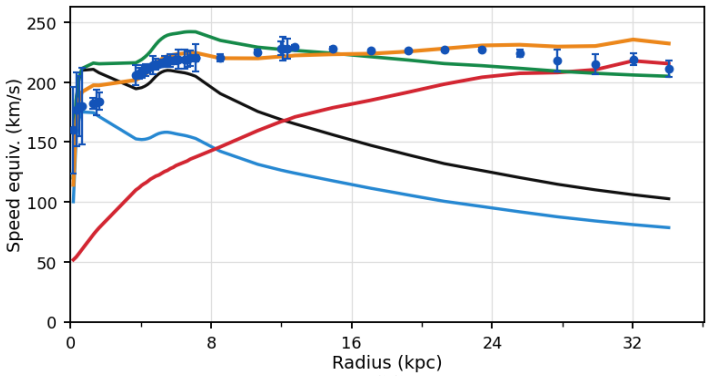


Figure 113: UGC 06786.

UGC 06787 71 radii

89.27% geometric; 10.73% material at $R_{\max} = 37.19$ kpc. Outer measured speed is 255.0 ± 4.2 km s^{-1}; CPTG gives 265.0 km s^{-1} and $\chi^2 = 1130.63$ for 71 points. Mass equivalents at that radius are $M_{\rm total} = 607.22$, $M_{\rm mat} = 65.169$, and $M_{\rm geom} = 542.05$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is below the last observed speed.

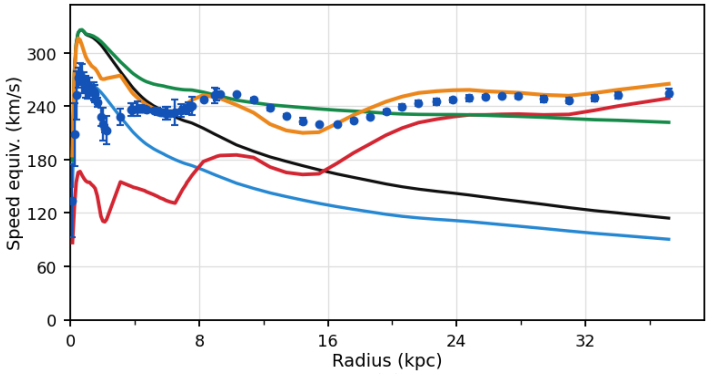


Figure 114: UGC 06787.

UGC 06818 8 radii

83.36% geometric; 16.64% material at $R_{\max} = 6.98$ kpc. Outer measured speed is 74.4 ± 6.0 km s^{-1}; CPTG gives 71.0 km s^{-1} and $\chi^2 = 2.39$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 8.1733$, $M_{\rm mat} = 1.3596$, and $M_{\rm geom} = 6.8136$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

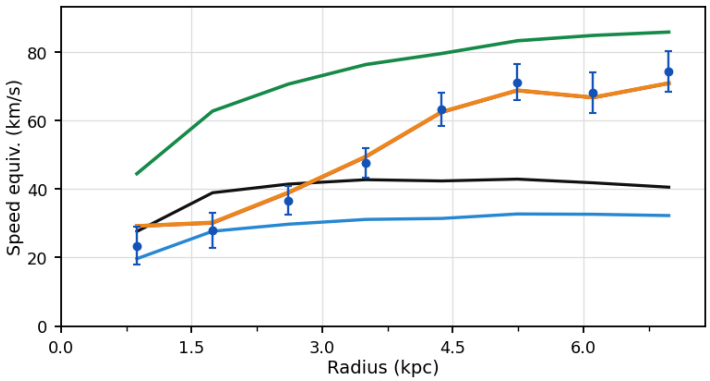


Figure 115: UGC 06818.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 06917 11 radii

79.83% geometric; 20.17% material at $R_{\max} = 10.47$ kpc. Outer measured speed is 111.0 ± 6.0 km s^{-1}; CPTG gives 112.5 km s^{-1} and $\chi^2 = 8.16$ for 11 points. Mass equivalents at that radius are $M_{\rm total} = 30.820$, $M_{\rm mat} = 6.2172$, and $M_{\rm geom} = 24.603$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

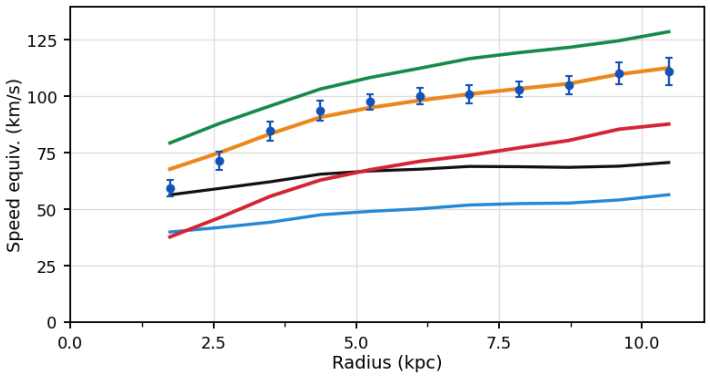


Figure 116: UGC 06917.

UGC 06923 6 radii

67.39% geometric; 32.61% material at $R_{\max} = 5.16$ kpc. Outer measured speed is 81.1 ± 5.1 km s^{-1}; CPTG gives 74.8 km s^{-1} and $\chi^2 = 5.54$ for 6 points. Mass equivalents at that radius are $M_{\rm total} = 6.7083$, $M_{\rm mat} = 2.1874$, and $M_{\rm geom} = 4.5208$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

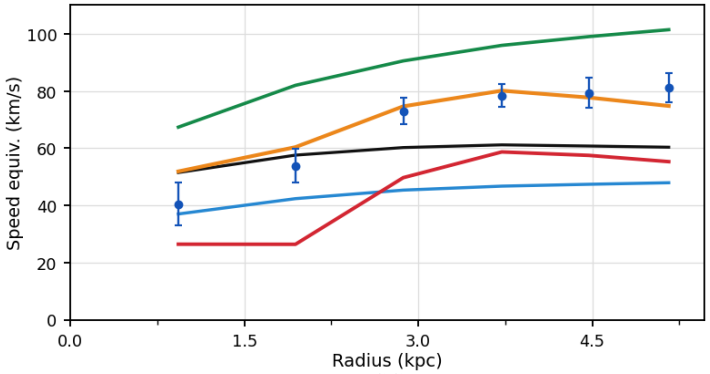


Figure 117: UGC 06923.

UGC 06930 10 radii

79.87% geometric; 20.13% material at $R_{\max} = 16.61$ kpc. Outer measured speed is 108.0 ± 7.4 km s^{-1}; CPTG gives 107.4 km s^{-1} and $\chi^2 = 1.87$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 44.525$, $M_{\rm mat} = 8.9634$, and $M_{\rm geom} = 35.561$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

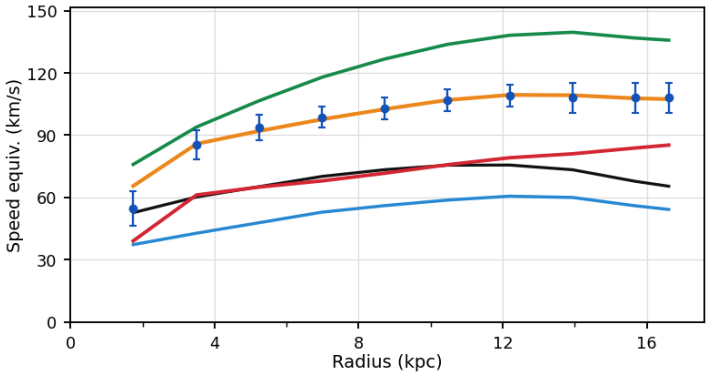


Figure 118: UGC 06930.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 06983 17 radii

85.96% geometric; 14.04% material at $R_{\max} = 15.68$ kpc. Outer measured speed is 109.0 ± 12.3 km s^{-1}; CPTG gives 115.5 km s^{-1} and $\chi^2 = 9.77$ for 17 points. Mass equivalents at that radius are $M_{\rm total} = 48.629$, $M_{\rm mat} = 6.8274$, and $M_{\rm geom} = 41.802$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

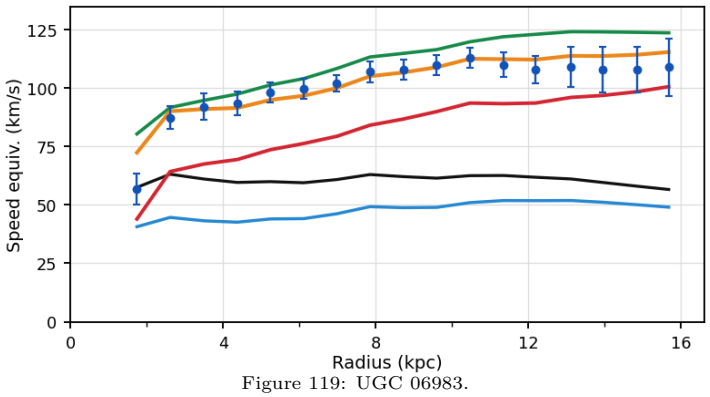


Figure 119: UGC 06983.

UGC 07089 12 radii

75.64% geometric; 24.36% material at $R_{\max} = 9.16$ kpc. Outer measured speed is 79.1 ± 7.3 km s^{-1}; CPTG gives 72.8 km s^{-1} and $\chi^2 = 8.43$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 11.279$, $M_{\rm mat} = 2.7475$, and $M_{\rm geom} = 8.5313$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

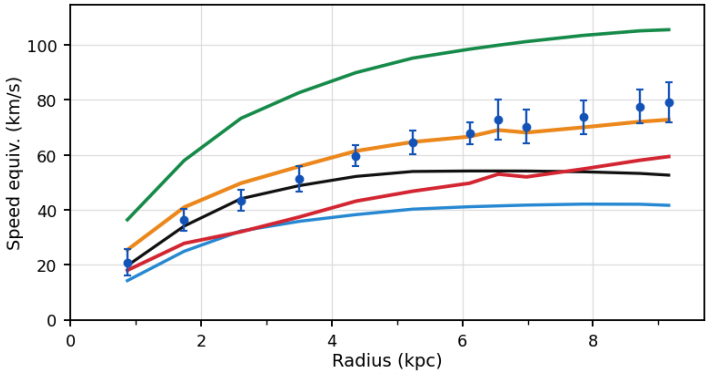


Figure 120: UGC 07089.

UGC 07125 13 radii

73.01% geometric; 26.99% material at $R_{\max} = 18.68$ kpc. Outer measured speed is 64.9 ± 3.5 km s^{-1}; CPTG gives 67.6 km s^{-1} and $\chi^2 = 4.77$ for 13 points. Mass equivalents at that radius are $M_{\rm total} = 19.820$, $M_{\rm mat} = 5.3491$, and $M_{\rm geom} = 14.471$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

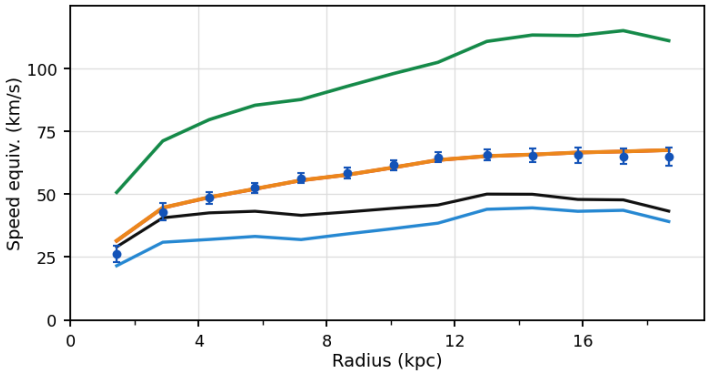


Figure 121: UGC 07125.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 07151 11 radii

73.60% geometric; 26.40% material at $R_{\max} = 5.50$ kpc. Outer measured speed is 76.2 ± 3.2 km s^{-1}; CPTG gives 76.8 km s^{-1} and $\chi^2 = 7.52$ for 11 points. Mass equivalents at that radius are $M_{\rm total} = 7.5506$, $M_{\rm mat} = 1.9933$, and $M_{\rm geom} = 5.5572$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

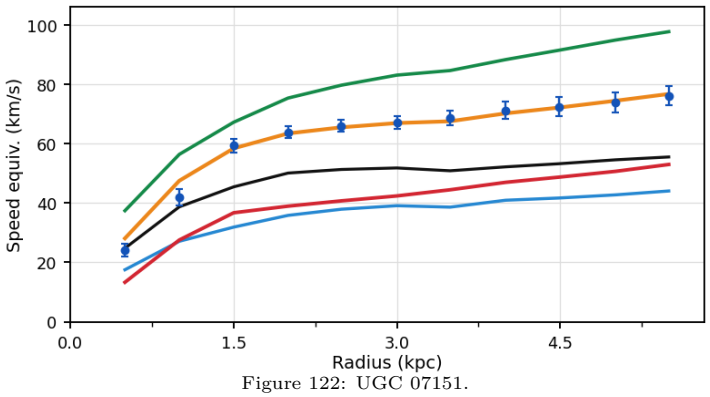


Figure 122: UGC 07151.

UGC 07232 4 radii

78.52% geometric; 21.48% material at $R_{\max} = 0.82$ kpc. Outer measured speed is 44.0 ± 3.3 km s^{-1}; CPTG gives 44.1 km s^{-1} and $\chi^2 = 0.38$ for 4 points. Mass equivalents at that radius are $M_{\rm total} = 0.3715$, $M_{\rm mat} = 0.0798$, and $M_{\rm geom} = 0.2917$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

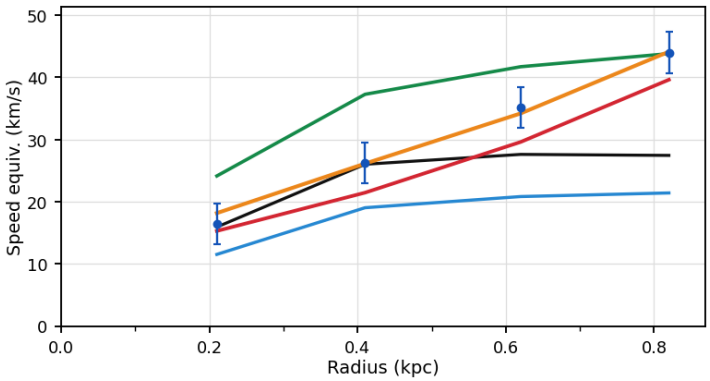


Figure 123: UGC 07232.

UGC 07261 7 radii

77.06% geometric; 22.94% material at $R_{\max} = 6.67$ kpc. Outer measured speed is 76.1 ± 5.7 km s^{-1}; CPTG gives 77.4 km s^{-1} and $\chi^2 = 0.12$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 9.2840$, $M_{\rm mat} = 2.1299$, and $M_{\rm geom} = 7.1540$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

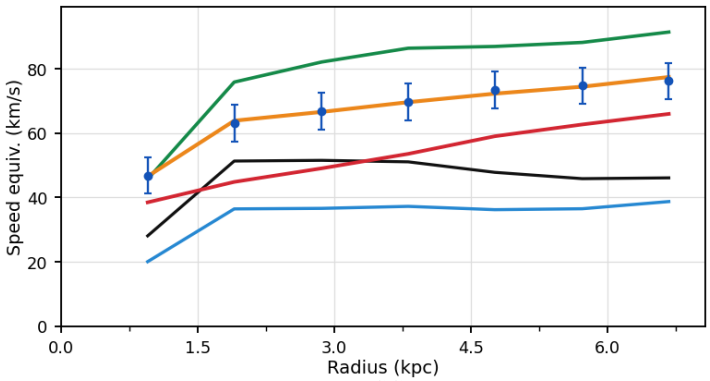


Figure 124: UGC 07261.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 07323 10 radii

69.67% geometric; 30.33% material at $R_{\max} = 5.82$ kpc. Outer measured speed is 85.6 ± 3.9 km s^{-1}; CPTG gives 82.6 km s^{-1} and $\chi^2 = 7.60$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 9.2384$, $M_{\rm mat} = 2.8019$, and $M_{\rm geom} = 6.4365$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

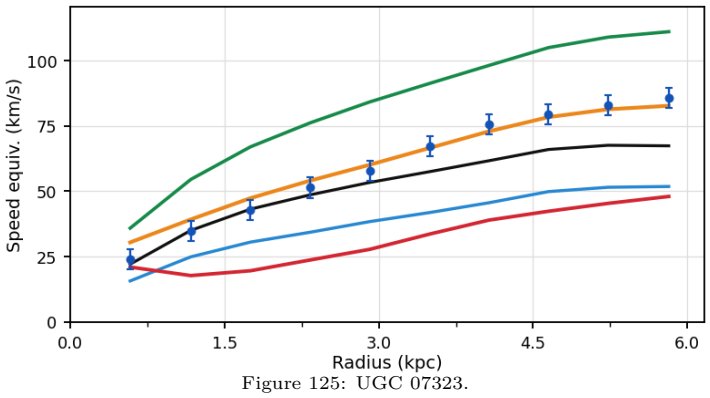


Figure 125: UGC 07323.

UGC 07399 10 radii

93.11% geometric; 6.89% material at $R_{\max} = 6.13$ kpc. Outer measured speed is 106.0 ± 5.0 km s^{-1}; CPTG gives 109.1 km s^{-1} and $\chi^2 = 7.64$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 16.973$, $M_{\rm mat} = 1.1702$, and $M_{\rm geom} = 15.803$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is below the last observed speed.

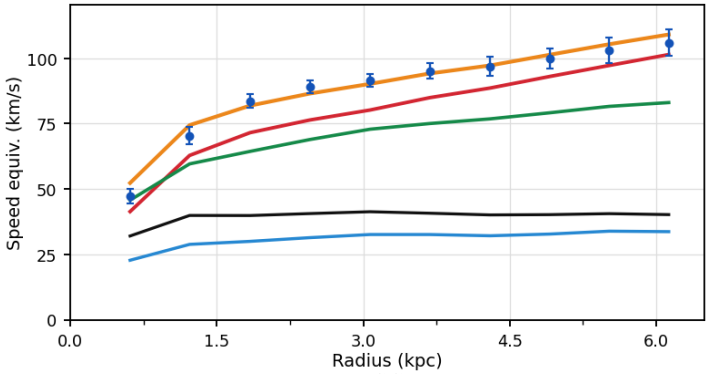


Figure 126: UGC 07399.

UGC 07524 31 radii

77.62% geometric; 22.38% material at $R_{\max} = 10.69$ kpc. Outer measured speed is 79.0 ± 2.8 km s^{-1}; CPTG gives 81.7 km s^{-1} and $\chi^2 = 8.76$ for 31 points. Mass equivalents at that radius are $M_{\rm total} = 16.606$, $M_{\rm mat} = 3.7158$, and $M_{\rm geom} = 12.891$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

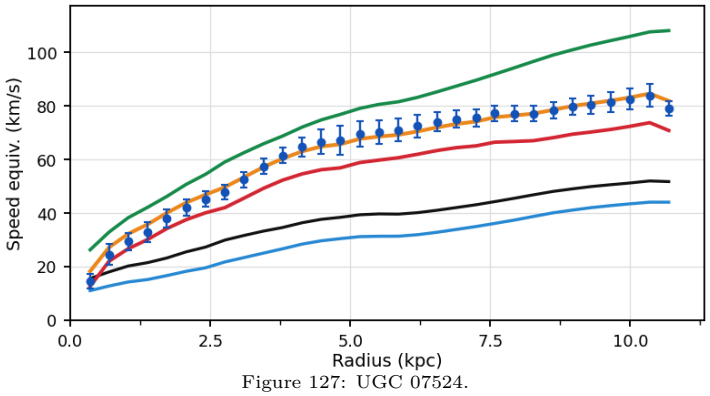


Figure 127: UGC 07524.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 07559 7 radii

68.07% geometric; 31.93% material at $R_{\max} = 2.53$ kpc. Outer measured speed is 32.1 ± 3.2 km s^{-1}; CPTG gives 31.8 km s^{-1} and $\chi^2 = 0.35$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 0.5949$, $M_{\rm mat} = 0.1899$, and $M_{\rm geom} = 0.4050$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

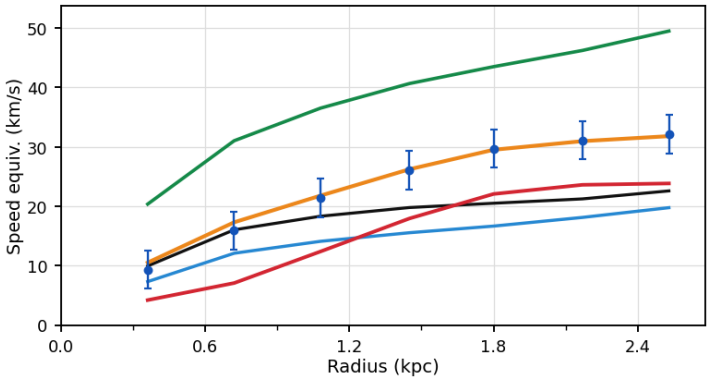


Figure 128: UGC 07559.

UGC 07577 9 radii

57.61% geometric; 42.39% material at $R_{\max} = 1.69$ kpc. Outer measured speed is 17.8 ± 3.2 km s^{-1}; CPTG gives 17.5 km s^{-1} and $\chi^2 = 0.06$ for 9 points. Mass equivalents at that radius are $M_{\rm total} = 0.1208$, $M_{\rm mat} = 0.0512$, and $M_{\rm geom} = 0.0696$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

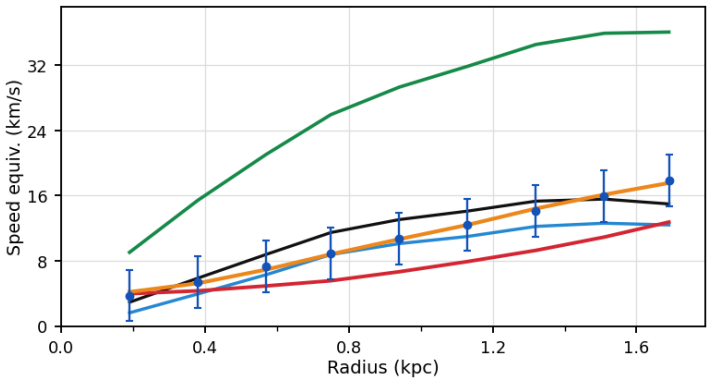


Figure 129: UGC 07577.

UGC 07603 12 radii

89.70% geometric; 10.30% material at $R_{\max} = 4.11$ kpc. Outer measured speed is 64.0 ± 2.9 km s^{-1}; CPTG gives 64.9 km s^{-1} and $\chi^2 = 10.29$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 4.0192$, $M_{\rm mat} = 0.4140$, and $M_{\rm geom} = 3.6052$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

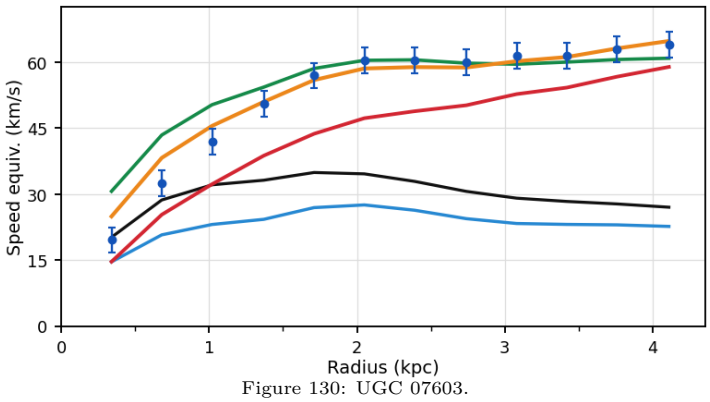


Figure 130: UGC 07603.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 07690 7 radii

74.23% geometric; 25.77% material at $R_{\max} = 4.13$ kpc. Outer measured speed is 55.9 ± 4.3 km s^{-1}; CPTG gives 56.9 km s^{-1} and $\chi^2 = 1.86$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 3.1085$, $M_{\rm mat} = 0.8011$, and $M_{\rm geom} = 2.3074$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

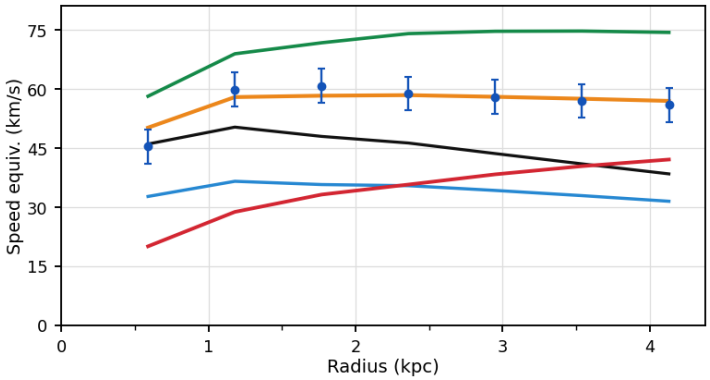


Figure 131: UGC 07690.

UGC 07866 7 radii

76.54% geometric; 23.46% material at $R_{\max} = 2.32$ kpc. Outer measured speed is 33.1 ± 4.1 km s^{-1}; CPTG gives 32.4 km s^{-1} and $\chi^2 = 0.13$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 0.5651$, $M_{\rm mat} = 0.1326$, and $M_{\rm geom} = 0.4325$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

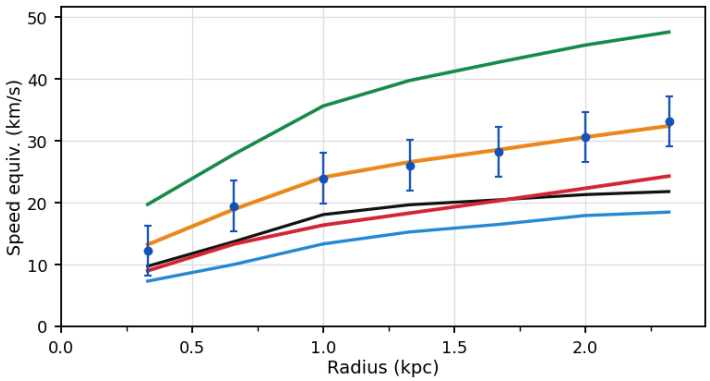


Figure 132: UGC 07866.

UGC 08286 17 radii

80.64% geometric; 19.36% material at $R_{\max} = 8.04$ kpc. Outer measured speed is 84.3 ± 2.0 km s^{-1}; CPTG gives 85.7 km s^{-1} and $\chi^2 = 3.24$ for 17 points. Mass equivalents at that radius are $M_{\rm total} = 13.730$, $M_{\rm mat} = 2.6585$, and $M_{\rm geom} = 11.071$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

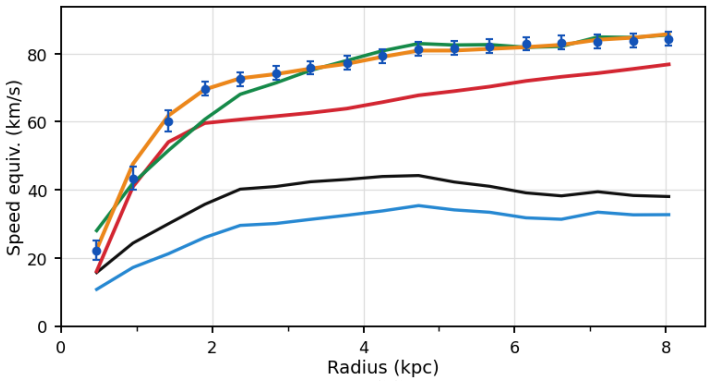


Figure 133: UGC 08286.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 08490 30 radii

90.16% geometric; 9.84% material at $R_{\max} = 10.15$ kpc. Outer measured speed is 77.6 ± 3.7 km s^{-1}; CPTG gives 82.2 km s^{-1} and $\chi^2 = 25.14$ for 30 points. Mass equivalents at that radius are $M_{\rm total} = 15.934$, $M_{\rm mat} = 1.5686$, and $M_{\rm geom} = 14.365$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

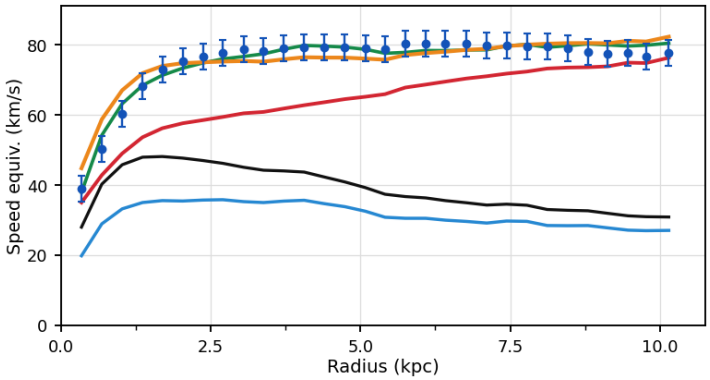


Figure 134: UGC 08490.

UGC 08550 11 radii

87.25% geometric; 12.75% material at $R_{\max} = 5.36$ kpc. Outer measured speed is 57.5 ± 2.0 km s^{-1}; CPTG gives 59.6 km s^{-1} and $\chi^2 = 2.95$ for 11 points. Mass equivalents at that radius are $M_{\rm total} = 4.4204$, $M_{\rm mat} = 0.5634$, and $M_{\rm geom} = 3.8570$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

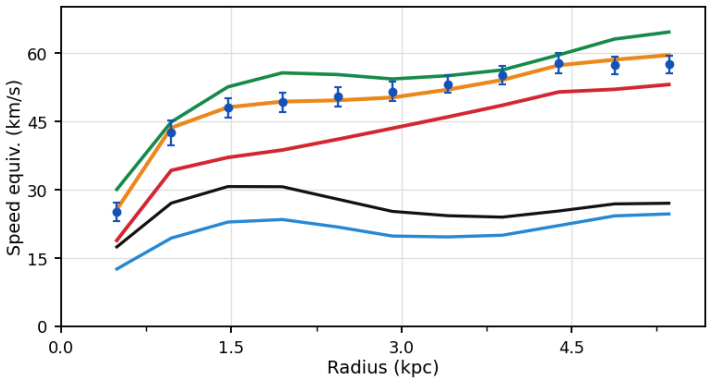


Figure 135: UGC 08550.

UGC 08699 41 radii

83.61% geometric; 16.39% material at $R_{\max} = 25.70$ kpc. Outer measured speed is 183.0 ± 7.5 km s^{-1}; CPTG gives 184.9 km s^{-1} and $\chi^2 = 6.15$ for 41 points. Mass equivalents at that radius are $M_{\rm total} = 204.25$, $M_{\rm mat} = 33.470$, and $M_{\rm geom} = 170.78$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

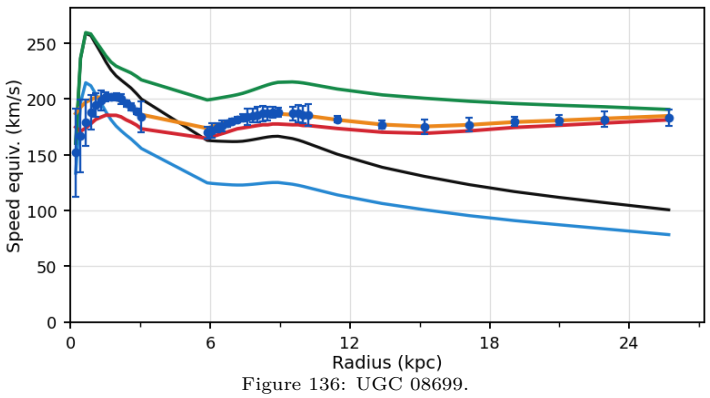


Figure 136: UGC 08699.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 08837 8 radii

67.16% geometric; 32.84% material at $R_{\max} = 4.20$ kpc. Outer measured speed is 48.0 ± 3.0 km s^{-1}; CPTG gives 45.7 km s^{-1} and $\chi^2 = 9.43$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 2.0355$, $M_{\rm mat} = 0.6684$, and $M_{\rm geom} = 1.3671$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

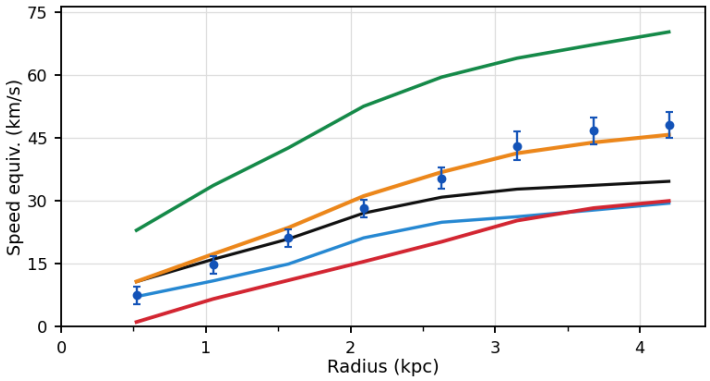


Figure 137: UGC 08837.

UGC 09037 22 radii

70.22% geometric; 29.78% material at $R_{\max} = 27.96$ kpc. Outer measured speed is 152.0 ± 16.0 km s^{-1}; CPTG gives 158.7 km s^{-1} and $\chi^2 = 85.69$ for 22 points. Mass equivalents at that radius are $M_{\rm total} = 163.68$, $M_{\rm mat} = 48.736$, and $M_{\rm geom} = 114.94$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

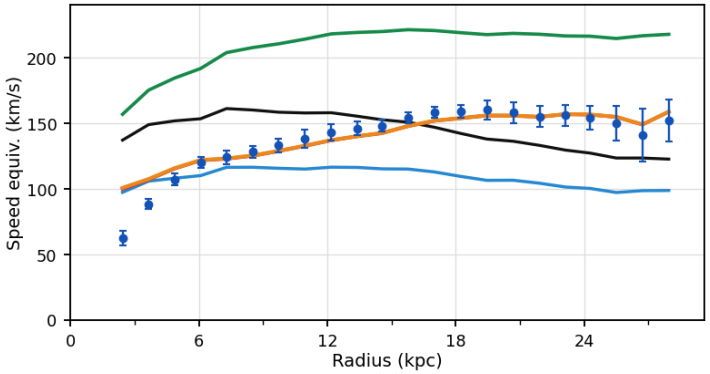


Figure 138: UGC 09037.

UGC 09133 68 radii

86.46% geometric; 13.54% material at $R_{\max} = 108.31$ kpc. Outer measured speed is 229.0 ± 4.1 km s^{-1}; CPTG gives 266.9 km s^{-1} and $\chi^2 = 1946.50$ for 68 points. Mass equivalents at that radius are $M_{\rm total} = 1,793.36$, $M_{\rm mat} = 242.84$, and $M_{\rm geom} = 1,550.52$ in $10^9 M_\odot$. This entry is also material/proxy flagged; the substantial χ^2 requires caution in interpreting its radial mass-equivalent fraction. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

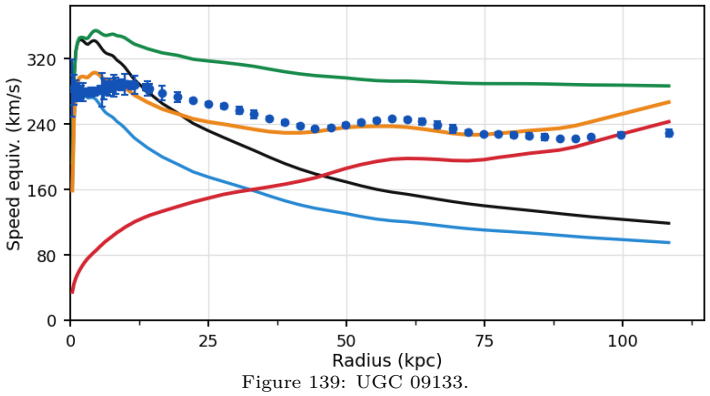


Figure 139: UGC 09133.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 09992 5 radii

65.08% geometric; 34.92% material at $R_{\max} = 3.89$ kpc. Outer measured speed is 34.3 ± 5.7 km s^{-1}; CPTG gives 34.2 km s^{-1} and $\chi^2 = 0.01$ for 5 points. Mass equivalents at that radius are $M_{\rm total} = 1.0551$, $M_{\rm mat} = 0.3684$, and $M_{\rm geom} = 0.6867$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

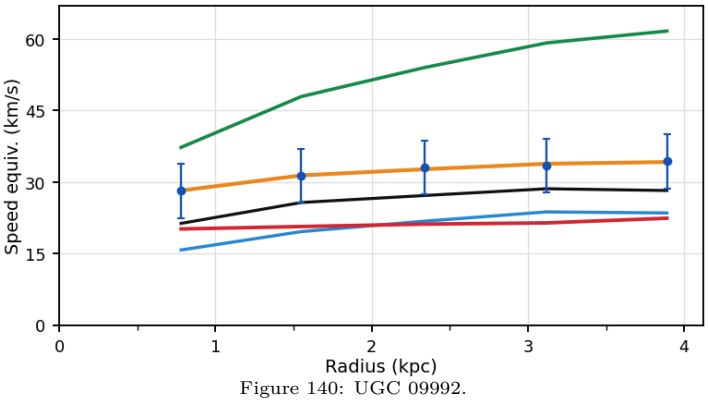


Figure 140: UGC 09992.

UGC 10310 7 radii

78.95% geometric; 21.05% material at $R_{\max} = 7.74$ kpc. Outer measured speed is 73.2 ± 5.1 km s^{-1}; CPTG gives 74.1 km s^{-1} and $\chi^2 = 0.58$ for 7 points. Mass equivalents at that radius are $M_{\rm total} = 9.8824$, $M_{\rm mat} = 2.0798$, and $M_{\rm geom} = 7.8026$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

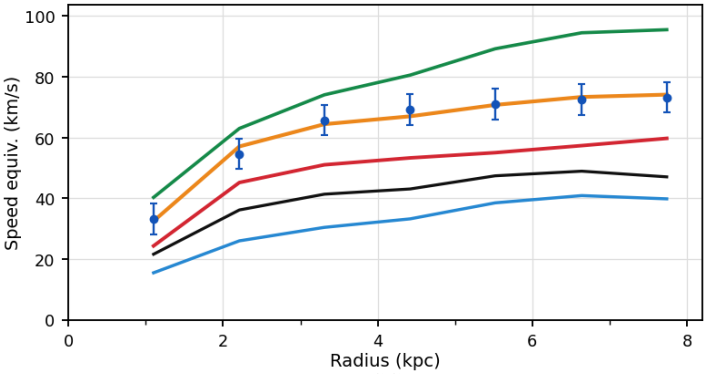


Figure 141: UGC 10310.

UGC 11455 36 radii

72.16% geometric; 27.84% material at $R_{\max} = 41.93$ kpc. Outer measured speed is 266.0 ± 8.8 km s^{-1}; CPTG gives 272.3 km s^{-1} and $\chi^2 = 470.64$ for 36 points. Mass equivalents at that radius are $M_{\rm total} = 723.05$, $M_{\rm mat} = 201.30$, and $M_{\rm geom} = 521.75$ in $10^9 M_\odot$. The fit has substantial error-weighted differences and is retained without exclusion. The fixed-a_0 MOND curve is above the last observed speed.

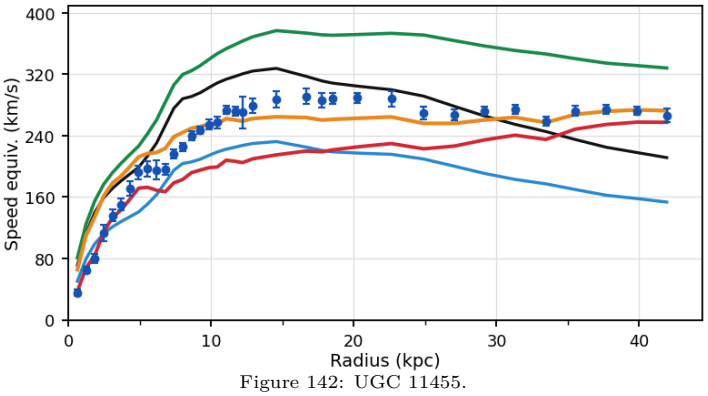


Figure 142: UGC 11455.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 11557 12 radii

40.32% geometric; 59.68% material at $R_{\max} = 10.56$ kpc. Outer measured speed is 84.5 ± 6.7 km s^{-1}; CPTG gives 72.4 km s^{-1} and $\chi^2 = 20.36$ for 12 points. Mass equivalents at that radius are $M_{\rm total} = 12.881$, $M_{\rm mat} = 7.6870$, and $M_{\rm geom} = 5.1940$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

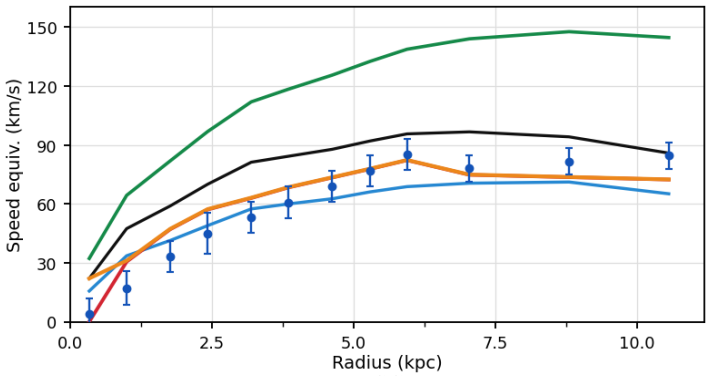


Figure 143: UGC 11557.

UGC 11820 10 radii

83.99% geometric; 16.01% material at $R_{\max} = 15.82$ kpc. Outer measured speed is 84.5 ± 1.3 km s^{-1}; CPTG gives 85.8 km s^{-1} and $\chi^2 = 5.28$ for 10 points. Mass equivalents at that radius are $M_{\rm total} = 27.068$, $M_{\rm mat} = 4.3322$, and $M_{\rm geom} = 22.736$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

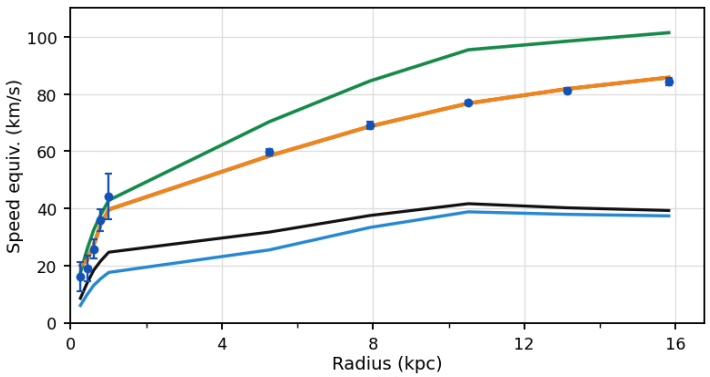


Figure 144: UGC 11820.

UGC 11914 65 radii

61.46% geometric; 38.54% material at $R_{\max} = 9.83$ kpc. Outer measured speed is 305.0 ± 5.1 km s^{-1}; CPTG gives 289.4 km s^{-1} and $\chi^2 = 99.61$ for 65 points. Mass equivalents at that radius are $M_{\rm total} = 191.37$, $M_{\rm mat} = 73.746$, and $M_{\rm geom} = 117.62$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

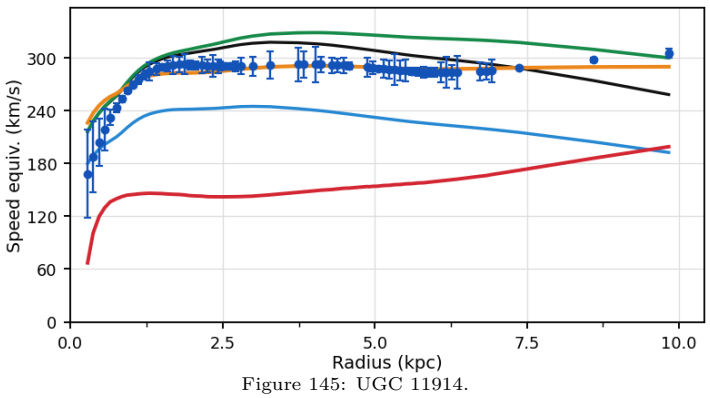


Figure 145: UGC 11914.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

UGC 12506 31 radii

74.08% geometric; 25.92% material at $R_{\max} = 49.99$ kpc. Outer measured speed is 225.0 ± 11.6 km s^{-1}; CPTG gives 226.3 km s^{-1} and $\chi^2 = 4.36$ for 31 points. Mass equivalents at that radius are $M_{\rm total} = 595.12$, $M_{\rm mat} = 154.23$, and $M_{\rm geom} = 440.88$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed. The modeled material aperture exceeds the catalog-wide photometric/H I proxy by over 30%; see Sec. 3.5.

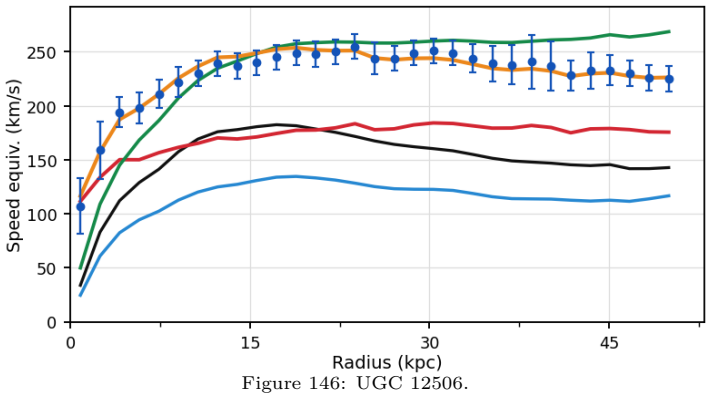


Figure 146: UGC 12506.

UGC 12632 15 radii

82.62% geometric; 17.38% material at $R_{\max} = 10.66$ kpc. Outer measured speed is 73.1 ± 2.8 km s^{-1}; CPTG gives 72.6 km s^{-1} and $\chi^2 = 2.33$ for 15 points. Mass equivalents at that radius are $M_{\rm total} = 13.060$, $M_{\rm mat} = 2.2702$, and $M_{\rm geom} = 10.790$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

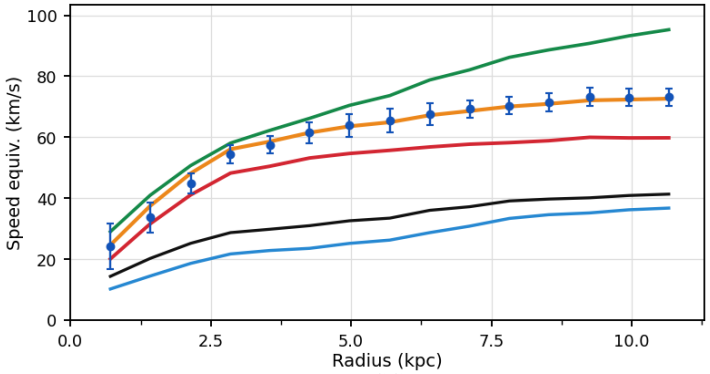


Figure 147: UGC 12632.

UGC 12732 16 radii

82.85% geometric; 17.15% material at $R_{\max} = 15.40$ kpc. Outer measured speed is 98.0 ± 3.2 km s^{-1}; CPTG gives 99.8 km s^{-1} and $\chi^2 = 4.14$ for 16 points. Mass equivalents at that radius are $M_{\rm total} = 35.679$, $M_{\rm mat} = 6.1183$, and $M_{\rm geom} = 29.561$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

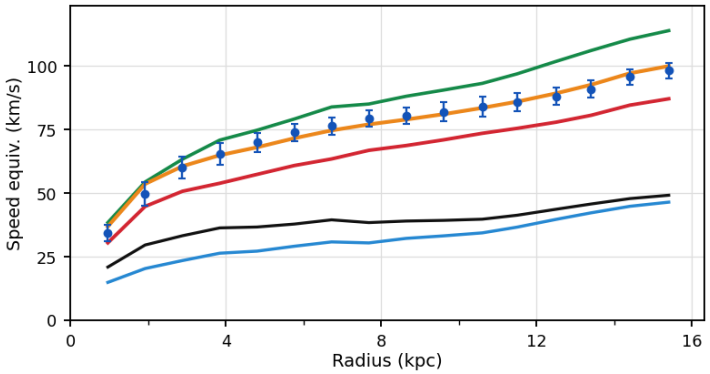


Figure 148: UGC 12732.

8 Remaining primary galaxies: CamB, D-series and UGCA

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

CamB 9 radii

58.57% geometric; 41.43% material at $R_{\max} = 1.79$ kpc. Outer measured speed is 20.1 ± 1.5 km s^{-1}; CPTG gives 17.4 km s^{-1} and $\chi^2 = 16.70$ for 9 points. Mass equivalents at that radius are $M_{\rm total} = 0.1255$, $M_{\rm mat} = 0.0520$, and $M_{\rm geom} = 0.0735$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND curve is above the last observed speed.

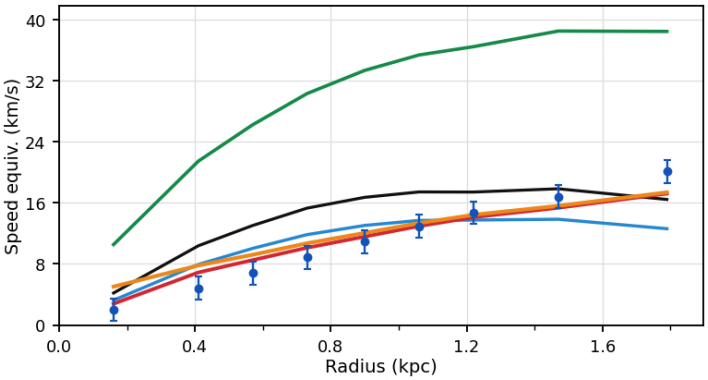


Figure 149: CamB.

D512-2 4 radii

85.18% geometric; 14.82% material at $R_{\max} = 3.83$ kpc. Outer measured speed is 35.9 ± 6.0 km s^{-1}; CPTG gives 36.7 km s^{-1} and $\chi^2 = 0.06$ for 4 points. Mass equivalents at that radius are $M_{\rm total} = 1.1986$, $M_{\rm mat} = 0.1776$, and $M_{\rm geom} = 1.0210$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

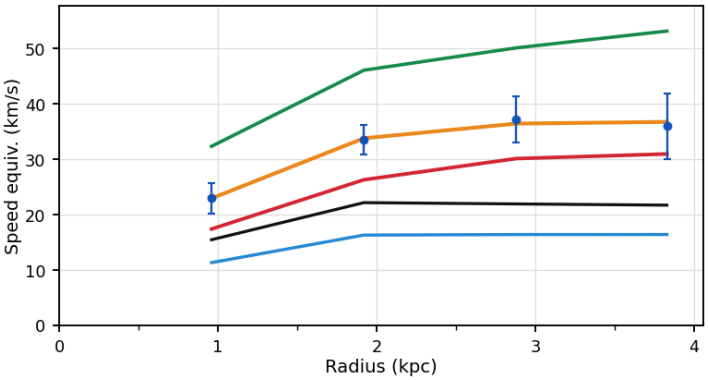


Figure 150: D512-2.

D564-8 6 radii

89.25% geometric; 10.75% material at $R_{\max} = 3.07$ kpc. Outer measured speed is 25.0 ± 2.0 km s^{-1}; CPTG gives 24.1 km s^{-1} and $\chi^2 = 0.65$ for 6 points. Mass equivalents at that radius are $M_{\rm total} = 0.4140$, $M_{\rm mat} = 0.0445$, and $M_{\rm geom} = 0.3695$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND curve is above the last observed speed.

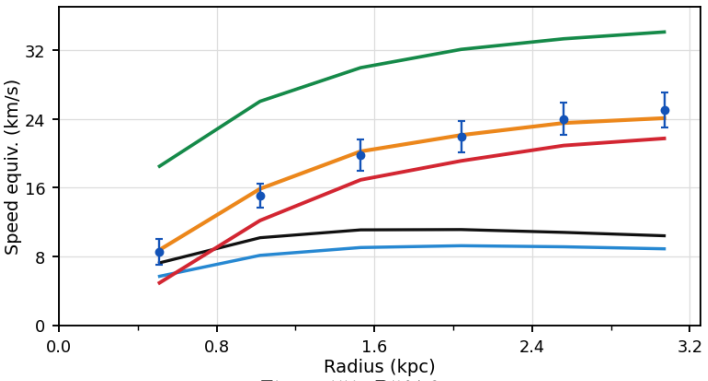


Figure 151: D564-8.

● ± Observed	Baryonic	Material	Geometric	CPTG	MOND

D631-7 16 radii

89.76% geometric; 10.24% material at $R_{\rm max} = 7.19$ kpc. Outer measured speed is 57.3 ± 3.9 km s^{-1}; CPTG gives 52.7 km s^{-1} and $\chi^2 = 60.18$ for 16 points. Mass equivalents at that radius are $M_{\rm total} = 4.6415$, $M_{\rm mat} = 0.4751$, and $M_{\rm geom} = 4.1664$ in $10^9 M_\odot$. Some measured radii retain significant model differences. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

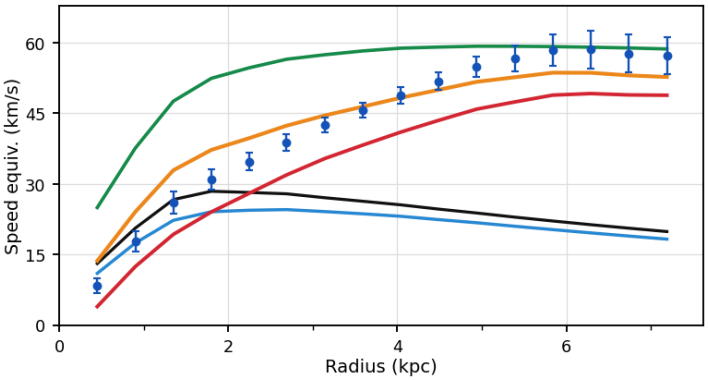


Figure 152: D631-7.

UGCA442 8 radii

90.68% geometric; 9.32% material at $R_{\rm max} = 6.33$ kpc. Outer measured speed is 56.5 ± 0.7 km s^{-1}; CPTG gives 56.6 km s^{-1} and $\chi^2 = 2.71$ for 8 points. Mass equivalents at that radius are $M_{\rm total} = 4.7219$, $M_{\rm mat} = 0.4402$, and $M_{\rm geom} = 4.2817$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

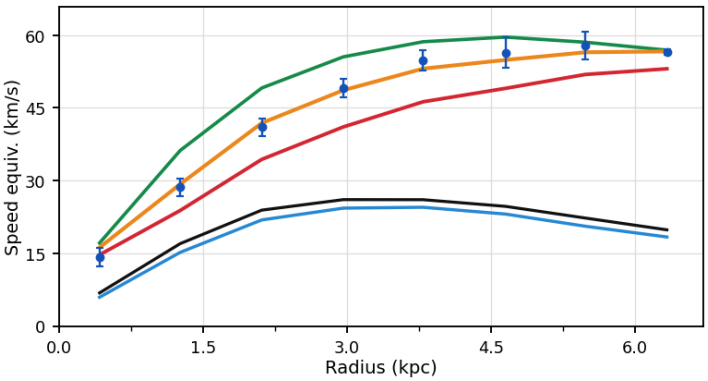


Figure 153: UGCA442.

UGCA444 36 radii

86.53% geometric; 13.47% material at $R_{\rm max} = 2.62$ kpc. Outer measured speed is 38.3 ± 7.8 km s^{-1}; CPTG gives 36.8 km s^{-1} and $\chi^2 = 0.57$ for 36 points. Mass equivalents at that radius are $M_{\rm total} = 0.8229$, $M_{\rm mat} = 0.1108$, and $M_{\rm geom} = 0.7120$ in $10^9 M_\odot$. The resolved velocity fit is close on average in the reported error scale. The fixed-a_0 MOND outer value lies within about 1.5σ of the observation.

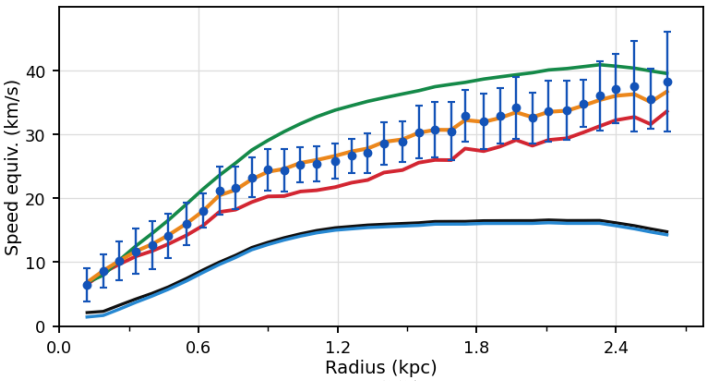


Figure 154: UGCA444.

9 Discussion and conclusion

A physical interpretation other than a material halo

The conventional mass-discrepancy question asks how much additional gravitating *matter* must be supplied to reproduce the orbital speeds of a disk galaxy. CPTG asks a different question: how much gravitational response can arise when the existing material source is coupled to an organized geometric polarization–transport field? In the parent theory, the source stays on the material side of equation (1), while the additional response belongs to the geometry. This distinction is fundamental. A large gravitational effect is not automatically evidence for a correspondingly large population of invisible material particles.

The present atlas makes that idea quantitative and inspectable for a large disk-galaxy sample. Each galaxy enters through its actual structural and kinematic data, the same registered radial CPTG equation is solved, and the corresponding modeled material inventory is constructed separately. The resulting complete gravitational field equivalent is decomposed into the specified material and geometric equivalents by equations (19)–(22). At the last observed radius, the sample has a 77.96% mean geometric-equivalent share, 80.33% median and 79.03% mass-equivalent-weighted share. These are substantial nonmaterial *gravitational equivalents* within the CPTG accounting; they do not imply that the same fraction of a galaxy's physical matter has been observed as a new substance.

The method's chief contribution is *uniformity and transparency*: one registered geometric field reduction and one independently specified material-reconstruction protocol are applied across 153 galaxies and 3,168 radii. The atlas does not introduce an object-by-object halo profile whose mass and scale must be selected to supply the reported geometric percentage. Nor does it impose a universal 80% fraction: galaxies return different values, as the worked examples demonstrate. Their registered structural parameters remain galaxy-specific and observation-conditioned, so the result should be described as a complete-field reconstruction, not as an untuned forward prediction of all rotation curves.

What the mass split means

The geometry–material separation is a direct property of the CPTG *field equation*; the numerical percentages are properties of a *declared galaxy mass-equivalent convention*. For a flattened disk, RV_C^2/G is a spherical reference-field equivalent, whereas the stellar and gas inventories are cylindrical aperture reconstructions with a declared bulge approximation. Their difference is therefore an informative geometric accounting remainder, not an invariant measurement of a new physical mass density. The 196 signed negative radial remainders in 49 galaxies are kept in the dataset and cannot be interpreted as negative material particles. This precise language strengthens rather than changes the core claim: CPTG offers a geometric explanation for part of the gravitational response that conventional halo interpretations attribute to additional matter.

The same geometric framework in galaxies and black holes. The parent CPTG equation also supplies the conceptual gravitational architecture used in compact-object applications: ordinary material stress–energy and the Einstein curvature remain present, while polarization and transport supply a coupled additional geometric response. The *realizations differ*. For compact objects the registered spacetime, boundary conditions and mass-state construction govern $M_\infty = M_c + M_{\rm geom}$ [7]; here the galaxy reduction operates on resolved circular support and defines $M_{\rm grav}^{\rm eq} = RV_C^2/G$ before comparing it with a separately reconstructed stellar/gas inventory. Thus the cross-scale claim concerns the same underlying geometric mechanism, not an identical enclosed-mass formula for a black hole and a flattened galaxy.

The atlas also retains observationally difficult objects, the 14 material/proxy flags and the six numerical replay differences without adjusting their fit coefficients or mass-to-light ratios to create agreement. Thirty-three galaxies have $\chi^2/N_{\rm radii} > 2$ under the registered observation-conditioned velocity reconstruction. Such cases show where the published reduced procedure remains incomplete in matching measured pointwise velocities; they are documented rather than concealed. Material-inventory normalizations, gas inversion assumptions, distances and inclinations likewise affect the precise percentages.

The contribution of this atlas

This paper supplies a reproducible, galaxy-resolved mass-equivalent inventory supporting CPTG as a *candidate geometric replacement for the added dark-halo component in rotation-curve modeling*. The alternative is specific: retain ordinary physical matter and the Einstein sector, compute the registered coupled curvature polarization–transport response, and report the completed gravitational field alongside the actual modeled material inventory. No new material halo is inserted

in the CPTG reconstruction. Establishing that the same dynamics also accounts for independent, held-out gravitational observations and other large-scale evidence remains an empirical question; the mass-equivalent closure alone is not that demonstration.

The 153-object atlas establishes the complete numerical starting point for that assessment. It turns a proposed physical distinction—*more gravitational response need not mean more physical matter*—into a clearly defined calculation that can be repeated, inspected and challenged one galaxy and one measured radius at a time.

Reproducibility and sample integrity

The numerical results reported in this study are reproducible using the companion archive `CPTG_SPARC_153_Numerical_Evidence.zip`. The archive contains the original SPARC observational inputs, catalog metadata, registered galaxy-specific fitted coefficients, CPTG gravitational-field calculations, material-mass reconstruction routines, and reference numerical results for all 153 galaxies and 3,168 observed radial measurements.

The numerical procedure reconstructs the CPTG polarization–transport gravitational field using the registered fixed-form equation and the supplied galaxy-specific coefficients. The physical material inventory is reconstructed separately from the stellar and gas distributions. The resulting gravitational and material mass equivalents determine the geometric mass equivalent at each measured radius.

The archive includes executable Python calculations, numerical reference tables, 4,754 inferred gas-density records, reproduction instructions, and SHA-256 checksums. Its contents are sufficient to reproduce and verify the reported results without re-optimizing the galaxy fits.

After extracting the archive and installing the documented Python dependencies, execute:

```
python calculate.py
python verify.py
```

The verification procedure checks the reconstructed numerical tables against the supplied reference results, including the 153-galaxy accounting, 3,168 radial measurements, and population mass-equivalent statistics. Complete reproduction instructions are provided in `README.md`.

The atlas follows the original SPARC rotation-model filenames, with 147 primary galaxies ordered from DDO 064 through UGC 12732, followed by CamB, D512-2, D564-8, D631-7, UGCA442, and UGCA444. The 22 systems outside the primary selection are excluded. The column `N` in the galaxy-summary data denotes the number of observed radial measurements, not Structural Mode N.

All calculations retain the registered gravitational equations, signed observational inputs, fitted coefficients, and geometric–material mass-equivalent definitions.

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